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Exercise list: *Poisson inference and Monte Carlo methods for inference*

Solutions

Instructions. Seven blocks, from Poisson likelihood theory to MCMC. Each block has (i) textbook exercises reproduced verbatim from the book PDF (statements from Portuguese-language books stay in Portuguese), (ii) an original math question (derivations and proofs, ending with a parameter analysis) and (iii) an original conceptual question in exam form. Every question carries a *Programming* part added to the book's text: solve it in Python (NumPy/SciPy/statsmodels) with the seeds given; the scaffold notebook `Listas/lista-poisson-monte-carlo.ipynb` lists the steps. The script `Listas/poisson-monte-carlo-check.py` reproduces and asserts every number printed in the solutions. Multiple choice (3 points): one correct option, no penalty. True/False (2 points): score = $\max\{0, \text{correct} - \text{wrong}/2\}\times$ item value; blank items do not count. Book abbreviations: B = Bolfarine & Sandoval (2010), G = Greenberg (2008, 1st ed.), I = ISLP (2023), MS = Morettin & Singer (2021). The coal-mining series used throughout (British mine disasters per year, 1851–1962, Jarrett 1979) is

```
1851-1878:  4 5 4 1 0 4 3 4 0 6 3 3 4 0 2 6 3 3 5 4 5 3 1 4 4 1 5 5
1879-1906:  3 4 2 5 2 2 3 4 2 1 3 2 2 1 1 1 1 3 0 0 1 0 1 1 0 0 3 1
1907-1934:  0 3 2 2 0 1 1 1 0 1 0 1 0 0 0 2 1 0 0 0 1 1 0 2 3 3 1 1
1935-1962:  2 1 1 1 1 2 4 2 0 0 0 1 4 0 0 0 1 0 0 0 0 0 1 0 0 1 0 1
```

($n = 112$ years, 191 disasters; 127 in the first 41 years, 1851–1891).

Block 1 — Poisson likelihood: sufficiency, information, Cramér–Rao, UMVU *(B 2.4, 2.6, 2.5; original)*

Question 1. B 2.4 — $X_1 + 2X_2$ is not sufficient for a Poisson mean.

Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 2, §2.6 Exercícios, Exercício 2.4, p. 32 [PDF p. 38]; the book gives no answer — reproduced verbatim.

2.4. Sejam X_1, X_2 uma amostra aleatória da variável aleatória $X \sim Poisson(\theta)$. Mostre que $T = X_1 + 2X_2$ não é suficiente para θ.

Programming (added, not in the book). Write `cond_pmf(t, theta, w2)` that enumerates every pair (x_1, x_2) with $x_1 + w_2 x_2 = t$ and returns $P(X_1 = x_1, X_2 = x_2 \mid X_1 + w_2 X_2 = t)$. Print it for $T = X_1 + 2X_2$ ($w_2 = 2$) at $t = 2$ and $t = 4$, for $\theta = 1$ and $\theta = 2$. Repeat for $S = X_1 + X_2$ ($w_2 = 1$) at $s = 4$ and check that the result is the Binomial$(4, 1/2)$ p.m.f. for both values of θ.

Solution.

Notation.

- X_1, X_2 i.i.d. Poisson(θ), $P(X = x) = \theta^x e^{-\theta}/x!$, $\theta > 0$; $T = X_1 + 2X_2$; $S = X_1 + X_2$.
- Sufficiency (Definição 2.2.1): $P(\mathbf{X} = \mathbf{x} \mid T = t)$ does not depend on θ.

Step — Pick a value of T. The pairs with $T = 2$ are

$$\{T = 2\} = \{(X_1, X_2) = (2, 0)\} \cup \{(X_1, X_2) = (0, 1)\}. \tag{1.1}$$

Step — Probabilities. By independence,

$$P(X_1 = 2, X_2 = 0) = \frac{\theta^2 e^{-\theta}}{2!} \cdot e^{-\theta} = \frac{\theta^2 e^{-2\theta}}{2}, \tag{1.2}$$

$$P(X_1 = 0, X_2 = 1) = e^{-\theta} \cdot \theta e^{-\theta} = \theta e^{-2\theta}. \tag{1.3}$$

Adding (1.2) and (1.3) as (1.1) requires:

$$P(T = 2) = e^{-2\theta}\Big(\frac{\theta^2}{2} + \theta\Big). \tag{1.4}$$

Step — Conditional probability. Dividing (1.2) by (1.4):

$$P(X_1 = 2, X_2 = 0 \mid T = 2) = \frac{\theta^2 e^{-2\theta}/2}{e^{-2\theta}(\theta^2/2 + \theta)} = \frac{\theta^2/2}{\theta^2/2 + \theta}.$$

Dividing numerator and denominator by $\theta/2$:

$$P(X_1 = 2, X_2 = 0 \mid T = 2) = \frac{\theta}{\theta + 2}, \tag{1.5}$$

which depends on θ ($1/3 = 0.3333$ at $\theta = 1$, $2/4 = 0.5$ at $\theta = 2$). So T is not sufficient.

Step — Contrast with S. For $S = X_1 + X_2$, $S \sim \text{Poisson}(2\theta)$, and for $0 \le x_1 \le s$

$$P(X_1 = x_1 \mid S = s) = \frac{\frac{\theta^{x_1} e^{-\theta}}{x_1!} \frac{\theta^{s-x_1} e^{-\theta}}{(s-x_1)!}}{\frac{(2\theta)^s e^{-2\theta}}{s!}} = \binom{s}{x_1} \left(\frac{1}{2}\right)^s, \tag{1.6}$$

free of θ: S is sufficient (as the factorization criterion also shows).

Code and output (programming part).

```
def cond_pmf(t, theta, w2):
    pts = [(t - w2 * x2, x2) for x2 in range(t // w2 + 1)]
    pr = np.array([poisson.pmf(a, theta) * poisson.pmf(b, theta) for a, b in pts])
    return pts, pr / pr.sum()
# T=2:  theta=1 -> P((2,0)|T=2) = 0.3333 ; theta=2 -> 0.5000
# T=4:  pairs (4,0),(2,1),(0,2): theta=1 -> 0.04, 0.48, 0.48
#                                theta=2 -> 0.10, 0.60, 0.30
# S=4:  theta=1 and theta=2 -> 0.0625, 0.25, 0.375, 0.25, 0.0625 = Binomial(4, 1/2)
```

The $t = 4$ row confirms (1.5) is not an accident of $t = 2$: the conditional law moves with θ, whereas (1.6) does not.

Intuition. The sufficient statistic here is $X_1 + X_2$. Weighting X_2 twice merges sample points with different totals: $T = 2$ mixes "two events" with "one event", and the split between these cases still carries information about θ.

Ref: Bolfarine & Sandoval (2010), §2.2 (estatísticas suficientes, pp. 20–23; Exemplo 2.2.4).

Question 2. B 2.6 — the information identity.

Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 2, §2.6 Exercícios, Exercício 2.6, p. 33 [PDF p. 39]; the book gives no answer — reproduced verbatim.

2.6. Seja $f(x|\theta)$ uma função densidade para a qual as condições de regularidade estão satisfeitas. Mostre que

$$E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \log f(X|\theta)}{\partial \theta^2}\right].$$

Programming (added, not in the book). (i) For $X \sim \text{Poisson}(\theta)$ with $\theta = 3$, compute $E[U]$, $E[U^2]$ and $-E[\partial^2 \log f/\partial\theta^2]$, with $U = \partial \log f(X|\theta)/\partial\theta$, by summing over $x = 0, \ldots, 399$, and compare with $1/\theta$. (ii) Do the same for $X \sim U(0, \theta)$ with $\theta = 2$, by numerical integration (`scipy.integrate.quad`), and explain the result.

Solution.

Notation.

- $f(x|\theta)$: density (sums replace integrals for a p.m.f.); θ in an open interval Θ.
- $U(\theta) = \partial \log f(X|\theta)/\partial\theta$: score of one observation; $I_F(\theta) = E[U^2]$: Fisher information.
- Regularity conditions (B §2.3): the support $A = \{x : f(x|\theta) > 0\}$ does not depend on θ, and $\int f(x|\theta)dx$ can be differentiated twice under the integral sign.

Step — The density integrates to one. For every $\theta \in \Theta$,

$$\int_A f(x|\theta)\, dx = 1. \tag{2.1}$$

Differentiating (2.1) in θ under the integral sign (regularity):

$$\int_A \frac{\partial f(x|\theta)}{\partial \theta}\, dx = 0. \tag{2.2}$$

Step — Score has mean zero. On A, the chain rule for the logarithm gives

$$\frac{\partial f(x|\theta)}{\partial \theta} = \frac{\partial \log f(x|\theta)}{\partial \theta}\, f(x|\theta). \tag{2.3}$$

Substituting (2.3) into (2.2):

$$\int_A \Big(\frac{\partial \log f(x|\theta)}{\partial \theta}\, f(x|\theta)\Big) dx = 0 \quad \Longleftrightarrow \quad E[U] = 0. \tag{2.4}$$

Step — Differentiate once more. Differentiating the left side of (2.4) under the integral sign and using the product rule:

$$\int_A \Big[\frac{\partial^2 \log f(x|\theta)}{\partial \theta^2} f(x|\theta) + \frac{\partial \log f(x|\theta)}{\partial \theta}\, \frac{\partial f(x|\theta)}{\partial \theta}\Big] dx = 0. \tag{2.5}$$

Substituting (2.3) for $\partial f/\partial\theta$ in the second term of (2.5):

$$\int_A \Big[\frac{\partial^2 \log f(x|\theta)}{\partial \theta^2} f(x|\theta) + \frac{\partial \log f(x|\theta)}{\partial \theta}\Big(\frac{\partial \log f(x|\theta)}{\partial \theta} f(x|\theta)\Big)\Big] dx = 0.$$

Writing both integrals as expectations:

$$E\Big[\frac{\partial^2 \log f(X|\theta)}{\partial \theta^2}\Big] + E\Big[\Big(\frac{\partial \log f(X|\theta)}{\partial \theta}\Big)^2\Big] = 0, \tag{2.6}$$

which rearranges to the identity. ■

Step — Poisson check. $\log f = x\log\theta - \theta - \log x!$, so

$$U = \frac{X}{\theta} - 1, \qquad \frac{\partial^2 \log f}{\partial \theta^2} = -\frac{X}{\theta^2}. \tag{2.7}$$

From (2.7) and $E(X) = \mathrm{Var}(X) = \theta$: $E[U^2] = \mathrm{Var}(X)/\theta^2 = 1/\theta$ and $-E[-X/\theta^2] = \theta/\theta^2 = 1/\theta$; both sides of (2.6) equal $1/\theta$.

Step — Where regularity fails. For $U(0,\theta)$, $\log f = -\log\theta$ on $(0,\theta)$, so the score is the constant $-1/\theta$ and $\partial^2 \log f/\partial\theta^2 = 1/\theta^2$. Then $E[U] = -1/\theta \neq 0$, $E[U^2] = 1/\theta^2$ but $-E[\partial^2 \log f/\partial\theta^2] = -1/\theta^2$. The step (2.2) is what breaks: by Leibniz's rule $0 = \frac{d}{d\theta}\int_0^\theta \frac{1}{\theta} dx = \frac{1}{\theta} + \int_0^\theta (-\frac{1}{\theta^2}) dx$, the boundary term $1/\theta$ appears because the support moves with θ.

Code and output (programming part).

```
th = 3.0; x = np.arange(400); f = poisson.pmf(x, th)
U, H = x / th - 1, -x / th**2
print(np.sum(f * U), np.sum(f * U**2), -np.sum(f * H))   # 0.000000  0.333333  0.333333
th = 2.0                                      # U(0, 2): score = -1/th on (0, th)
EU  = quad(lambda u: (-1 / th) / th, 0, th)[0]           # -0.5
EU2 = quad(lambda u: (1 / th**2) / th, 0, th)[0]         #  0.25
mEH = -quad(lambda u: (1 / th**2) / th, 0, th)[0]        # -0.25
```

The Poisson values equal $1/\theta = 0.333333$; for the uniform $E[U^2] = 0.25$ but $-E[\partial^2 \log f/\partial\theta^2] = -0.25$ and $E[U] = -0.5$, as derived.

Intuition. The score has mean zero because the model's total probability cannot change with θ; information measures how much the score fluctuates, and the curvature of the expected log-likelihood measures the same thing. When the support depends on θ, the data can rule values out sharply, the likelihood is not smooth at its peak, and both identities fail.

Ref: Bolfarine & Sandoval (2010), §2.3 (desigualdade da informação, pp. 23–28; Exemplo 2.3.1, Poisson); §3.1 (Exemplo 3.1.1, the uniform).

Question 3. B 2.5 — the Cramér–Rao bound for $g(\theta)$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 2, §2.6 Exercícios, Exercício 2.5, p. 33 [PDF p. 39]; the book gives no answer — reproduced verbatim.

2.5. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória X com função de densidade (ou de probabilidade) $f(x|\theta)$ para a qual as condições de regularidade estão satisfeitas. Seja $\hat{\gamma}$ um estimador não viciado para $g(\theta)$. Mostre que

$$Var(\hat{\gamma}) \geq \frac{(g'(\theta))^2}{nE\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right]}.$$

Programming (added, not in the book). Take $X \sim \text{Poisson}(\theta)$, $g(\theta) = \theta$, $\theta = 2$, $n = 10$. With `np.random.default_rng(3)` draw 200,000 samples of size n; for each compute $\overline{X}$ and $S^2 = \sum_i (X_i - \overline{X})^2/(n-1)$ (both unbiased for θ). Print the simulated means and variances of both estimators next to the bound θ/n and to the exact $\text{Var}(S^2) = \theta/n + 2\theta^2/(n-1)$.

Solution.

Notation.

- $\mathbf{X} = (X_1, \ldots, X_n)$ i.i.d. $f(x|\theta)$; $L(\theta; \mathbf{x}) = \prod_i f(x_i|\theta)$.
- $U_n = \sum_{i=1}^n \partial \log f(X_i|\theta)/\partial\theta$: score of the sample; $I_F(\theta) = E[(\partial \log f(X|\theta)/\partial\theta)^2]$.
- $\hat{\gamma} = \hat{\gamma}(\mathbf{X})$ with $E_\theta(\hat{\gamma}) = g(\theta)$ for all θ; g differentiable.

Step — Unbiasedness, differentiated. Unbiasedness reads

$$\int \hat{\gamma}(\mathbf{x})\, L(\theta; \mathbf{x})\, d\mathbf{x} = g(\theta). \tag{3.1}$$

Differentiating (3.1) under the integral (regularity) and substituting $\partial L/\partial\theta = U_n(\mathbf{x})L$, the sample analogue of (2.3):

$$\int \hat{\gamma}(\mathbf{x})\, (U_n(\mathbf{x})L(\theta; \mathbf{x}))\, d\mathbf{x} = g'(\theta) \quad \Longleftrightarrow \quad E[\hat{\gamma}\, U_n] = g'(\theta). \tag{3.2}$$

Step — It is a covariance. Each term of U_n has mean zero by (2.4) (Question 2), so $E[U_n] = 0$ and (3.2) becomes

$$\text{Cov}(\hat{\gamma}, U_n) = E[\hat{\gamma} U_n] - E[\hat{\gamma}]\,(0) = g'(\theta). \tag{3.3}$$

Step — Variance of the score. The terms of U_n are i.i.d. with mean zero, so

$$\text{Var}(U_n) = \sum_{i=1}^n \text{Var}\left(\frac{\partial \log f(X_i|\theta)}{\partial\theta}\right) = nE\Big[\Big(\frac{\partial \log f(X|\theta)}{\partial\theta}\Big)^2\Big] = nI_F(\theta). \tag{3.4}$$

Step — Cauchy–Schwarz. For any two variables with finite variance, $\text{Cov}(A, B)^2 \leq \text{Var}(A)\,\text{Var}(B)$. With $A = \hat{\gamma}$, $B = U_n$, substituting (3.3) and (3.4):

$$(g'(\theta))^2 \leq \text{Var}(\hat{\gamma})\,(nI_F(\theta)).$$

Dividing by $nI_F(\theta) > 0$:

$$\text{Var}(\hat{\gamma}) \geq \frac{(g'(\theta))^2}{nI_F(\theta)}. \qquad \blacksquare \tag{3.5}$$

Equality in Cauchy–Schwarz holds iff $U_n = a(\theta)\,(\hat{\gamma} - g(\theta))$ with probability one: the bound is attained only by estimators that are affine in the score.

Step — Poisson, $g(\theta) = \theta$**.** From (2.7), $I_F(\theta) = 1/\theta$, so (3.5) gives the bound θ/n. $U_n = \sum_i (X_i/\theta - 1) = \frac{n}{\theta}(\overline{X} - \theta)$ is affine in $\overline{X}$, so $\overline{X}$ attains it: $\text{Var}(\overline{X}) = \theta/n$. The sample variance S^2

is also unbiased ($\operatorname{Var} X = \theta$), but by the standard formula $\operatorname{Var}(S^2) = \mu_4/n - \sigma^4(n-3)/(n(n-1))$ with the Poisson moments $\sigma^2 = \theta$, $\mu_4 = \theta + 3\theta^2$:

$$\operatorname{Var}(S^2) = \frac{\theta + 3\theta^2}{n} - \frac{\theta^2(n-3)}{n(n-1)} = \frac{\theta}{n} + \frac{2\theta^2}{n-1} > \frac{\theta}{n}. \tag{3.6}$$

At $\theta = 2$, $n = 10$: bound $= 2/10 = 0.2$, and (3.6) gives $0.2 + 2(4)/9 = 0.2 + 0.888889 = 1.088889$.

Code and output (programming part).

```
rng = np.random.default_rng(3)
X = rng.poisson(2.0, size=(200_000, 10))
xb, s2 = X.mean(1), X.var(1, ddof=1)
print(xb.mean(), xb.var(ddof=1))   # 2.0014  0.1997   (bound 0.2)
print(s2.mean(), s2.var(ddof=1))   # 1.9995  1.0871   (exact 1.088889)
```

Both estimators are unbiased; only $\overline{X}$ sits at the bound, and S^2 is more than five times as variable, as (3.6) predicts.

Intuition. An unbiased estimator must move one-for-one with $g(\theta)$; the score is the direction in which the data move most when θ moves. The covariance between the two is pinned at $g'(\theta)$, so an estimator can only be precise if it loads heavily on the score. $\overline{X}$ is the score up to scale; S^2 also reacts to the spread of the data, which adds noise.

Ref: Bolfarine & Sandoval (2010), §2.3, Teorema 2.3.1 and Corolário (pp. 24–27).

Question 4. Original math question — estimating the probability of an accident-free day.
Source: Original question (not from a book).

The Federal Highway Police (PRF) counts serious accidents per day on a stretch of the BR-116. Model the daily counts $X_1, \ldots, X_n$ as i.i.d. Poisson(θ), $\theta > 0$, and let $T = \sum_{i=1}^n X_i$. The indicator the PRF reports is $\psi = P(X = 0) = e^{-\theta}$.

(a) Find the maximum likelihood estimator $\hat{\psi}$ of ψ, its exact expectation, and the sign of its bias for every $n \geq 1$ and $\theta > 0$.

(b) Find the uniformly minimum-variance unbiased estimator (UMVU) $\tilde{\psi}$ of ψ. Prove every step (sufficiency and completeness of T may be quoted from B, Exemplo 2.5.4).

(c) Find $\operatorname{Var}(\tilde{\psi})$ in closed form and compare it with the Cramér–Rao bound for $g(\theta) = e^{-\theta}$. Is there an efficient unbiased estimator of ψ for finite n? Prove your answer.

(d) Parameter analysis: let $\rho(\theta, n) = \operatorname{Var}(\tilde{\psi})/\text{CR bound}$. Prove the signs of $\partial\rho/\partial\theta$ and of the change in ρ with n, find the limits $n \to \infty$, $\theta \to 0$ and the value at $n = 1$, and interpret them for the PRF.

Programming (added, not in the book). With $\theta = 1.5$, $n = 10$ and `np.random.default_rng(4)`, simulate 200,000 samples; print the simulated mean of $\hat{\psi}$ against (a), the mean and variance of $\tilde{\psi}$ against (b)–(c), the bound, and the mean squared error of $\hat{\psi}$. Then print $\rho(1.5, n)$ for $n = 1, 2, 5, 10, 50, 1000$.

Solution.

Notation.

- $X_i \sim$ Poisson(θ) i.i.d., $\theta > 0$; $T = \sum_i X_i \sim$ Poisson($n\theta$); $\overline{X} = T/n$.
- $\psi = e^{-\theta} \in (0, 1)$; $\hat{\psi}$: MLE; $\tilde{\psi}$: UMVU; $W = \mathbf{1}\{X_1 = 0\}$.
- $G_T(s) = E[s^T]$, $s \geq 0$: probability generating function of T.
- $\rho(\theta, n) = \operatorname{Var}(\tilde{\psi})/\text{CR}(\theta, n)$; $x = \theta/n > 0$.

Step — (a) The generating function of T. For $\mu = n\theta$,

$$G_T(s) = \sum_{t=0}^{\infty} s^t \frac{e^{-\mu}\mu^t}{t!} = e^{-\mu}\sum_{t=0}^{\infty}\frac{(s\mu)^t}{t!} = e^{-\mu}e^{s\mu},$$

that is,

$$G_T(s) = \exp\{n\theta(s-1)\}. \tag{4.1}$$

Step — (a) MLE and its mean. The MLE of θ is $\overline{X}$ (Question 14, (14.5)); by invariance of the MLE (B, §3.1), $\hat{\psi} = e^{-\overline{X}} = (e^{-1/n})^T$. Substituting $s = e^{-1/n}$ into (4.1):

$$E[\hat{\psi}] = \exp\{n\theta(e^{-1/n}-1)\}. \tag{4.2}$$

Step — (a) Sign of the bias. e^{-u} is strictly convex, so it lies above its tangent at 0: $e^{-u} > 1-u$ for $u \neq 0$. With $u = 1/n$:

$$e^{-1/n} - 1 > -\frac{1}{n}. \tag{4.3}$$

Multiplying (4.3) by $n\theta > 0$ and exponentiating (an increasing map), (4.2) gives

$$E[\hat{\psi}] > \exp\{n\theta(-1/n)\} = e^{-\theta} = \psi.$$

The MLE is biased upwards for every n and θ.

Step — (b) An unbiased start. $E[W] = P(X_1 = 0) = e^{-\theta} = \psi$, so W is unbiased. By Rao–Blackwell, $E[W \mid T]$ is unbiased with variance no larger; by Lehmann–Scheffé (B, Teorema 2.5.3), with T complete and sufficient, it is the UMVU.

Step — (b) Conditional law of X_1 given T. X_1 and $\sum_{i\geq 2} X_i \sim \text{Poisson}((n-1)\theta)$ are independent, so for $0 \leq x_1 \leq t$:

$$P(X_1 = x_1 \mid T = t) = \frac{\dfrac{\theta^{x_1}e^{-\theta}}{x_1!}\cdot\dfrac{((n-1)\theta)^{t-x_1}e^{-(n-1)\theta}}{(t-x_1)!}}{\dfrac{(n\theta)^t e^{-n\theta}}{t!}}.$$

Cancelling $\theta^t e^{-n\theta}$:

$$P(X_1 = x_1 \mid T = t) = \binom{t}{x_1}\Big(\frac{1}{n}\Big)^{x_1}\Big(1-\frac{1}{n}\Big)^{t-x_1}, \tag{4.4}$$

a Binomial$(t, 1/n)$, free of θ.

Step — (b) The UMVU. Setting $x_1 = 0$ in (4.4):

$$\tilde{\psi} = E[W \mid T] = P(X_1 = 0 \mid T) = \Big(1-\frac{1}{n}\Big)^T. \tag{4.5}$$

Check: substituting $s = 1 - 1/n$ into (4.1), $E[\tilde{\psi}] = \exp\{n\theta(-1/n)\} = e^{-\theta}$.

Step — (c) Variance of the UMVU. $\tilde{\psi}^2 = ((1-1/n)^2)^T$; substituting $s = (1-1/n)^2$ into (4.1):

$$E[\tilde{\psi}^2] = \exp\Big\{n\theta\Big(\Big(1-\frac{1}{n}\Big)^2 - 1\Big)\Big\} = \exp\Big\{n\theta\Big(-\frac{2}{n}+\frac{1}{n^2}\Big)\Big\} = e^{-2\theta+\theta/n}.$$

Subtracting $(E\tilde{\psi})^2 = e^{-2\theta}$:

$$\text{Var}(\tilde{\psi}) = e^{-2\theta}(e^{\theta/n}-1). \tag{4.6}$$

Step — (c) Cramér–Rao bound. With $g(\theta) = e^{-\theta}$, $(g'(\theta))^2 = e^{-2\theta}$, and $I_F = 1/\theta$ (Question 2), (3.5) gives

$$\text{CR}(\theta, n) = \frac{e^{-2\theta}}{n(1/\theta)} = \frac{\theta e^{-2\theta}}{n}. \tag{4.7}$$

Dividing (4.6) by (4.7), with $x = \theta/n$:

$$\rho(\theta, n) = \frac{e^x - 1}{x} = \frac{1}{x}\sum_{k\geq 1}\frac{x^k}{k!} = 1 + \frac{x}{2} + \frac{x^2}{6} + \cdots > 1. \tag{4.8}$$

Step — (c) No efficient estimator. By (4.8) the UMVU, which has the smallest variance among unbiased estimators, stays strictly above the bound; hence no unbiased estimator of $e^{-\theta}$ attains it for finite n. Equivalently, by the equality condition after (3.5), an efficient estimator would be affine in $U_n = \frac{n}{\theta}(\overline{X} - \theta)$, i.e. $a + b\overline{X}$, whose mean $a + b\theta$ is linear in θ and cannot equal $e^{-\theta}$. So a UMVU exists while an efficient estimator does not.

Step — (d) Signs. Let $r(x) = (e^x - 1)/x$. Its derivative is

$$r'(x) = \frac{xe^x - (e^x - 1)}{x^2} = \frac{h(x)}{x^2}, \qquad h(x) = xe^x - e^x + 1. \tag{4.9}$$

$h(0) = 0$ and $h'(x) = e^x + xe^x - e^x = xe^x > 0$ for $x > 0$, so $h(x) > 0$ and, by (4.9), $r'(x) > 0$. Since $x = \theta/n$:

$$\frac{\partial \rho}{\partial \theta} = \frac{1}{n}\, r'(x) > 0, \qquad \rho \text{ decreases as } n \text{ grows.}$$

Step — (d) Limits. From (4.8): $\rho \to 1$ as $x \to 0$, i.e. as $n \to \infty$ (fixed θ) or $\theta \to 0$ (fixed n); at $n = 1$, $\rho = (e^\theta - 1)/\theta$ and $\tilde{\psi} = (1 - 1)^{X_1} = \mathbf{1}\{X_1 = 0\} = W$ (with $0^0 = 1$): with one day of data the only unbiased estimator based on T is the indicator itself.

Step — (d) Reading. The bound is nearly attained when accidents are rare relative to the sample (small θ/n): the PRF loses little by using any reasonable estimator. On busy stretches (large θ) with few days of data, the bound is far from attainable, and it understates the real uncertainty of the indicator. Asymptotically both $\hat{\psi}$ and $\tilde{\psi}$ are efficient.

Step — Numbers. $\theta = 1.5$, $n = 10$: $\psi = e^{-1.5} = 0.223130$; (4.2): $\exp\{15(e^{-0.1} - 1)\} = 0.239923$, bias 0.016792; (4.6): $e^{-3}(e^{0.15} - 1) = 0.008057$; (4.7): $1.5e^{-3}/10 = 0.007468$; (4.8): $\rho = 0.008057/0.007468 = 1.078895$.

Code and output (programming part).

```
rng = np.random.default_rng(4)
T = rng.poisson(1.5, size=(200_000, 10)).sum(1)
mle, umvu = np.exp(-T / 10), (1 - 1 / 10) ** T
print(mle.mean(), umvu.mean(), umvu.var(ddof=1))   # 0.239825  0.223033  0.008042
print(np.mean((mle - np.exp(-1.5))**2))            # 0.008638 (exact MSE 0.008657)
rho = lambda n: (np.exp(1.5 / n) - 1) / (1.5 / n)
# n = 1, 2, 5, 10, 50, 1000 -> 2.321126 1.489333 1.166196 1.078895 1.015151 1.000750
```

The simulated values match (a)–(c) within Monte Carlo error; the ρ column falls monotonically to 1, as (d) proves.

Intuition. The MLE plugs $\overline{X}$ into a convex function, so Jensen pushes it up. The UMVU replaces $e^{-1/n}$ by its first-order approximation $1 - 1/n$, exactly the correction that removes the bias. The Cramér–Rao bound is a local, linear benchmark: it is attainable only for parameters that are linear in the sufficient statistic's mean, which is why $e^{-\theta}$ misses it.

Ref: Bolfarine & Sandoval (2010), §2.3 (Cramér–Rao), §2.5 (Teorema 2.5.3 and Exemplo 2.5.4), §3.1 (invariance); Exercício 2.10 for the same Rao–Blackwell step with Bernoulli data.

Grading rubric.

- (a) MLE by invariance and $E[\hat{\psi}]$ via the p.g.f.: 2 pts; bias sign proved (convexity): 1 pt.
- (b) W unbiased and Rao–Blackwell/Lehmann–Scheffé named: 1 pt; (4.4) derived: 2 pts; (4.5): 1 pt.
- (c) (4.6): 2 pts; (4.7): 1 pt; $\rho > 1$ and no efficient estimator, proved: 2 pts.
- (d) sign of r' proved: 2 pts; limits: 1 pt; reading: 1 pt. Signs stated without proof: max. half of (d).
- Programming: correct simulation and comparison with (a)–(c): 2 pts.

Question 5. Original conceptual question — sufficiency, information and optimal estimators (10 points).
Source: Original question (not from a book).

The PRF models the daily number of accidents on a stretch of road as $X_1, \ldots, X_n$ i.i.d. Poisson(θ), $T = \sum_i X_i$.

a) (3 points) Which statement about T is correct?

A. T is sufficient for θ because its expectation, $n\theta$, depends on θ.

B. T is sufficient for θ because the law of the sample given $T = t$ is multinomial $(t; 1/n, \ldots, 1/n)$, free of θ.

C. T is sufficient for θ only for large n, when $\overline{X}$ is approximately normal by the CLT.

D. $\overline{X}$ is sufficient for θ but T is not, because only $\overline{X}$ is unbiased for θ.

Key: B

Step — Why. B: by the computation of (4.4) for all coordinates, $P(\mathbf{X} = \mathbf{x} \mid T = t) = \frac{t!}{\prod x_i!} n^{-t}$, free of θ (Definição 2.2.1). A: a statistic whose mean depends on θ need not be sufficient (X_1 alone is a counterexample). C: sufficiency is an exact finite-sample property, unrelated to the CLT. D: T and $\overline{X} = T/n$ are one-to-one, so they carry the same information; sufficiency does not involve unbiasedness.

Ref: B §2.2 (Definição 2.2.1, Exemplo 2.2.4).

b) (3 points) The Cramér–Rao lower bound for the variance of unbiased estimators of θ^2 is:

A. θ/n (the bound for θ itself).

B. $4\theta^2/n$ (squared derivative, unit information).

C. $2\theta^3/n$ (derivative not squared).

D. $4\theta^3/n$ (squared derivative over n/θ).

E. $4\theta^3/n^2$ (information counted twice).

Key: D

Step — Why. By (3.5) with $g(\theta) = \theta^2$, $(g')^2 = 4\theta^2$ and $nI_F = n/\theta$: bound $= 4\theta^2\theta/n = 4\theta^3/n$ (D). A ignores g; B drops $I_F = 1/\theta$; C forgets to square g'; E divides by n twice.

c) (2 points, T or F) If an unbiased estimator of $g(\theta)$ attains the Cramér–Rao bound at every θ, then it is the UMVU of $g(\theta)$.

Key: T

Step — Why. Every unbiased estimator has variance at least the bound (3.5); one that attains it has the minimum variance at every θ, which is the definition of UMVU.

d) (2 points, T or F) If a UMVU estimator of $g(\theta)$ exists, then it attains the Cramér–Rao bound.

Key: F

Step — Why. The converse of (c) fails. Question 4: $\tilde{\psi} = (1 - 1/n)^T$ is the UMVU of $e^{-\theta}$, yet $\mathrm{Var}(\tilde{\psi})/\mathrm{CR} = \rho > 1$ by (4.8).

Ref: B §2.3–2.5.

Programming (added, not in the book). For $\theta = 2$, $n = 10$: $T(T-1)/n^2 = \overline{X}^2 - \overline{X}/n$ is unbiased for θ^2 and a function of the complete sufficient T, hence the UMVU. Compute its exact variance by summing over $T \sim \mathrm{Poisson}(n\theta)$ $(t = 0, \ldots, 399)$ and compare it with the bound of (b).

Code and output (programming part).

```
t = np.arange(400); pt = poisson.pmf(t, 20); est = t * (t - 1) / 100
np.sum(pt * est), np.sum(pt * est**2) - 16      # 4.0 (unbiased)  3.28
4 * 2**3 / 10                                   # bound 3.2 ; ratio 1.025
```

The exact variance is $4\theta^3/n + 2\theta^2/n^2 = 3.2 + 0.08 = 3.28 > 3.2$: a second UMVU that misses the bound, confirming that (d) is false.

Block 2 — Poisson tests and intervals *(B 6.5, 6.11; original)*

Question 6. B 6.5 — UMP test for a Poisson mean and its power function.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 6, §6.7 Exercícios, Exercício 6.5, p. 115 [PDF p. 121]; the book gives no answer — reproduced verbatim.

6.5. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim Poisson(\theta)$.
i) Encontre o teste UMP para testar $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$.
ii) Seja $\alpha = 0,05$, faça o gráfico da função poder para $\theta_0 = 1$ e $n = 25$ (use o Teorema do limite central).

Programming (added, not in the book). For $\theta_0 = 1$, $n = 25$, $\alpha = 0.05$: (1) find the smallest integer c with $P(T > c \mid \theta_0) \le \alpha$, $T = \sum_i X_i$, and the randomization probability γ that makes the size exactly 0.05; (2) print, for $\theta \in \{1, 1.2, 1.5, 2\}$, the CLT power from (ii), the exact power of the test "reject if $T > c$" and that of the randomized test; (3) plot the three power curves on $\theta \in [0.8, 2.2]$.

Solution.

Notation.

- $X_i \sim$ Poisson(θ) i.i.d.; $T = \sum_i X_i \sim$ Poisson($n\theta$); $\overline{X} = T/n$.
- $\phi(T) \in [0,1]$: test function (probability of rejecting H_0); $\pi(\theta) = E_\theta[\phi(T)]$: power.
- $c \in \{0, 1, \ldots\}$, $\gamma \in [0,1)$: critical value and randomization; $z_{0.95} = 1.644854$.

Step — (i) Likelihood ratio. For a fixed $\theta_1 > \theta_0$,

$$\frac{L(\theta_1; \mathbf{x})}{L(\theta_0; \mathbf{x})} = \frac{\theta_1^t e^{-n\theta_1} / \prod x_i!}{\theta_0^t e^{-n\theta_0} / \prod x_i!} = \Big(\frac{\theta_1}{\theta_0}\Big)^t e^{-n(\theta_1 - \theta_0)}, \tag{6.1}$$

strictly increasing in t because $\theta_1/\theta_0 > 1$.

Step — (i) Neyman–Pearson. The most powerful size-α test of θ_0 against θ_1 rejects when (6.1) exceeds a constant (B, Lema de Neyman–Pearson); by monotonicity, when T exceeds a constant. With the randomization needed for a discrete T:

$$\phi(T) = \begin{cases} 1, & T > c, \\ \gamma, & T = c, \\ 0, & T < c, \end{cases} \qquad P_{\theta_0}(T > c) + \gamma P_{\theta_0}(T = c) = \alpha. \tag{6.2}$$

c and γ depend only on θ_0, n, α, not on θ_1: the same test is most powerful against every $\theta_1 > \theta_0$, hence UMP for $H_1 : \theta > \theta_0$ (B, §6.3: families with monotone likelihood ratio in T).

Step — (ii) CLT version. Under θ, $\overline{X} \approx N(\theta, \theta/n)$. The size-$\alpha$ rule rejects when $\overline{X} > k$ with

$$k = \theta_0 + z_{0.95}\sqrt{\theta_0/n}. \tag{6.3}$$

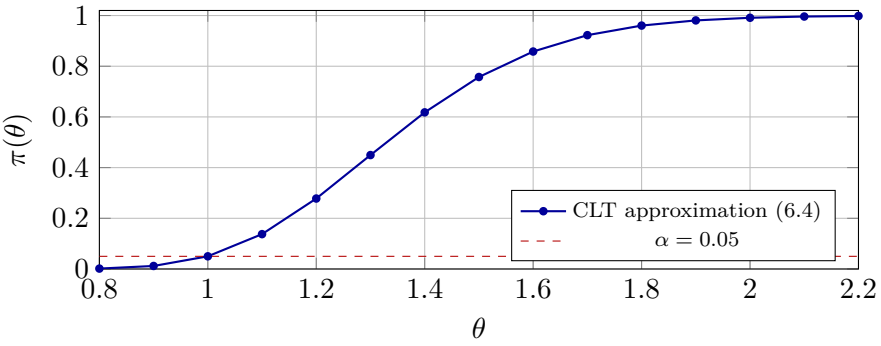
Substituting $\theta_0 = 1$, $n = 25$ into (6.3):

$$k = 1 + (1.644854)\sqrt{1/25} = 1 + 1.644854\,(0.2) = 1.328971.$$

The power function is

$$\pi(\theta) = P_\theta(\overline{X} > k) \approx 1 - \Phi\Big(\frac{k - \theta}{\sqrt{\theta/n}}\Big). \tag{6.4}$$

Substituting $\theta = 1.5$: $\pi(1.5) \approx 1 - \Phi((1.328971 - 1.5)/\sqrt{1.5/25}) = 1 - \Phi(-0.698223) = 0.757481$; likewise $\pi(1) = 0.05$, $\pi(1.2) = 0.278042$, $\pi(2) = 0.991165$.



Step — Exact test (programming). $T \sim \text{Poisson}(25)$ under H_0: the smallest c with $P(T > c) \leq 0.05$ is $c = 33$, size $P(T > 33) = 0.049780$; (6.2) gives $\gamma = (0.05 - 0.049780)/P(T = 33) = 0.010133$. Since $25k = 33.22$, the CLT rule $\overline{X} > k$ is the same as $T > 33$: its true size is 0.049780, and its true power is the exact column below.

Code and output (programming part).

```
c = 33; size = poisson.sf(c, 25)                          # 0.049780
gam = (0.05 - size) / poisson.pmf(c, 25)                  # 0.010133
for th in (1, 1.2, 1.5, 2):
    clt = 1 - norm.cdf((1.328971 - th) / np.sqrt(th / 25))
    ex = poisson.sf(c, 25 * th); rnd = ex + gam * poisson.pmf(c, 25 * th)
# theta   CLT       exact     randomized
# 1.0     0.050000  0.049780  0.050000
# 1.2     0.278042  0.255551  0.256158
# 1.5     0.757481  0.738112  0.738642
# 2.0     0.991165  0.993021  0.993047
```

The CLT curve overstates the power near θ_0 by about two points (the Poisson is right-skewed) and understates it slightly at $\theta = 2$; randomization adds at most $\gamma P(T = 33)$.

Intuition. All the information about θ is in the total count, and larger totals always point to larger rates, so "reject when the count is large" is optimal against every larger rate at once. The discreteness of counts is why an exact 5% test needs a coin flip at the boundary.

Ref: Bolfarine & Sandoval (2010), §6.2 (Lema de Neyman–Pearson), §6.3 (testes UMP), Exemplo 6.3.2.

Question 7. B 6.11 — likelihood ratio test for two Poisson means.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 6, §6.7 Exercícios, Exercício 6.11, p. 116 [PDF p. 122]; the book gives no answer — reproduced verbatim.

6.11. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim Poisson(\theta_1)$ e sejam $Y_1, \ldots, Y_m$ uma amostra aleatória da variável aleatória $Y \sim Poisson(\theta_2)$ sendo as amostras independentes.
i) Encontre o teste da RVG(aproximado) para testar $H_0 : \theta_1 = \theta_2$ versus $H_1 : \theta_1 \neq \theta_2$.
ii) Sendo $n = 5$, $\sum x_i = 3,8$; $m = 8$; $\sum y_i = 4,8$, qual a sua conclusão a um nível de significância de 5%?

Programming (added, not in the book). Write `lr2(sx, sy, n, m)` returning $-2\log\lambda$ from the two totals. Apply it to the printed data. Then estimate the true size of the approximate test (reject when $-2\log\lambda > 3.841$) for $n = 5$, $m = 8$ by simulating 100,000 pairs of samples under H_0 with $\theta_1 = \theta_2 = 0.66$ and with $\theta_1 = \theta_2 = 3$ (`np.random.default_rng(7)`).

Solution.

Notation.

- $s_x = \sum_i x_i$, $s_y = \sum_j y_j$; $\overline{x} = s_x/n$, $\overline{y} = s_y/m$.
- $\hat{\theta}_0 = (s_x + s_y)/(n + m)$: MLE of the common mean under H_0.

- $\lambda(\mathbf{x}, \mathbf{y})$: generalized likelihood ratio (RVG); $\chi^2_{1,0.95} = 3.841459$.

Step — (i) Likelihood. By independence,

$$L(\theta_1, \theta_2) = \frac{\theta_1^{s_x} e^{-n\theta_1}\, \theta_2^{s_y} e^{-m\theta_2}}{\prod_i x_i! \prod_j y_j!}. \tag{7.1}$$

Maximizing $\log L$ in each rate separately gives $\hat\theta_1 = \overline{x}$ and $\hat\theta_2 = \overline{y}$ (as in (14.5)). Under H_0, (7.1) becomes

$$L(\theta, \theta) = \frac{\theta^{s_x + s_y} e^{-(n+m)\theta}}{\prod_i x_i! \prod_j y_j!},$$

maximized at $\hat\theta_0$.

Step — (i) The statistic. Dividing the restricted by the unrestricted maximum (factorials cancel):

$$\lambda = \frac{\hat\theta_0^{\,s_x+s_y} e^{-(n+m)\hat\theta_0}}{\overline{x}^{\,s_x} e^{-n\overline{x}}\, \overline{y}^{\,s_y} e^{-m\overline{y}}}.$$

Taking $-2\log$:

$$-2\log\lambda = 2\Big[s_x \log\frac{\overline{x}}{\hat\theta_0} + s_y \log\frac{\overline{y}}{\hat\theta_0}\Big] - 2\big[n\overline{x} + m\overline{y} - (n+m)\hat\theta_0\big].$$

The second bracket is $(s_x + s_y) - (s_x + s_y) = 0$, so

$$-2\log\lambda = 2\Big[s_x \log\frac{\overline{x}}{\hat\theta_0} + s_y \log\frac{\overline{y}}{\hat\theta_0}\Big]. \tag{7.2}$$

H_1 has two free parameters and H_0 one, so under H_0, $-2\log\lambda \to \chi^2_1$ in distribution (B, Teorema 6.5.1): reject at 5% when (7.2) exceeds 3.841459.

Step — (ii) The printed data. *Book error:* a sum of Poisson counts is an integer, but the book prints $\sum x_i = 3{,}8$ and $\sum y_i = 4{,}8$ (reading them as means does not help either: $m\overline{y} = 38.4$). Statistic (7.2) only needs the totals, so we evaluate it with the printed values: $\overline{x} = 3.8/5 = 0.76$, $\overline{y} = 4.8/8 = 0.6$, $\hat\theta_0 = 8.6/13 = 0.661538$. Substituting into (7.2):

$$-2\log\lambda = 2\Big[(3.8)\log\frac{0.76}{0.661538} + (4.8)\log\frac{0.6}{0.661538}\Big] = 2\,[0.527251 + (-0.468665)] = 0.117173.$$

Since $0.117173 < 3.841459$ (p-value 0.7321), $H_0 : \theta_1 = \theta_2$ is not rejected at 5%.

Code and output (programming part).

```
def lr2(sx, sy, n, m):
    th0 = (sx + sy) / (n + m)
    t = lambda s, k: np.where(s > 0, s * np.log(np.where(s > 0, s, 1) / (k * th0)), 0.)
    return 2 * (t(sx, n) + t(sy, m))           # 0 log 0 = 0
lr2(3.8, 4.8, 5, 8)                            # 0.117173 ; p = chi2.sf(., 1) = 0.7321
rng = np.random.default_rng(7)
for th in (0.66, 3.0):
    sx, sy = rng.poisson(5 * th, 100_000), rng.poisson(8 * th, 100_000)
    print(np.mean(lr2(sx, sy, 5, 8) > 3.841459))   # 0.06949 (th=0.66), 0.05167 (th=3)
```

With expected totals of only 3.3 and 5.3 events the χ^2_1 approximation over-rejects (size about 7%); with means near 3 per observation it is close to 5%.

Intuition. The statistic compares how well one common rate explains both samples against two separate rates; it is twice the gain in log-likelihood from letting the rates differ. Wilks' χ^2_1 is a large-count result: with a handful of events its tail is too light.

Ref: Bolfarine & Sandoval (2010), §6.5 (razão de verossimilhanças generalizada, Teorema 6.5.1, Exemplo 6.5.5 for one Poisson sample).

Question 8. Original math question — three confidence intervals for a Poisson rate.
Source: Original question (not from a book).

The National Civil Defence records the yearly number of dam incidents in a region, modelled as $X_1, \dots, X_n$ i.i.d. Poisson(θ); $T = \sum_i X_i$, $z = z_{1-\alpha/2}$.

(a) Derive the Wald interval from the CLT and Slutsky's theorem. Show that it degenerates when $T = 0$ and prove that, for fixed n, its coverage tends to 0 as $\theta \to 0$.

(b) Derive the score interval by inverting $|\overline{X} - \theta| \le z\sqrt{\theta/n}$. Prove that both endpoints are non-negative and find the interval when $T = 0$.

(c) Prove that $P(T \le t; \mu) = P(G_{t+1} > \mu)$ for $T \sim$ Poisson(μ) and $G_{t+1} \sim$ Gamma($t+1, 1$). Use it to write the exact (Garwood) interval with gamma quantiles and prove that its coverage is at least $1 - \alpha$ for every θ.

(d) Parameter analysis: sign of the shift of the score interval's centre relative to $\overline{X}$; ratio of the score and Wald half-widths and its limit as $n \to \infty$; limits of the three coverages as $\theta \to 0$.

Programming (added, not in the book). Write `intervals(t, n)` returning the three 95% intervals and `coverage(theta, n)` computing their *exact* coverage by summing $P(T = t)$ over the t whose interval contains θ (no simulation). Print the coverages for $n = 10$ and $\theta \in \{0.1, 1, 5\}$, and the three intervals for the coal data 1851–1891 ($n = 41$, $T = 127$).

Solution.

Notation.

- $T \sim$ Poisson($n\theta$), $\overline{X} = T/n$; $\alpha \in (0,1)$, $z = z_{1-\alpha/2}$ (1.959964 for 95%).
- $F_t(\mu) = P(T \le t; \mu)$ for $T \sim$ Poisson(μ); $G_a \sim$ Gamma($a, 1$) with quantile $q_p(a)$.
- $C(\theta)$: coverage probability $P_\theta(\theta \in \text{interval})$.

Step — (a) Wald. By the CLT, $\sqrt{n}(\overline{X} - \theta)/\sqrt{\theta} \to N(0,1)$; $\overline{X} \to \theta$ in probability, so by Slutsky

$$\frac{\overline{X} - \theta}{\sqrt{\overline{X}/n}} \xrightarrow{d} N(0,1), \qquad \text{interval} \quad \overline{X} \pm z\sqrt{\overline{X}/n}. \tag{8.1}$$

If $T = 0$, (8.1) gives $[0, 0]$, which excludes every $\theta > 0$. Hence

$$C_W(\theta) \le 1 - P_\theta(T = 0) = 1 - e^{-n\theta} \xrightarrow[\theta \to 0]{} 0. \tag{8.2}$$

Step — (b) Score. Squaring the inequality, $(\overline{X} - \theta)^2 \le z^2\theta/n$; expanding:

$$\theta^2 - \Big(2\overline{X} + \frac{z^2}{n}\Big)\theta + \overline{X}^2 \le 0. \tag{8.3}$$

The roots of (8.3) are

$$\theta_\pm = \overline{X} + \frac{z^2}{2n} \pm z\sqrt{\frac{\overline{X}}{n} + \frac{z^2}{4n^2}}. \tag{8.4}$$

Their product is $\overline{X}^2 \ge 0$ and their sum $2\overline{X} + z^2/n > 0$, so neither root is negative. With $T = 0$, (8.4) gives $[0,\ z^2/n]$.

Step — (c) The Poisson–gamma identity. Differentiate $F_t(\mu) = \sum_{j=0}^{t} e^{-\mu}\mu^j/j!$:

$$F_t'(\mu) = \sum_{j=0}^{t} \Big(- \frac{e^{-\mu}\mu^j}{j!} \Big) + \sum_{j=1}^{t} \frac{e^{-\mu}\mu^{j-1}}{(j-1)!}.$$

The second sum is the first without its last term, so the sum telescopes:

$$F_t'(\mu) = -\frac{e^{-\mu}\mu^t}{t!} < 0. \tag{8.5}$$

Since $F_t(\mu) \to 0$ as $\mu \to \infty$, integrating (8.5) from μ to ∞:

$$F_t(\mu) = \int_\mu^\infty \frac{u^t e^{-u}}{t!}\, du = P(G_{t+1} > \mu). \qquad \blacksquare \tag{8.6}$$

Step — (c) Garwood limits. The upper limit solves $F_T(n\theta_U) = \alpha/2$; by (8.6), $P(G_{T+1} > n\theta_U) = \alpha/2$. The lower limit solves $P(T' \geq T; n\theta_L) = \alpha/2$, i.e. $1 - F_{T-1}(n\theta_L) = \alpha/2$, and by (8.6) $P(G_T \leq n\theta_L) = \alpha/2$. So

$$\theta_L = \frac{q_{\alpha/2}(T)}{n} \ (\theta_L = 0 \text{ if } T = 0), \qquad \theta_U = \frac{q_{1-\alpha/2}(T+1)}{n}. \tag{8.7}$$

Step — (c) Coverage. Fix θ and let t^* be the largest t with $F_t(n\theta) \leq \alpha/2$. By (8.5), F_t is decreasing in μ, so $\theta > \theta_U(t) \iff F_t(n\theta) < F_t(n\theta_U(t)) = \alpha/2$; and $F_t(n\theta)$ increases in t, so this happens only for $t \leq t^*$. Hence $P_\theta(\theta > \theta_U) \leq P_\theta(T \leq t^*) = F_{t^*}(n\theta) \leq \alpha/2$. The same argument on the upper tail gives $P_\theta(\theta < \theta_L) \leq \alpha/2$, so $C_G(\theta) \geq 1 - \alpha$ for every θ. $\blacksquare$

Step — (d) Parameter analysis. By (8.4) the score interval is centred at $\overline{X} + z^2/(2n)$: shifted *up* (sign +), towards the side where the Poisson is skewed, and the shift vanishes at rate $1/n$. The half-width ratio is

$$\frac{z\sqrt{\overline{X}/n + z^2/(4n^2)}}{z\sqrt{\overline{X}/n}} = \sqrt{1 + \frac{z^2}{4n\overline{X}}} > 1, \qquad \to 1 \ (n \to \infty). \tag{8.8}$$

As $\theta \to 0$: $C_W \to 0$ by (8.2); $C_S \geq P(T = 0) = e^{-n\theta} \to 1$, because the $T = 0$ interval $[0, z^2/n]$ contains every $\theta \leq z^2/n$; and $C_G \geq 0.95$ always, tending to 1 since the $T = 0$ interval $[0, q_{1-\alpha/2}(1)/n]$ covers small θ. Reading: for rare events the Wald interval is unusable, the score interval is a good default, and the exact interval is the safe (conservative) choice.

Code and output (programming part).

```
def intervals(t, n, z=1.959964, al=0.05):
    xb = t / n
    wald = (xb - z*np.sqrt(xb/n), xb + z*np.sqrt(xb/n))
    c, h = xb + z**2/(2*n), z*np.sqrt(xb/n + z**2/(4*n**2))
    exact = (0.0 if t == 0 else gamma.ppf(al/2, t)/n, gamma.ppf(1-al/2, t+1)/n)
    return wald, (c - h, c + h), exact
# exact coverage, n = 10:   theta   Wald      score     exact
#                           0.1     0.631526  0.919699  0.981012
#                           1       0.925728  0.962622  0.975386
#                           5       0.948781  0.944398  0.952638
# coal 1851-1891 (n=41, T=127): Wald [2.558837, 3.636285]
#     score [2.603651, 3.685165]  exact [2.582297, 3.685510]
```

At $\theta = 0.1$ the Wald coverage 0.6315 is close to the bound (8.2), $1 - e^{-1} = 0.6321$; the exact interval is above 0.95 at every θ, as proved.

Intuition. Wald uses the estimated variance $\overline{X}/n$, which is zero exactly when the data are least informative; the score interval uses the variance at the hypothesized θ and never collapses. Inverting exact tail probabilities trades length for a guaranteed coverage.

Ref: Bolfarine & Sandoval (2010), §5.2 (quantidades pivotais), §5.3 (intervalos aproximados); Class 1 deck, *Poisson* (asymptotic interval).

Grading rubric.

- (a) pivot (8.1) with CLT and Slutsky named: 2 pts; (8.2): 2 pts.
- (b) (8.3)–(8.4): 2 pts; non-negativity proved: 1 pt; $T = 0$ case: 1 pt.
- (c) telescoping (8.5) and identity (8.6): 3 pts; limits (8.7): 2 pts; coverage proof: 2 pts.
- (d) shift sign, (8.8) and the three limits with reading: 3 pts (limits without proof: max. half).
- Programming: exact coverage by enumeration (not simulation) and coal intervals: 2 pts.

Question 9. Original conceptual question — tests and intervals for counts (10 points).
Source: Original question (not from a book).

a) (3 points) To test $H_0 : \theta = 5$ against $H_1 : \theta \neq 5$ with a Poisson sample (B, Exemplo 6.5.5), one uses $-2\log\lambda = 2n[\overline{x}\log(\overline{x}/5) - (\overline{x} - 5)]$. For large n, under H_0 this statistic is approximately:

A. χ^2_n, one degree of freedom for each observation in the sample.

B. $N(0,1)$, because $\overline{X}$ is asymptotically normal under H_0.

C. χ^2_1, the number of free parameters in H_1 minus that in H_0.

D. χ^2_2, because the alternative $\theta \neq 5$ has two sides.

Key: C

Step — Why. C: Wilks (B, Teorema 6.5.1), $1 - 0 = 1$ degree of freedom. A confuses sample size with dimension; B: the signed root is $N(0,1)$, the statistic itself is its square; D: two-sidedness is already built into squaring, not into extra degrees of freedom.

b) (3 points) Over $n = 12$ years a dam recorded no incident ($\overline{x} = 0$). The 95% Wald interval for the yearly rate θ is:

A. $[0, 0]$, a degenerate interval of zero length.

B. [0, 3.84/12], from inverting the score statistic.

C. [0, 1.96/12], from the normal quantile over n.

D. $[-0.16, 0.16]$, from a symmetric normal band.

Key: A

Step — Why. A: by (8.1), $0 \pm 1.96\sqrt{0/12}$. B is the *score* interval $[0, z^2/n] = [0, 0.320122]$ of (8.4), a correct answer to a different question; C and D have no derivation behind them.

c) (2 points, T or F) For Poisson data, a UMP test of $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$ exists because the likelihood ratio is monotone in $T = \sum X_i$.

Key: T

Step — Why. (6.1) is increasing in t for $\theta_1 > \theta_0$; the test (6.2) does not depend on θ_1 (B §6.3).

d) (2 points, T or F) Because a UMP test exists for the one-sided alternative, a UMP test also exists for $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$.

Key: F

Step — Why. The most powerful test against $\theta_1 > \theta_0$ rejects for large T, the one against $\theta_1 < \theta_0$ for small T; no single test is most powerful on both sides. Two-sided problems use the GLR test of (a) instead.

Ref: B §6.3–6.5.

Programming (added, not in the book). Check (a) by simulation. Draw 100,000 samples of size $n = 50$ from Poisson(5), through their totals $T \sim$ Poisson(250), using the generator `np.random.default_rng(9)`; compute the statistic of (a) for each and report the fraction above $\chi^2_{1,0.95} = 3.841$.

Code and output (programming part).

```
rng = np.random.default_rng(9); xb = rng.poisson(250, 100_000) / 50
lr = 2 * 50 * (xb * np.log(xb / 5) - (xb - 5))
np.mean(lr > 3.841459)                          # 0.04866
```

The rejection rate 0.0487 is close to 0.05: with $n\theta_0 = 250$ expected events the χ^2_1 approximation is accurate (compare Question 7, where 3 to 5 events gave about 7%).

Block 3 — Two Poisson regimes: equal rates and the change point *(G 3.8, 7.3; original)*

Question 10. G 3.8 — Bayes factor for two typists.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 3, §3.5 Exercises, Exercise 3.8, p. 40 [PDF p. 55]; the book gives no answer — reproduced verbatim. Table 3.2, to which it refers, is reproduced from §3.2.4, p. 38 [PDF p. 53].

3.8 Suppose the number of typographical errors per page has a Poisson distribution $P(\theta)$. Sam types the first m pages of a manuscript and makes e_1 errors in total, and Levi types the last m pages and makes e_2 errors. Let Sam's error rate be θ_1 and Levi's error rate be θ_2. To compare model $M_1 : \theta_1 = \theta_2 = \theta$ with model $M_2 : \theta_1 \neq \theta_2$, produce a table like Table 3.2 for $m = 100$ with 10 total errors and for $m = 200$ with 20 total errors and Sam's error proportions of 0.9, 0.8, 0.7, 0.6, and 0.5. Take $G(1, 1)$ as a prior for θ_1 and θ_2.

Referred to (Greenberg 2008, p. 38): **Table 3.2.** Bayes Factors for Selected Possible Outcomes

Michaela Proportion heads	Lila Proportion heads	$\log_{10}$(Bayes factor) $m = 10$	$m = 100$
0.1	0.9	−2.506	−30.775
0.2	0.8	−1.200	−15.793
0.3	0.7	−0.348	−6.316
0.4	0.6	0.138	−0.975
0.5	0.5	0.297	0.756

Programming (added, not in the book). Compute the table from the closed form. Then check the cell $m = 100$, $(e_1, e_2) = (9, 1)$ by Monte Carlo integration over the prior: with `np.random.default_rng(10)` draw 10^6 values from $G(1, 1)$ for each of the three integrals (M_1; M_2's two factors), average the integrands, and report $\log_{10} \hat{B}_{12}$ with its delta-method standard error.

Solution.

Notation.

- y_{1i}, y_{2i} $(i = 1, \ldots, m)$: errors on page i typed by Sam and by Levi; $e_1 = \sum_i y_{1i}$, $e_2 = \sum_i y_{2i}$, $e = e_1 + e_2$.
- $\theta, \theta_1, \theta_2 > 0$: error rates per page; prior $G(1, 1)$, i.e. $\pi(\theta) = e^{-\theta}$.
- $P = \prod_i y_{1i}! \prod_i y_{2i}!$; $m_j(y)$: marginal likelihood of M_j; $B_{12} = m_1(y)/m_2(y)$.

Step — The gamma integral. For $a, b > 0$ (Greenberg App. A.1.7),

$$\int_0^\infty \theta^{a-1} e^{-b\theta} d\theta = \frac{\Gamma(a)}{b^a}. \tag{10.1}$$

Step — Marginal likelihood of M_1. All $2m$ pages share θ:

$$m_1(y) = \int_0^\infty \prod_i \frac{\theta^{y_{1i}} e^{-\theta}}{y_{1i}!} \prod_i \frac{\theta^{y_{2i}} e^{-\theta}}{y_{2i}!} \, e^{-\theta} \, d\theta \tag{10.2}$$

$$= \frac{1}{P} \int_0^\infty \theta^e e^{-(2m+1)\theta} d\theta. \tag{10.3}$$

Substituting (10.1) with $a = e + 1$, $b = 2m + 1$:

$$m_1(y) = \frac{1}{P} \frac{\Gamma(e + 1)}{(2m + 1)^{e+1}}. \tag{10.4}$$

Step — Marginal likelihood of M_2. The two rates have independent priors, so the integral factorises:

$$m_2(y) = \frac{1}{P} \int_0^\infty \theta_1^{e_1} e^{-(m+1)\theta_1} d\theta_1 \int_0^\infty \theta_2^{e_2} e^{-(m+1)\theta_2} d\theta_2 \tag{10.5}$$

$$= \frac{1}{P} \frac{\Gamma(e_1 + 1)}{(m + 1)^{e_1+1}} \frac{\Gamma(e_2 + 1)}{(m + 1)^{e_2+1}}. \tag{10.6}$$

Step — Bayes factor. Dividing (10.4) by (10.6); P cancels, so only the totals matter:

$$B_{12} = \frac{\Gamma(e+1)\,(m+1)^{e+2}}{(2m+1)^{e+1}\,\Gamma(e_1+1)\,\Gamma(e_2+1)}, \tag{10.7}$$

$$\log_{10} B_{12} = \frac{\ln\Gamma(e+1) + (e+2)\ln(m+1) - (e+1)\ln(2m+1) - \ln\Gamma(e_1+1) - \ln\Gamma(e_2+1)}{\ln 10}. \tag{10.8}$$

Step — One entry worked out. $m = 100$, $e = 10$, Sam's proportion 0.9, so $e_1 = 9$, $e_2 = 1$. The pieces of (10.8): $\ln\Gamma(11) = \ln 10! = 15.104413$; $12\ln 101 = 12(4.615121) = 55.381446$; $11\ln 201 = 11(5.303305) = 58.336354$; $\ln\Gamma(10) = \ln 9! = 12.801827$; $\ln\Gamma(2) = 0$. Substituting:

$$\log_{10} B_{12} = \frac{(15.104413) + (55.381446) - (58.336354) - (12.801827) - (0)}{2.302585} \tag{10.9}$$

$$= \frac{-0.652323}{2.302585} = -0.283. \tag{10.10}$$

Step — The table. The same computation for every cell:

	$m = 100$, $e = 10$		$m = 200$, $e = 20$	
Sam's proportion	(e_1, e_2)	$\log_{10} B_{12}$	(e_1, e_2)	$\log_{10} B_{12}$
0.9	(9, 1)	−0.283	(18, 2)	−1.717
0.8	(8, 2)	0.370	(16, 4)	−0.310
0.7	(7, 3)	0.796	(14, 6)	0.593
0.6	(6, 4)	1.039	(12, 8)	1.105
0.5	(5, 5)	1.118	(10, 10)	1.271

Step — Reading it. On Jeffreys' scale (Greenberg Table 3.1), with 10 errors even a 9–1 split favours M_2 only weakly ($\log_{10} B_{21} = 0.283 < 1/2$), while an even split gives strong support to M_1 ($1 < 1.118 < 3/2$). Doubling the evidence (20 errors) sharpens everything: an 18–2 split now gives very strong evidence for M_2 ($3/2 < 1.717 < 2$), and equal shares strengthen M_1 (1.271).

Step — Monte Carlo check (programming). Each marginal likelihood in (10.3)–(10.5) is a prior expectation, e.g. $P\,m_1(y) = E_\pi[\theta^e e^{-2m\theta}]$ with $\theta \sim G(1,1)$, so it can be estimated by averaging the integrand over prior draws (Greenberg §5.4 with $h = \pi$). If $\hat{m} = \bar{d}$ averages G draws d_g, the delta method gives $\mathrm{Var}(\ln\hat{m}) \approx \mathrm{Var}(d)/(G\bar{d}^2)$, and the three estimates are independent.

Code and output (programming part).

```
rng = np.random.default_rng(10); G, m, e1, e2 = 10**6, 100, 9, 1
h = lambda t, k, r: np.exp(k * np.log(t) - r * t)
d1 = h(rng.exponential(1, G), e1 + e2, 2 * m)          # M1 integrand
d2a, d2b = h(rng.exponential(1, G), e1, m), h(rng.exponential(1, G), e2, m)
lB = np.log10(d1.mean() / (d2a.mean() * d2b.mean()))   # -0.28295
cv2 = lambda d: d.var(ddof=1) / (G * d.mean()**2)
se = np.sqrt(cv2(d1) + cv2(d2a) + cv2(d2b)) / np.log(10)   # 0.00312
```

-0.28295 ± 0.00312 brackets the exact -0.28330. Prior sampling works here because the prior $G(1,1)$ still puts mass near the likelihood's peak ($\theta \approx 0.05$); with a more concentrated likelihood most draws would contribute nothing.

Intuition. The Bayes factor automatically penalises the extra parameter of M_2: it pays off only when the split is lopsided enough to beat that Occam penalty, and the same proportions carry more weight when they rest on more errors (compare the two m columns, as in Table 3.2).

Ref: Greenberg (2008), §3.2.4 (Bayes factors, Tables 3.1–3.2, pp. 33–38) and §5.4 (Monte Carlo integration).

Question 11. G 7.3 — Gibbs sampler for the coal-mining switch point (computer exercise).
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 7, §7.5 Exercises, Exercise 7.3, p. 106 [PDF p. 121]; the book gives no answer — reproduced verbatim. Equations (7.3) and (7.4), to which the exercise refers, are reproduced verbatim from §7.1.1, pp. 93–94 [PDF pp. 108–109].

Equations referred to (Greenberg 2008, pp. 93–94):

$$p(y_i) = \begin{cases} \dfrac{e^{-\theta_1}\theta_1^{y_i}}{y_i!}, & \text{for } i = 1, \ldots, k, \\ \dfrac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, & \text{for } i = k+1, \ldots, n, \end{cases} \tag{7.3}$$

"where $y_i = 0, 1, \ldots,$ and the switch point k is unknown. The specification is completed by assigning the conditionally conjugate priors,

$$\theta_1 \sim G(\alpha_{10}, \beta_{10}), \quad \theta_2 \sim G(\alpha_{20}, \beta_{20}), \quad \pi(k = j) = 1/n, \quad j = 1, \ldots, n." \tag{7.4}$$

7.3 Construct a Gibbs algorithm to analyze the Poisson model with unknown switch point. Given the specification in Equations (7.3) and (7.4), show that

$$\pi(\theta_1, \theta_2, k \mid y) \propto \theta_1^{\alpha_{10}-1} e^{-\beta_{10}\theta_1} \theta_2^{\alpha_{20}-1} e^{-\beta_{20}\theta_2} \prod_{i=1}^{k} e^{-\theta_1}\theta_1^{y_i} \prod_{i=k+1}^{n} e^{-\theta_2}\theta_2^{y_i},$$

and verify that

$$\theta_1 \mid y, k \sim G\Big(\alpha_{10} + \sum_{1}^{k} y_i,\ \beta_{10} + k\Big),$$

$$\theta_2 \mid k, y \sim G\Big(\alpha_{20} + \sum_{k+1}^{n} y_i,\ \beta_{20} + n - k\Big),$$

$$\pi(k \mid y, \theta_1, \theta_2) = \frac{e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{\sum_1^k y_i}}{\sum_{k=1}^{n} e^{k(\theta_2-\theta_1)}\,(\theta_1/\theta_2)^{\sum_1^k y_i}}, \qquad k = 1, \ldots, n.$$

Apply your algorithm to the mining disaster data analyzed by Carlin et al. (1992a); the data may be found in the article. Your analysis should generate marginal posterior distributions for θ_1, θ_2, $p(k)$, and the marginal likelihood computed by the method of Chib (1995). For simplicity, set $\alpha_{10} = \alpha_{20} = 0.5$ and $\beta_{10} = \beta_{20} = 1.0$. (Curiously, although these data have been analyzed from the Bayesian viewpoint in several papers, none of them present historical material that might shed light on when changes in parameters took place. For example, changes in government regulation, technology, or the location of mines might be expected to affect the probability of a disaster. A more general version of the model permits multiple switch points if necessary.)

Programming (added, not in the book). Beyond the book's tasks (use the coal series of the Instructions, `np.random.default_rng(2026)`, start $k^{(0)} = 56$, 1,000 burn-in and $G = 10{,}000$ kept draws): (1) compute the *exact* $p(k \mid y)$ by integrating θ_1, θ_2 out analytically and report the total-variation distance between it and the Gibbs frequencies of k; (2) report the numerical standard error of the posterior mean of θ_1 by batch means (50 batches of 200), the i.i.d. standard error $s/\sqrt{G}$, and their squared ratio (inefficiency factor).

Solution.

Notation.

- y_i: number of British coal-mining disasters in year i, $i = 1$ (1851) to $n = 112$ (1962); the series of Carlin, Gelfand & Smith (1992) is Jarrett (1979), total $S = \sum_i y_i = 191$.
- $k \in \{1, \ldots, n\}$: switch point, the last year of regime 1 (year $1850 + k$); $k = n$ means no change.

- $\theta_1, \theta_2 > 0$: disaster rates before and after the switch; $S_k = \sum_{i=1}^{k} y_i$ (so $\sum_{k+1}^{n} y_i = S - S_k$).
- $G(a, b)$: gamma with shape a, rate b; $\alpha_{10} = \alpha_{20} = \alpha = 0.5$, $\beta_{10} = \beta_{20} = \beta = 1$.
- $g = 1, \ldots, G$: Gibbs iteration; $\theta^* = (\theta_1^*, \theta_2^*)$: point where Chib's identity is evaluated; $m(y)$: marginal likelihood.

Step — Joint posterior. By (7.3) and independence, the likelihood is

$$p(y \mid \theta_1, \theta_2, k) = \prod_{i=1}^{k} \frac{e^{-\theta_1}\theta_1^{y_i}}{y_i!} \prod_{i=k+1}^{n} \frac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, \tag{11.1}$$

and by (7.4) the prior is

$$\pi(\theta_1, \theta_2, k) = \frac{\beta_{10}^{\alpha_{10}}}{\Gamma(\alpha_{10})}\theta_1^{\alpha_{10}-1}e^{-\beta_{10}\theta_1} \cdot \frac{\beta_{20}^{\alpha_{20}}}{\Gamma(\alpha_{20})}\theta_2^{\alpha_{20}-1}e^{-\beta_{20}\theta_2} \cdot \frac{1}{n}. \tag{11.2}$$

Multiplying (11.1) by (11.2) and dropping the factors free of (θ_1, θ_2, k) (the $y_i!$, the gamma constants, $1/n$) gives the expression in the statement:

$$\pi(\theta_1, \theta_2, k \mid y) \propto \theta_1^{\alpha_{10}-1}e^{-\beta_{10}\theta_1}\theta_2^{\alpha_{20}-1}e^{-\beta_{20}\theta_2} \prod_{i=1}^{k} e^{-\theta_1}\theta_1^{y_i} \prod_{i=k+1}^{n} e^{-\theta_2}\theta_2^{y_i}. \tag{11.3}$$

Step — Full conditional of θ_1. Keeping in (11.3) only the factors with θ_1:

$$\pi(\theta_1 \mid y, k, \theta_2) \propto \theta_1^{\alpha_{10}-1}e^{-\beta_{10}\theta_1}\big(e^{-k\theta_1}\theta_1^{S_k}\big).$$

Adding exponents:

$$\pi(\theta_1 \mid y, k) \propto \theta_1^{\alpha_{10}+S_k-1}e^{-(\beta_{10}+k)\theta_1} \implies \theta_1 \mid y, k \sim G(\alpha_{10} + S_k,\ \beta_{10} + k). \tag{11.4}$$

Step — Full conditional of θ_2. In the same way, the θ_2 factors of (11.3) are

$$\pi(\theta_2 \mid y, k, \theta_1) \propto \theta_2^{\alpha_{20}-1}e^{-\beta_{20}\theta_2}\big(e^{-(n-k)\theta_2}\theta_2^{S-S_k}\big),$$

so, adding exponents,

$$\theta_2 \mid y, k \sim G(\alpha_{20} + S - S_k,\ \beta_{20} + n - k). \tag{11.5}$$

Neither (11.4) nor (11.5) involves the other rate: given k, θ_1 and θ_2 are independent.

Step — Full conditional of k. The factors of (11.3) that change with k are the two products:

$$\pi(k \mid y, \theta_1, \theta_2) \propto e^{-k\theta_1}\theta_1^{S_k}\, e^{-(n-k)\theta_2}\theta_2^{S-S_k}. \tag{11.6}$$

Dividing (11.6) by the k-free factor $e^{-n\theta_2}\theta_2^{S}$:

$$\pi(k \mid y, \theta_1, \theta_2) \propto e^{k(\theta_2-\theta_1)}\Big(\frac{\theta_1}{\theta_2}\Big)^{S_k}, \qquad k = 1, \ldots, n, \tag{11.7}$$

and normalising over $k = 1, \ldots, n$ gives the formula in the statement.

Step — Algorithm. Start at $k^{(0)} = 56$. For $g = 1, \ldots, 11{,}000$: draw $\theta_1^{(g)}$ from (11.4) and $\theta_2^{(g)}$ from (11.5) given $k^{(g-1)}$; compute the n weights (11.7) on the log scale, normalise, and draw $k^{(g)}$. Discard 1,000 burn-in draws and keep $G = 10{,}000$ (NumPy seed 2026).

Step — Exact benchmark. Because the priors are conjugate given k, θ_1, θ_2 integrate out of (11.1)×(11.2) in closed form, by (10.1):

$$p(y \mid k) = \frac{1}{\prod_i y_i!} \cdot \frac{\beta^{\alpha}\,\Gamma(\alpha + S_k)}{\Gamma(\alpha)\,(\beta + k)^{\alpha+S_k}} \cdot \frac{\beta^{\alpha}\,\Gamma(\alpha + S - S_k)}{\Gamma(\alpha)\,(\beta + n - k)^{\alpha+S-S_k}}, \tag{11.8}$$

so the exact marginal likelihood and posterior of k are

$$m(y) = \sum_{k=1}^{n} \frac{1}{n}\, p(y \mid k), \qquad p(k \mid y) = \frac{p(y \mid k)/n}{m(y)}, \tag{11.9}$$

giving $\log m(y) = -179.4590$, $\mathbb{E}(\theta_1 \mid y) = 3.0500$, $\mathbb{E}(\theta_2 \mid y) = 0.9147$. These check the sampler.

Step — Marginal posteriors. Gibbs output (exact values from (11.9) for k):

	Posterior mean	Posterior sd	95% interval
θ_1 (before)	3.0506	0.2809	[2.530, 3.635]
θ_2 (after)	0.9166	0.1163	[0.701, 1.156]

k	year	$p(k \mid y)$ Gibbs	exact
41	1891	0.2471	0.2482
40	1890	0.1849	0.1848
39	1889	0.1433	0.1414
42	1892	0.0987	0.1010
37	1887	0.0881	0.0898
36	1886	0.0742	0.0745

The posterior of k has mean 40.14 and mode $k = 41$ (1891); $P(36 \le k \le 46 \mid y) = 0.98$, i.e. the switch happened between 1886 and 1896 with high probability. Disasters fell from about 3.05 to about 0.92 per year.

Step — Marginal likelihood by Chib (1995). For any θ^*, Bayes' theorem rearranged gives the identity

$$\log m(y) = \log p(y \mid \theta^*) + \log \pi(\theta^*) - \log \pi(\theta^* \mid y). \tag{11.10}$$

Here k is a latent parameter, so the likelihood term sums it out with its prior:

$$p(y \mid \theta^*) = \sum_{k=1}^{n} \frac{1}{n}\, p(y \mid \theta_1^*, \theta_2^*, k). \tag{11.11}$$

For the ordinate, conditional independence of θ_1, θ_2 given k (from (11.4) and (11.5)) gives the Rao–Blackwell estimate

$$\hat\pi(\theta^* \mid y) = \frac{1}{G} \sum_{g=1}^{G} G\big(\theta_1^* \mid \alpha + S_{k^{(g)}}, \beta + k^{(g)}\big)\, G\big(\theta_2^* \mid \alpha + S - S_{k^{(g)}}, \beta + n - k^{(g)}\big), \tag{11.12}$$

with $G(\cdot \mid a, b)$ the gamma density. At the posterior means $\theta^* = (3.0506,\ 0.9166)$: $\log p(y \mid \theta^*) = -172.2563$ from (11.11), $\log \pi(\theta^*) = -5.6261$ from (7.4), and $\log \hat\pi(\theta^* \mid y) = 1.5765$ from (11.12). Substituting into (11.10):

$$\log \hat m(y) = (-172.2563) + (-5.6261) - (1.5765) = -179.4589,$$

against the exact -179.4590 from (11.9).

Step — Added: exactness and numerical standard error. The total-variation distance $\frac{1}{2}\sum_k |\hat p(k) - p(k \mid y)|$ between the Gibbs frequencies and (11.9) is 0.0077. For $\mathbb{E}(\theta_1 \mid y)$, batch means give n.s.e. $= 0.003035$ against the i.i.d. value $0.2809/\sqrt{10{,}000} = 0.002809$; the inefficiency factor $(0.003035/0.002809)^2 = 1.167$ says the chain is almost as informative as independent draws (Greenberg §7.3).

Code and output (programming part).

```
cs, ks = np.cumsum(y), np.arange(1, n + 1)        # S_k and k = 1..n
rng, k = np.random.default_rng(2026), n // 2
for g in range(11000):
    t1 = rng.gamma(a0 + cs[k-1], 1 / (b0 + k))          # (11.4) theta1 | k
    t2 = rng.gamma(a0 + S - cs[k-1], 1 / (b0 + n - k))  # (11.5) theta2 | k
    lp = ks * (t2 - t1) + cs * np.log(t1 / t2)          # (11.7) log pi(k | ...)
    p = np.exp(lp - lp.max()); k = rng.choice(ks, p=p / p.sum())
# theta1 3.0506 (0.2809), theta2 0.9166 (0.1163), mode k = 41 (1891)
# Chib: -172.2563 + (-5.6261) - 1.5765 = -179.4589 ; exact -179.4590
bm = th1.reshape(50, 200).mean(1)
nse = bm.std(ddof=1) / np.sqrt(50)                  # 0.003035 (iid: 0.002809)
```

Intuition. The data choose a sharp break: about three disasters a year until around 1890, under one afterwards, with the switch year pinned down within a few years. The Gibbs sampler works because each block is easy given the others: rates are conjugate gammas given k, and k is a discrete draw over 112 years given the rates. On the historical question Greenberg raises, the posterior mass near

1887–1891 sits close to the Coal Mines Regulation Act of 1887, a candidate explanation that this model does not test. Chib's identity recovers the exact marginal likelihood to the fourth decimal.

Ref: Greenberg (2008), §7.1.1 (pp. 91–95), §7.3 (n.s.e., batch means) and §7.4 (Chib 1995); Class 1 deck *Poisson with breakpoint.*

Question 12. Original math question — likelihood ratio for a change point.
Source: Original question (not from a book).

Counts $y_1, \dots, y_n$ are independent with $y_i \sim \text{Poisson}(\theta_1)$ for $i \le \tau$ and $\text{Poisson}(\theta_2)$ for $i > \tau$, $1 \le \tau < n$. Let $S = \sum_{i=1}^{n} y_i > 0$ and $S_\tau = \sum_{i \le \tau} y_i$.

(a) For known τ, derive the MLEs and the statistic $\Lambda_\tau = -2 \log \lambda_\tau$ for $H_0 : \theta_1 = \theta_2$. Show that $\Lambda_\tau = 2S\,\mathrm{KL}(p \,\|\, \pi)$, the Kullback–Leibler divergence between Bernoulli(p) and Bernoulli(π) with $p = S_\tau/S$ and $\pi = \tau/n$, and prove that $\Lambda_\tau \ge 0$ with equality iff $p = \pi$.

(b) Prove that under H_0, conditionally on S, $S_\tau \sim \text{Binomial}(S, \tau/n)$ whatever the common rate, and use it to build an exact test for known τ.

(c) For unknown τ, let $M = \max_{1 \le \tau < n} \Lambda_\tau$ (the profile-likelihood statistic) and $\hat\tau$ its argmax. Prove that, under H_0, $P(M > c) \ge P(\Lambda_{\tau_0} > c)$ for any fixed τ_0 and c, and conclude what happens to a test that compares M with the χ^2_1 critical value.

(d) Parameter analysis: for fixed S and π, prove the sign of $\partial \Lambda_\tau / \partial p$ and the convexity of Λ_τ in p; find its limits as $p \to 0$ and $p \to 1$; say how Λ_τ scales with S at a fixed p. Interpret.

Programming (added, not in the book). For the coal series ($n = 112$): compute Λ_τ for $\tau = 1, \dots, 111$, $\hat\tau$ and M; the conditional binomial p-value of (b) at $\tau = \hat\tau$ (and why it is not a valid p-value once τ was chosen by the data); and the null distribution of M by a parametric bootstrap (2,000 series of 112 i.i.d. Poisson(S/n) counts, `np.random.default_rng(12)`): its 95% quantile, the rejection rate of the rule "$M > 3.841$" and the bootstrap p-value of the observed M.

Solution.

Notation.

- $n_1 = \tau$, $n_2 = n - \tau$; $\hat\theta_1 = S_\tau/\tau$, $\hat\theta_2 = (S - S_\tau)/(n - \tau)$, $\hat\theta = S/n$.
- $p = S_\tau/S \in [0, 1]$: share of events before τ; $\pi = \tau/n \in (0, 1)$: share of time before τ.
- $\mathrm{KL}(p\|\pi) = p \log \frac{p}{\pi} + (1-p) \log \frac{1-p}{1-\pi}$ (with $0 \log 0 = 0$).

Step — (a) MLEs. The log-likelihood is $\ell = S_\tau \log \theta_1 - \tau\theta_1 + (S - S_\tau) \log \theta_2 - (n - \tau)\theta_2 + \text{const}$; its two score equations separate and give $\hat\theta_1$, $\hat\theta_2$ of the Notation (as in (14.5)); under H_0 the pooled MLE is $\hat\theta = S/n$.

Step — (a) The statistic. As in (7.2) of Question 7, with (n, m) replaced by $(\tau, n - \tau)$ (the linear terms cancel because $\tau\hat\theta_1 + (n-\tau)\hat\theta_2 = S = n\hat\theta$):

$$\Lambda_\tau = 2\Big[S_\tau \log \frac{\hat\theta_1}{\hat\theta} + (S - S_\tau) \log \frac{\hat\theta_2}{\hat\theta}\Big]. \tag{12.1}$$

Step — (a) KL form. The ratios in (12.1) are

$$\frac{\hat\theta_1}{\hat\theta} = \frac{S_\tau/\tau}{S/n} = \frac{S_\tau/S}{\tau/n} = \frac{p}{\pi}, \qquad \frac{\hat\theta_2}{\hat\theta} = \frac{(S - S_\tau)/S}{(n-\tau)/n} = \frac{1-p}{1-\pi}. \tag{12.2}$$

Substituting (12.2) and $S_\tau = Sp$ into (12.1):

$$\Lambda_\tau = 2\Big[(Sp) \log \frac{p}{\pi} + (S(1-p)) \log \frac{1-p}{1-\pi}\Big] = 2S\,\mathrm{KL}(p\|\pi). \tag{12.3}$$

Step — (a) Non-negativity. $-\log$ is strictly convex, so by Jensen's inequality over the two-point law $(p, 1-p)$:

$$\mathrm{KL}(p\|\pi) = -\Big[p \log \frac{\pi}{p} + (1-p) \log \frac{1-\pi}{1-p}\Big] \ge -\log\Big[p\frac{\pi}{p} + (1-p)\frac{1-\pi}{1-p}\Big] = -\log 1 = 0,$$

with equality iff $\pi/p = (1-\pi)/(1-p)$, i.e. $p = \pi$. By (12.3), $\Lambda_\tau \geq 0$, zero iff the share of events equals the share of time.

Step — (b) Conditional law. Under H_0 with common rate θ, $S_\tau \sim \text{Poisson}(\tau\theta)$ and $S - S_\tau \sim \text{Poisson}((n-\tau)\theta)$ independently, so for $0 \leq s \leq S$:

$$P(S_\tau = s \mid S) = \frac{\frac{(\tau\theta)^s e^{-\tau\theta}}{s!}\,\frac{((n-\tau)\theta)^{S-s} e^{-(n-\tau)\theta}}{(S-s)!}}{\frac{(n\theta)^S e^{-n\theta}}{S!}} = \binom{S}{s}\Big(\frac{\tau}{n}\Big)^s\Big(1-\frac{\tau}{n}\Big)^{S-s},$$

the same cancellation as (4.4). An exact level-α test for known τ rejects when S_τ is in the tails of Binomial(S, π); its size is at most α for every θ because the conditional law does not involve θ.

Step — (c) The maximum dominates. For every sample, $M = \max_\tau \Lambda_\tau \geq \Lambda_{\tau_0}$. Hence $\{\Lambda_{\tau_0} > c\} \subseteq \{M > c\}$ and

$$P_{H_0}(M > c) \geq P_{H_0}(\Lambda_{\tau_0} > c). \tag{12.4}$$

For fixed τ_0, $\Lambda_{\tau_0} \to \chi^2_1$ under H_0 (one restriction; B Teorema 6.5.1), so with $c = 3.841$ the right side of (12.4) tends to 0.05: comparing M with χ^2_1 gives a test whose size is at least 5% and, since the maximum runs over many correlated statistics, in practice far larger. Searching for the break and then testing it as if it were known is a multiple-testing error; the null law of M must be simulated.

Step — (d) Signs and convexity. Differentiating (12.3) in p:

$$\frac{\partial \Lambda_\tau}{\partial p} = 2S\Big[\log\frac{p}{\pi} + 1 - \log\frac{1-p}{1-\pi} - 1\Big] = 2S\log\frac{p(1-\pi)}{(1-p)\pi}, \tag{12.5}$$

negative for $p < \pi$, zero at $p = \pi$, positive for $p > \pi$ (the argument of the log exceeds 1 iff $p > \pi$). Differentiating (12.5) again:

$$\frac{\partial^2 \Lambda_\tau}{\partial p^2} = 2S\Big(\frac{1}{p} + \frac{1}{1-p}\Big) = \frac{2S}{p(1-p)} > 0.$$

Limits from (12.3): $p \to 0$ gives $2S\log\frac{1}{1-\pi}$; $p \to 1$ gives $2S\log\frac{1}{\pi}$. At fixed p, Λ_τ is linear in S.

Step — (d) Reading. Evidence of a break grows with the gap between the share of disasters and the share of years before τ, symmetrically in the KL sense, and doubles when the number of events doubles at the same shares: a 66%–37% split is weak with 10 events and overwhelming with 191.

Step — Coal data (programming). The maximum is at $\hat\tau = 41$ (1891): $S_{41} = 127$, $p = 127/191$, $\pi = 41/112$; $\hat\theta_1 = 127/41 = 3.097561$, $\hat\theta_2 = 64/71 = 0.901408$, $\hat\theta = 191/112 = 1.705357$, and $M = \Lambda_{41} = 69.99$. Under H_0, $E(S_{41} \mid S) = 191 \cdot 41/112 = 69.92$, while 127 were observed: the binomial tail $P(S_{41} \geq 127) = 5.4 \times 10^{-17}$. That number is not a valid p-value, since $\tau = 41$ was chosen because it maximizes the evidence. The bootstrap null law of M has 95% quantile 10.08 (not 3.84); the rule $M > 3.841$ rejects in 56.75% of null series, which quantifies (c); none of the 2,000 null values reaches 69.99 (bootstrap p-value $< 1/2000$).

Code and output (programming part).

```
def lr_tau(y):
    n, S = len(y), y.sum(); tau = np.arange(1, n); s1 = np.cumsum(y)[:-1]
    xlogy = lambda a, b: np.where(a > 0, a * np.log(np.where(a > 0, a, 1) / b), 0.)
    return 2 * (xlogy(s1, tau * S / n) + xlogy(S - s1, (n - tau) * S / n))
L = lr_tau(y); L.argmax() + 1, L.max()          # 41, 69.99
binom.sf(126, 191, 41 / 112)                    # 5.4e-17
rng = np.random.default_rng(12)
M = np.array([lr_tau(rng.poisson(191 / 112, 112)).max() for _ in range(2000)])
np.quantile(M, .95), np.mean(M > 3.841), np.mean(M >= 69.99)   # 10.08 0.5675 0.0
```

Intuition. A change point turns a count series into a split of events between two periods; the likelihood ratio measures how far that split is from the split of time. Choosing the period that looks most

different and then testing it as if fixed inflates the evidence, the same trap as reporting the best of many specifications.

Ref: Class 1 deck *Poisson with breakpoint* (LR at 1891 = 69.99); Bolfarine & Sandoval (2010), §6.5; Greenberg (2008), §7.1.1 for the Bayesian treatment.

Grading rubric.

- (a) MLEs and (12.1): 3 pts; (12.2)–(12.3): 2 pts; Jensen proof: 2 pts.
- (b) conditional binomial derived and the exact test: 3 pts.
- (c) (12.4) and the conclusion about the χ^2_1 rule: 3 pts (claim without the inclusion argument: max. 1).
- (d) (12.5) with sign, convexity, limits, scaling and reading: 4 pts.
- Programming: $\hat{\tau}$, M, binomial tail and bootstrap quantile: 3 pts.

Question 13. Original conceptual question — switch points and Bayes factors (10 points).
Source: Original question (not from a book).

a) (3 points) In the Gibbs sampler of Question 11, the full conditional distribution of the switch point k is:

A. uniform on $\{1, \ldots, n\}$, its prior, because k does not appear in the gamma conditionals.

B. a gamma distribution, because the gamma prior is conjugate to the Poisson likelihood.

C. Binomial$(S, k/n)$, the conditional law of the first-segment count given the total.

D. discrete on $\{1, \ldots, n\}$ with weights proportional to $e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{S_k}$.

Key: D

Step — Why. D is (11.7). A: k enters the likelihood through which years use which rate, so the data update it. B: k is discrete; conjugacy concerns the rates. C confuses the conditional law of the *data* S_k under H_0 (Question 12(b)) with the posterior of the *parameter* k.

b) (3 points) The Bayes factor B_{12} of Question 10 depends on the page-level data only through:

A. the individual page counts, through the product of factorials $\prod_i y_{1i}! \prod_i y_{2i}!$.

B. the totals e_1 and e_2 (given m), because the factorials cancel in the ratio.

C. the grand total $e = e_1 + e_2$, because model M_1 pools the two typists' pages.

D. the proportion e_1/e alone, whatever the number of errors behind it.

Key: B

Step — Why. B: (10.7). A: P appears in both marginal likelihoods and cancels. C: M_2 uses e_1 and e_2 separately. D: the table shows that the same proportion gives different factors for $e = 10$ and $e = 20$.

c) (2 points, T or F) In the switch-point model with independent gamma priors, θ_1 and θ_2 are independent a posteriori *given* k.

Key: T

Step — Why. (11.4) does not involve θ_2 and (11.5) does not involve θ_1; the joint (11.3) factorizes for fixed k.

d) (2 points, T or F) Hence θ_1 and θ_2 are also independent in their marginal posterior, after k is integrated out.

Key: F

Step — Why. The marginal posterior is a mixture over k of product densities, and mixing products creates dependence. Here a later k lowers both conditional means ($k = 36$: 3.18 and 0.97; $k = 44$: 2.90 and 0.89, from (11.4)–(11.5)), so averaging over $p(k \mid y)$ gives an exact posterior correlation of 0.060, not 0. Conditional independence does not imply marginal independence.

Programming (added, not in the book). Using the Gibbs draws of Question 11, compute the sample correlation of $(\theta_1^{(g)}, \theta_2^{(g)})$; compute the exact posterior correlation from the mixture over $p(k \mid y)$ of the moments of the two conditional gammas $\theta_j \mid y, k$ given in the statement of Question 11.

Code and output (programming part).

```
np.corrcoef(th1, th2)[0, 1]                          # 0.0708 (Gibbs)
m1, m2 = (a0 + cs) / (b0 + ks), (a0 + S - cs) / (b0 + n - ks)     # E(theta_j | k)
v1 = np.sum(pk * ((a0 + cs) / (b0 + ks)**2 + m1**2)) - np.sum(pk * m1)**2
v2 = np.sum(pk * ((a0 + S - cs) / (b0 + n - ks)**2 + m2**2)) - np.sum(pk * m2)**2
(np.sum(pk * m1 * m2) - np.sum(pk * m1) * np.sum(pk * m2)) / np.sqrt(v1 * v2)   # 0.0600
```

Both are clearly positive (the Gibbs value carries Monte Carlo error of about 0.01), so (d) is false.

Block 4 — Bayesian gamma–Poisson: posterior, predictive, Bayes estimators *(G 2.2, 3.2, 3.5; B 4.9/5.12; original)*

Question 14. G 2.2 — gamma–Poisson conjugacy.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 2, §2.5 Exercises, Exercise 2.2, p. 19 [PDF p. 34]; the book gives no answer — reproduced verbatim.

2.2 The Poisson distribution has probability mass function

$$p(y_i \mid \theta) = \frac{\theta^{y_i} e^{-\theta}}{y_i!}, \qquad \theta > 0,\ y_i = 0, 1, \ldots,$$

and let $y_1, \ldots, y_n$ be a random sample from this distribution.

(a) Show that the gamma distribution $G(\alpha, \beta)$ is a conjugate prior distribution for the Poisson distribution.

(b) Show that $\bar{y}$ is the MLE for θ.

(c) Write the mean of the posterior distribution as a weighted average of the mean of the prior distribution and the MLE.

(d) What happens to the weight on the prior mean as n becomes large?

Programming (added, not in the book). Use the coal counts 1851–1891 ($n = 41$) and the prior $G(0.5, 1)$ of G 7.3. Evaluate the unnormalized posterior $\theta^{\alpha+s-1}e^{-(\beta+n)\theta}$ on a grid of 200,001 points in $(0, 10]$ (work on the log scale), normalize it with the trapezoidal rule, and compare its mean with the closed form of (c). Print the prior weight w for $n = 1, 10, 41, 112, 1000$.

Solution.

Notation.

- $y_i \in \{0, 1, \ldots\}$: counts; $\theta > 0$: Poisson rate; $s = \sum_{i=1}^{n} y_i$, $\bar{y} = s/n$.
- $G(\alpha, \beta)$, $\alpha, \beta > 0$: gamma with shape α and rate β (Greenberg's convention, App. A), $\pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\theta^{\alpha-1}e^{-\beta\theta}$, mean α/β.
- $\ell(\theta)$: log-likelihood; $\hat{\theta}$: MLE; w: weight on the prior mean.

Step — (a) Likelihood. Independence gives

$$p(y \mid \theta) = \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} = \frac{\theta^{s} e^{-n\theta}}{\prod_i y_i!}. \tag{14.1}$$

Step — (a) Posterior. Bayes' theorem, $\pi(\theta \mid y) \propto p(y \mid \theta)\pi(\theta)$, with (14.1) and the gamma prior, dropping factors free of θ:

$$\pi(\theta \mid y) \propto \left(\theta^{s} e^{-n\theta}\right)\left(\theta^{\alpha-1} e^{-\beta\theta}\right). \tag{14.2}$$

Adding exponents in (14.2):

$$\pi(\theta \mid y) \propto \theta^{\alpha+s-1} e^{-(\beta+n)\theta}, \tag{14.3}$$

the kernel of $G(\alpha + s,\ \beta + n)$. Prior and posterior are both gamma, so the gamma family is conjugate for the Poisson likelihood.

Step — (b) MLE. Taking logs of (14.1):

$$\ell(\theta) = s \log \theta - n\theta - \sum_i \log y_i!. \tag{14.4}$$

Differentiating (14.4) and setting the score to zero:

$$\ell'(\theta) = \frac{s}{\theta} - n = 0 \implies \hat{\theta} = \frac{s}{n} = \bar{y}. \tag{14.5}$$

The second derivative $\ell''(\theta) = -s/\theta^2 < 0$ when $s > 0$, so (14.5) is a maximum. (If $s = 0$, $\ell(\theta) = -n\theta$ decreases and the supremum is at the boundary $0 = \bar{y}$.)

Step — (c) Posterior mean as a weighted average. The mean of the gamma (14.3) is

$$\mathbb{E}(\theta \mid y) = \frac{\alpha + s}{\beta + n}. \tag{14.6}$$

Substituting $\alpha = \beta(\alpha/\beta)$ and $s = n\bar{y}$ into (14.6):

$$\mathbb{E}(\theta \mid y) = \frac{\beta\,(\alpha/\beta) + n\,(\bar{y})}{\beta + n}.$$

Splitting the fraction:

$$\mathbb{E}(\theta \mid y) = \underbrace{\frac{\beta}{\beta + n}}_{w} \cdot \frac{\alpha}{\beta} + \underbrace{\frac{n}{\beta + n}}_{1-w} \cdot \bar{y}. \tag{14.7}$$

Step — (d) Large n. From (14.7),

$$w = \frac{\beta}{\beta + n} \xrightarrow[n\to\infty]{} 0, \tag{14.8}$$

so the posterior mean converges to the MLE $\bar{y}$; the posterior variance $(\alpha + s)/(\beta + n)^2$ also goes to zero at rate $1/n$.

Step — Numerical illustration. Coal-mining disasters 1851–1891 ($n = 41$, $s = 127$, $\bar{y} = 3.097561$) with the prior of G 7.3, $\alpha = 0.5$, $\beta = 1$. By (14.8), $w = 1/(1 + 41) = 0.023810$; by (14.7),

$$\mathbb{E}(\theta \mid y) = 0.023810\,(0.5) + 0.976190\,(3.097561) = 0.011905 + 3.023810 = 3.035714,$$

which equals (14.6): $(0.5 + 127)/(1 + 41) = 127.5/42 = 3.035714$.

Code and output (programming part).

```
grid = np.linspace(1e-6, 10, 200_001)
ker = np.exp((0.5 + 127 - 1) * np.log(grid) - (1 + 41) * grid - 300)  # -300: scale
pm = trapezoid(grid * ker, grid) / trapezoid(ker, grid)                 # 3.035714
w = {n: 1 / (1 + n) for n in (1, 10, 41, 112, 1000)}
# 0.500000  0.090909  0.023810  0.008850  0.000999
```

Intuition. The prior acts like β extra periods of data containing α events in total. After n real periods it weighs $\beta/(\beta + n)$; with 41 years of data a prior worth one year moves the estimate only from 3.098 to 3.036.

Ref: Greenberg (2008), §2.2 (pp. 12–18) and App. A (gamma distribution); the Class 1 deck *first Bayesian example* (Poisson with an exponential prior, the case $\alpha = 1$).

Question 15. G 3.2 — Poisson–gamma predictive distribution.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 3, §3.5 Exercises, Exercise 3.2, p. 39 [PDF p. 54]; the book gives no answer — reproduced verbatim.

3.2 Compute the predictive distribution for y_{n+1} if the y_i have independent Poisson distributions with parameter θ, given that the first n experiments yielded $y_1, y_2, \ldots, y_n$ events, respectively. Assume the prior distribution $G(\alpha, \beta)$.

Programming (added, not in the book). For the coal data 1851–1891 with prior $G(0.5, 1)$, simulate the predictive by composition (Greenberg §5.2): draw $\theta^{(g)}$ from the posterior and then $y^{(g)} \sim \text{Poisson}(\theta^{(g)})$, $g = 1, \ldots, 10^6$, with `np.random.default_rng(15)`. Compare the simulated frequencies of $y = 0, 1, 2, 3$, mean and variance with your closed form, and check the closed form at $y = 0, \ldots, 3$ against numerical integration of the predictive integral.

Solution.

Notation.

- y_i, $s = \sum_{i \le n} y_i$, θ, α, β as in Question 14; $y \in \{0, 1, \ldots\}$: value of y_{n+1}.
- $\alpha_1 = \alpha + s$, $\beta_1 = \beta + n$: posterior parameters.

Step — Posterior. By (14.3) of Question 14, $\theta \mid y_{1:n} \sim G(\alpha_1, \beta_1)$, and y_{n+1} is independent of $y_{1:n}$ given θ.

Step — Predictive integral. Integrating the Poisson p.m.f. against the posterior:

$$p(y_{n+1} = y \mid y_{1:n}) = \int_0^\infty \frac{\theta^y e^{-\theta}}{y!} \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)} \theta^{\alpha_1 - 1} e^{-\beta_1 \theta} \, d\theta \tag{15.1}$$

$$= \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)\, y!} \int_0^\infty \theta^{\alpha_1 + y - 1} e^{-(\beta_1 + 1)\theta} d\theta. \tag{15.2}$$

Substituting the gamma integral (10.1) with $a = \alpha_1 + y$, $b = \beta_1 + 1$ into (15.2):

$$p(y_{n+1} = y \mid y_{1:n}) = \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)\, y!} \cdot \frac{\Gamma(\alpha_1 + y)}{(\beta_1 + 1)^{\alpha_1 + y}} \tag{15.3}$$

$$= \frac{\Gamma(\alpha_1 + y)}{\Gamma(\alpha_1)\, y!} \left(\frac{\beta_1}{\beta_1 + 1}\right)^{\alpha_1} \left(\frac{1}{\beta_1 + 1}\right)^{y}, \qquad y = 0, 1, \ldots \tag{15.4}$$

This is the negative binomial with $r = \alpha_1$ and success probability $p = \beta_1/(\beta_1 + 1)$.

Step — Moments. For that negative binomial, $E = r(1-p)/p$ and $\text{Var} = r(1-p)/p^2$; substituting $r = \alpha_1$, $p = \beta_1/(\beta_1 + 1)$:

$$E(y_{n+1} \mid y_{1:n}) = \frac{\alpha_1}{\beta_1}, \tag{15.5}$$

$$\text{Var}(y_{n+1} \mid y_{1:n}) = \frac{\alpha_1}{\beta_1}\Big(1 + \frac{1}{\beta_1}\Big). \tag{15.6}$$

Step — Numbers. Coal data 1851–1891 with $G(0.5, 1)$: $\alpha_1 = 127.5$, $\beta_1 = 42$. From (15.4), $P(y_{n+1} = 0) = (42/43)^{127.5} = 0.049780$ and $P(y_{n+1} = 3) = \frac{(129.5)(128.5)(127.5)}{3!}(42/43)^{127.5}(1/43)^3 = 0.221403$; from (15.5)–(15.6), $E = 127.5/42 = 3.035714$ and $\text{Var} = (3.035714)(1 + 1/42) = 3.107993$.

Code and output (programming part).

```
a1, b1 = 127.5, 42.0
pmf = lambda y: nbinom.pmf(y, a1, b1 / (b1 + 1))  # (15.4)
[pmf(k) for k in range(4)]                         # 0.049780 0.147604 0.220548 0.221403
quad(lambda t: poisson.pmf(3, t) * gamma.pdf(t, a1, scale=1/b1), 0, 10)[0]   # 0.221403
rng = np.random.default_rng(15)
yp = rng.poisson(rng.gamma(a1, 1 / b1, 10**6))    # composition
yp.mean(), yp.var(ddof=1)                          # 3.035098  3.101461
[np.mean(yp == k) for k in range(4)]               # 0.049581 0.147294 0.221058 0.221849
```

The simulated frequencies differ from (15.4) by at most 0.0006, about one Monte Carlo standard error ($\sqrt{p(1-p)/10^6} \le 0.0005$).

Intuition. The predictive variance exceeds the mean (a Poisson with known θ would have them equal): the factor $1 + 1/\beta_1$ is the extra spread coming from not knowing θ, and it vanishes as n (hence β_1) grows. Composition sampling is exactly how the predictive is defined: draw a parameter, then data given that parameter.

Ref: Greenberg (2008), §3.2.3 (prediction), §5.2 (composition), App. A.1.3 (negative binomial) and A.1.7.

Question 16. G 3.5 — the posterior median minimizes expected absolute loss.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 3, §3.5 Exercises, Exercise 3.5, p. 39 [PDF p. 54]; the book gives no answer — reproduced verbatim.

3.5 Show that the median of the posterior distribution minimizes loss under the absolute value loss function.

Programming (added, not in the book). For the coal posterior $G(127.5, 42)$ of Question 15, write the posterior expected loss $\rho(d) = \int |\theta - d|\, \pi(\theta \mid y)\, d\theta$ with `scipy.integrate.quad`, minimize it with `minimize_scalar` on $[2.5, 3.5]$, and compare the minimizer with the posterior median (`gamma.median`). Print ρ at the median, the mean and the mode, and check numerically that $\rho'(d) = 2F(d) - 1$ at $d = 3$.

Solution.

Notation.

- $\pi(\theta \mid y)$: posterior density, c.d.f. F, assumed continuous with finite mean.
- $L(\hat\theta, \theta) = |\hat\theta - \theta|$: absolute loss; $d \in \mathbb{R}$: a candidate estimate.
- $\rho(d) = \mathbb{E}[|\theta - d| \mid y]$: posterior expected loss; m: a posterior median, $F(m) = 1/2$.

Step — Split the loss. On $\theta < d$ the loss is $d - \theta$, on $\theta > d$ it is $\theta - d$:

$$\rho(d) = \int_{-\infty}^{d} (d - \theta)\, \pi(\theta \mid y)\, d\theta + \int_{d}^{\infty} (\theta - d)\, \pi(\theta \mid y)\, d\theta. \tag{16.1}$$

Step — Differentiate. By Leibniz's rule on each integral of (16.1) (the boundary terms are $(d-d)\pi(d \mid y) = 0$):

$$\rho'(d) = \int_{-\infty}^{d} \pi(\theta \mid y)\, d\theta - \int_{d}^{\infty} \pi(\theta \mid y)\, d\theta.$$

Writing both integrals with F:

$$\rho'(d) = F(d) - \big(1 - F(d)\big) = 2F(d) - 1. \tag{16.2}$$

Step — Minimum. By (16.2), $\rho'(d) < 0$ for $F(d) < 1/2$ and $\rho'(d) > 0$ for $F(d) > 1/2$: ρ decreases up to the median and increases after it. Hence ρ is minimized at $d = m$ with $F(m) = 1/2$. Equivalently, $\rho''(d) = 2\pi(d \mid y) \ge 0$: ρ is convex and (16.2) is the first-order condition of a global minimum. ■ (If F is flat at $1/2$ on an interval, every point of it is a median and minimizes ρ.)

Step — Coal posterior. $G(127.5, 42)$ has median 3.027781, mean $127.5/42 = 3.035714$ and mode $126.5/42 = 3.011905$: the right skew of the gamma orders them mode $<$ median $<$ mean.

Code and output (programming part).

```
post = gamma(127.5, scale=1/42)
rho = lambda d: quad(lambda t: abs(t - d) * post.pdf(t), 0, 10, points=[d])[0]
res = minimize_scalar(rho, bounds=(2.5, 3.5), method="bounded")
res.x, post.median()                    # 3.027781  3.027781
rho(post.median()), rho(post.mean()), rho(126.5 / 42)   # 0.214276 0.214369 0.214650
(rho(3 + 1e-5) - rho(3 - 1e-5)) / 2e-5, 2 * post.cdf(3) - 1   # equal, as in (16.2)
```

The numerical minimizer is the median; using the mean or the mode instead costs little here because the posterior is nearly symmetric.

Intuition. Moving the estimate up by a small step lowers the loss for every draw above it and raises it for every draw below, each by the same amount. The estimate stops moving when half the posterior mass lies on each side. Squared loss weights far draws more and pulls towards the mean.

Ref: Greenberg (2008), §3.2.1 (point estimates and loss functions, pp. 29–31).

Question 17. B 5.12 (with B 4.9) — Bayes estimator of $P(X = 0)$ and a Bayesian interval.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, Chapter 5, §5.6 Exercícios, Exercício 5.12, p. 89 [PDF p. 95]; it refers to Exercício 4.9, Chapter 4, §4.5 Exercícios, p. 72 [PDF p. 78], reproduced first. The book gives no answer to either — both reproduced verbatim.

Exercício referido: **4.9.** Em uma área de reflorestamento, o número de árvores de determinada espécie, por hectare, com certa doença tem uma distribuição *Poisson*(θ). A distribuição a priori de θ é exponencial com média igual a 1. Encontre o estimador de Bayes de $P_\theta(X = 0)$ com relação à perda quadrática..

5.12. Considere o Exercício 4.9. Obtenha um intervalo de confiança Bayesiano para θ com coeficiente de confiança $\gamma = 0,95$. Como fica seu intervalo se $x = 4$?

Programming (added, not in the book). For $x = 4$: (1) compute the equal-tailed 95% interval with `scipy.stats.gamma` and check it against the χ^2_{10} quantiles; (2) compute the HPD interval numerically (minimize the length of $[F^{-1}(p), F^{-1}(p + 0.95)]$ over p) and verify that the density is equal at both ends; (3) compute the exact frequentist coverage of the equal-tailed rule $x \mapsto I(x)$ at $\theta \in \{0.5, 1, 4, 8\}$ by summing Poisson probabilities.

Solution.

Notation.

- $X \mid \theta \sim$ Poisson(θ): one observation x; prior $\pi(\theta) = e^{-\theta}$, $\theta > 0$ (exponential, mean $1 = G(1, 1)$).
- $G(a, b)$: shape a, rate b; q_p: posterior p-quantile; $\chi^2_{\nu,p}$: chi-square quantile.

Step — Posterior (4.9). By Bayes' theorem,

$$\pi(\theta \mid x) \propto \frac{\theta^x e^{-\theta}}{x!} \cdot e^{-\theta} \propto \theta^x e^{-2\theta}, \tag{17.1}$$

the kernel of $G(x + 1, 2)$ (the case $\alpha = \beta = 1$, $n = 1$ of (14.3)):

$$\pi(\theta \mid x) = \frac{2^{x+1}}{x!}\theta^x e^{-2\theta}. \tag{17.2}$$

Step — Bayes estimator (4.9). Under quadratic loss the Bayes estimator of $g(\theta) = P_\theta(X = 0) = e^{-\theta}$ is its posterior mean. Using (17.2):

$$\mathbb{E}[e^{-\theta} \mid x] = \int_0^\infty e^{-\theta} \frac{2^{x+1}}{x!}\theta^x e^{-2\theta} d\theta = \frac{2^{x+1}}{x!} \int_0^\infty \theta^x e^{-3\theta} d\theta.$$

Substituting (10.1) with $a = x + 1$, $b = 3$:

$$\mathbb{E}[e^{-\theta} \mid x] = \frac{2^{x+1}}{x!} \cdot \frac{x!}{3^{x+1}} = \Big(\frac{2}{3}\Big)^{x+1}. \tag{17.3}$$

At $x = 4$: $(2/3)^5 = 0.131687$. (It is not $e^{-\mathbb{E}(\theta|x)} = e^{-5/2}$: the Bayes estimator of a function is the posterior mean of the function.)

Step — Pivot for the interval (5.12). If $\theta \mid x \sim G(x + 1, 2)$ then $4\theta \mid x \sim G(x + 1, 1/2)$, which is $\chi^2_{2(x+1)}$. The symmetric (equal-tailed) interval with $\gamma = 0.95$ is

$$\Big[\frac{\chi^2_{2x+2,\,0.025}}{4},\ \frac{\chi^2_{2x+2,\,0.975}}{4}\Big]. \tag{17.4}$$

For $x = 4$, $\chi^2_{10,0.025} = 3.246973$ and $\chi^2_{10,0.975} = 20.483177$; substituting into (17.4):

$$\Big[\frac{3.246973}{4},\ \frac{20.483177}{4}\Big] = [0.811743,\ 5.120794].$$

Step — HPD (programming). The posterior is right-skewed, so the shortest 95% interval is shifted left: [0.603480, 4.715108], length 4.111628 against 4.309051 for (17.4), with equal density at both ends.

Code and output (programming part).

```
pg = gamma(5, scale=0.5)
pg.ppf([.025, .975])                        # 0.811743 5.120794 = chi2(10) quantiles / 4
L = lambda p: pg.ppf(p + .95) - pg.ppf(p)
p = minimize_scalar(L, bounds=(1e-9, .05 - 1e-9), method="bounded").x
pg.ppf([p, p + .95])                        # HPD 0.603480 4.715108
xs = np.arange(200); lo, hi = gamma.ppf([[.025], [.975]], xs + 1, scale=.5)
cov = lambda th: np.sum(poisson.pmf(xs, th) * ((lo <= th) & (th <= hi)))
# coverage at theta = 0.5, 1, 4, 8:  0.985612  0.996340  0.761820  0.407449
```

A 95% *Bayesian* interval need not have 95% *frequentist* coverage: with a single observation the prior (mean 1) dominates, so the rule over-covers near $\theta = 1$ and badly under-covers large θ (40.7% at $\theta = 8$).

Intuition. With one count the posterior is a compromise between the observation and a prior that expects about one diseased tree per hectare; the interval is a statement about θ given this x, not a promise about repeated sampling.

Ref: Bolfarine & Sandoval (2010), §4.3 (estimador de Bayes, perda quadrática), §5.5 (intervalos Bayesianos, Exemplo 5.5.1).

Question 18. Original math question — Bayes estimators and their frequentist risk.
Source: Original question (not from a book).

An insurer prices a policy on the number of claims per client and year, $X_1, \ldots, X_n$ i.i.d. Poisson(θ), with a prior $\theta \sim G(\alpha, \beta)$ (shape, rate) from past portfolios; $S = \sum_i X_i$.

(a) Derive the Bayes estimator under squared loss, d_1, and under the weighted loss $L(\theta, d) = (\theta - d)^2/\theta$, d_2, stating when d_2 exists. Relate d_2 to a feature of the posterior.

(b) Derive the frequentist risk $R_1(\theta) = E_\theta[(d_1 - \theta)^2]$ and prove that $R_1(\theta) < \theta/n$ (the risk of the MLE $\overline{X}$) exactly on an interval of θ that contains the prior mean α/β; give the interval.

(c) Compute the Bayes risks $r = \int R(\theta)\pi(\theta)d\theta$ of d_1 and of $\overline{X}$ and prove $r(d_1) < r(\overline{X})$.

(d) Parameter analysis: sign of the bias of d_1 as a function of θ; the variance ratio $\mathrm{Var}(d_1)/\mathrm{Var}(\overline{X})$; the limits of the interval of (b) as $n \to \infty$ and as $\beta \to 0$ with α fixed. Interpret for the insurer.

Programming (added, not in the book). With $\alpha = 2$, $\beta = 1$, $n = 10$: print the interval of (b), its $n \to \infty$ limit and the two Bayes risks; with `np.random.default_rng(18)` simulate 400,000 values of S at $\theta \in \{1, 2, 5, 6\}$ and compare the simulated mean squared errors of d_1 and $\overline{X}$ with the formulas.

Solution.

Notation.

- $S \sim \mathrm{Poisson}(n\theta)$; posterior $\theta \mid \mathbf{x} \sim G(a, b)$, $a = \alpha + S$, $b = \beta + n$ (by (14.3)).
- $\rho_j(d)$: posterior expected loss; $R(\theta)$: frequentist risk (MSE); r: Bayes risk.
- $\alpha, \beta > 0$, $n \geq 1$.

Step — (a) Squared loss. $\rho_1(d) = \mathbb{E}[(\theta - d)^2 \mid \mathbf{x}] = \mathrm{Var}(\theta \mid \mathbf{x}) + (\mathbb{E}(\theta \mid \mathbf{x}) - d)^2$ is minimized at the posterior mean, so by (14.6)

$$d_1 = \frac{\alpha + S}{\beta + n}. \tag{18.1}$$

Step — (a) Weighted loss. Expanding the square,

$$\rho_2(d) = \mathbb{E}\Big[\frac{(\theta - d)^2}{\theta}\Big|\,\mathbf{x}\Big] = \mathbb{E}(\theta \mid \mathbf{x}) - 2d + d^2\,\mathbb{E}(\theta^{-1} \mid \mathbf{x}), \tag{18.2}$$

a convex quadratic in d with minimum where $-2 + 2d\,\mathbb{E}(\theta^{-1} \mid \mathbf{x}) = 0$:

$$d_2 = \frac{1}{\mathbb{E}(\theta^{-1} \mid \mathbf{x})}. \tag{18.3}$$

For $G(a, b)$, using (10.1) with exponent $a - 1$: $\mathbb{E}(\theta^{-1}) = \frac{b^a}{\Gamma(a)} \int_0^\infty \theta^{a-2} e^{-b\theta} d\theta = \frac{b^a}{\Gamma(a)} \cdot \frac{\Gamma(a-1)}{b^{a-1}} = \frac{b}{a-1}$, finite iff $a > 1$. Substituting into (18.3):

$$d_2 = \frac{a - 1}{b} = \frac{\alpha + S - 1}{\beta + n}, \qquad \text{if } \alpha + S > 1, \tag{18.4}$$

which is the posterior *mode*: dividing the loss by θ makes errors at small θ costly and pulls the estimate down by $1/(\beta + n)$.

Step — (b) Bias and variance of d_1. From (18.1) with $E_\theta S = \mathrm{Var}_\theta\, S = n\theta$:

$$E_\theta d_1 - \theta = \frac{\alpha + n\theta - (\beta + n)\theta}{\beta + n} = \frac{\alpha - \beta\theta}{\beta + n}, \qquad \mathrm{Var}_\theta(d_1) = \frac{n\theta}{(\beta + n)^2}. \tag{18.5}$$

Adding variance and squared bias:

$$R_1(\theta) = \frac{n\theta + (\alpha - \beta\theta)^2}{(\beta + n)^2}. \tag{18.6}$$

Step — (b) Where d_1 beats the MLE. $R_1(\theta) < \theta/n$; multiplying by $n(\beta + n)^2 > 0$ and substituting (18.6):

$$n\big[n\theta + (\alpha - \beta\theta)^2\big] < \theta(\beta + n)^2.$$

Expanding both sides and collecting powers of θ:

$$Q(\theta) \equiv n\beta^2\theta^2 - (\beta^2 + 2n\beta + 2n\alpha\beta)\theta + n\alpha^2 < 0. \tag{18.7}$$

Q is an upward parabola, so (18.7) holds exactly between its roots $\theta_- < \theta_+$. At the prior mean, substituting $\theta = \alpha/\beta$ into (18.7):

$$Q(\alpha/\beta) = n\alpha^2 - (\beta^2 + 2n\beta + 2n\alpha\beta)\frac{\alpha}{\beta} + n\alpha^2 = -\alpha\beta - 2n\alpha < 0,$$

so the roots are real and distinct and $\alpha/\beta \in (\theta_-, \theta_+)$. ■

Step — (c) Bayes risks. Under the prior, $E\theta = \alpha/\beta$ and $E(\alpha - \beta\theta)^2 = \beta^2\,\mathrm{Var}(\theta) = \beta^2(\alpha/\beta^2) = \alpha$. Averaging (18.6):

$$r(d_1) = \frac{n\alpha/\beta + \alpha}{(\beta + n)^2} = \frac{\alpha(n + \beta)}{\beta(\beta + n)^2} = \frac{\alpha}{\beta(\beta + n)}, \qquad r(\overline{X}) = E\Big[\frac{\theta}{n}\Big] = \frac{\alpha}{\beta n}. \tag{18.8}$$

Since $\beta + n > n$, $r(d_1) < r(\overline{X})$ (as it must: d_1 minimizes the Bayes risk).

Step — (d) Signs and limits. By (18.5) the bias is positive for $\theta < \alpha/\beta$, zero at the prior mean and negative above it: d_1 shrinks towards α/β. The variance ratio is $\mathrm{Var}(d_1)/\,\mathrm{Var}(\overline{X}) =$

$(n/(\beta+n))^2 < 1$. Dividing (18.7) by n: $\beta^2\theta^2 - (\beta^2/n + 2\beta + 2\alpha\beta)\theta + \alpha^2 < 0$; as $n \to \infty$ this tends to $\beta^2\theta^2 - 2\beta(1+\alpha)\theta + \alpha^2 < 0$, with roots

$$\theta_{\pm}^{\infty} = \frac{(1+\alpha) \pm \sqrt{(1+\alpha)^2 - \alpha^2}}{\beta} = \frac{(1+\alpha) \pm \sqrt{1+2\alpha}}{\beta} : \tag{18.9}$$

the region does not vanish with n (both risks shrink like $1/n$ and their ratio stays informative). As $\beta \to 0$ with α fixed, $\theta_- \approx n\alpha^2/(\beta^2 + 2n\beta(1+\alpha)) \to \infty$ and the region disappears: (18.6) tends to $\theta/n + \alpha^2/n^2 > \theta/n$, a pure bias with no shrinkage.

Step — (d) Reading. The prior helps the insurer when the true claim rate is near the portfolio history and hurts when the client is very different; the band (18.9) measures how far is "very different". A flat-but-shifted prior ($\beta \to 0$) is the worst of both worlds.

Step — Numbers. $\alpha = 2$, $\beta = 1$, $n = 10$: (18.7) is $10\theta^2 - (1 + 20 + 40)\theta + 40 = 10\theta^2 - 61\theta + 40 < 0$, discriminant $61^2 - 4(10)(40) = 2121$, roots $(61 \mp \sqrt{2121})/20 = (0.747284,\ 5.352716)$; (18.9): $3 \mp \sqrt{5} = (0.763932,\ 5.236068)$; (18.8): $r(d_1) = 2/11 = 0.181818 < r(\overline{X}) = 2/10 = 0.2$.

Code and output (programming part).

```
al, be, n = 2., 1., 10
rB = lambda t: (n * t + (al - be * t)**2) / (be + n)**2
rng = np.random.default_rng(18)
for th in (1, 2, 5, 6):
    S = rng.poisson(n * th, 400_000)
    print(th, rB(th), np.mean(((al + S)/(be + n) - th)**2), th/n, np.mean((S/n - th)**2))
# theta  R1 formula  R1 sim    MLE formula  MLE sim
# 1      0.090909    0.091203  0.1          0.100222
# 2      0.165289    0.164913  0.2          0.199545
# 5      0.487603    0.487363  0.5          0.499468
# 6      0.628099    0.629657  0.6          0.602826
```

Inside $(0.747, 5.353)$ (at $\theta = 1, 2, 5$) the Bayes estimator wins; at $\theta = 6$ it loses, as (b) predicts.

Intuition. Shrinkage trades a little bias for less variance. The trade pays off near the prior mean and on average over the prior, never uniformly: by (b), d_1 loses to the MLE outside a bounded interval of θ, whatever n.

Ref: Bolfarine & Sandoval (2010), §4.2–4.3 (risco, risco de Bayes, estimador de Bayes); Greenberg (2008), §3.2.1 (loss functions).

Grading rubric.

- (a) d_1: 2 pts; (18.2)–(18.4) with the existence condition and the mode: 3 pts.
- (b) (18.5)–(18.6): 3 pts; (18.7) and $Q(\alpha/\beta) < 0$: 3 pts.
- (c) (18.8) and the inequality: 2 pts.
- (d) bias sign, variance ratio, (18.9), $\beta \to 0$, reading: 4 pts (limits without derivation: max. half).
- Programming: interval, Bayes risks and simulated risks: 3 pts.

Question 19. Original conceptual question — gamma–Poisson Bayes (10 points).
Source: Original question (not from a book).

An analyst at ANS (the health-insurance regulator) models claims per beneficiary as Poisson(θ) with prior $G(\alpha, \beta)$ and observes n beneficiaries with total s.

a) (3 points) In the posterior mean written as a weighted average of the prior mean α/β and the MLE $\bar{y}$, the weight on $\bar{y}$ is:

A. $n/(\beta+n)$, the share of real observations in the total "sample size".

B. $\beta/(\beta+n)$, the share of the prior's pseudo-observations.

C. $n/(\alpha+n)$, with the prior's events counted as observations.

D. $\alpha/(\alpha+\beta)$, which does not depend on the sample.

Key: A

Step — Why. A: (14.7). B is the weight on the *prior* mean; C mixes up the prior's events α with its periods β; D ignores the data entirely.

b) (3 points) The predictive distribution of a new beneficiary's count y_{n+1} is:

A. Poisson with mean $(\alpha+s)/(\beta+n)$, plugging in the posterior mean.

B. gamma $G(\alpha+s, \beta+n)$, the posterior of θ.

C. negative binomial, with variance larger than its mean.

D. negative binomial, with variance smaller than its mean.

Key: C

Step — Why. C: (15.4), with (15.6) = mean $\times(1+1/\beta_1)$. A ignores the uncertainty about θ (plug-in); B is a distribution for the parameter, not for a count; D has the inequality reversed.

c) (2 points, T or F) Under absolute-error loss the Bayes estimator of θ is the posterior median.

Key: T

Step — Why. Question 16, (16.2).

d) (2 points, T or F) Since the Bayes estimator under squared loss is the posterior mean, it is unbiased for θ in repeated sampling.

Key: F

Step — Why. Its frequentist bias is $(\alpha-\beta\theta)/(\beta+n)$ by (18.5), zero only at $\theta=\alpha/\beta$. "Mean" refers to the posterior, not to the sampling distribution.

Programming (added, not in the book). Check (d) with $\alpha=2$, $\beta=1$, $n=10$, $\theta=1$: with `np.random.default_rng(19)` simulate 400,000 totals $S\sim$ Poisson(10) and report the average of $(\alpha+S)/(\beta+n)-\theta$.

Code and output (programming part).

```
rng = np.random.default_rng(19); S = rng.poisson(10, 400_000)
np.mean((2 + S) / 11) - 1                          # 0.090347 ; formula 1/11 = 0.090909
```

Block 5 — Monte Carlo for inference: integration, importance sampling, accept–reject *(G 5.6, 5.7; MS B.15, B.18; original)*

Question 20. G 5.6 — the GHK importance-sampling estimator (5.3).

Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 5, §5.8 Exercises, Exercise 5.6, p. 75 [PDF p. 90]; the book gives no answer — reproduced verbatim. The text it refers to (§5.4, pp. 71–72 [PDF pp. 86–87], including Equation (5.3)) and Equation (5.1) (§5.1, p. 65 [PDF p. 80]) are reproduced first.

Referred to (Greenberg 2008, p. 65): "we generate $U\sim U(0,1)$ and set $U=\frac{F(X)-F(c_1)}{F(c_2)-F(c_1)}$, which implies that

$$X=F^{-1}(F(c_1)+U[F(c_2)-F(c_1)]) \tag{5.1}$$

is a drawing from the truncated distribution."

Referred to (Greenberg 2008, pp. 71–72): "As a second example, we approximate $P(a_1<X_1\le b_1, a_2<X_2\le b_2)$, where $(X_1,X_2)\sim N_2(\mu,\Sigma)$, $\mu=(\mu_1,\mu_2)'$, and $\Sigma=[\sigma_{ij}]$. The desired integral is

$$P(a_1<X_1\le b_1, a_2<X_2\le b_2)=\int_{a_1}^{b_1}\int_{a_2}^{b_2} f(x_1,x_2)\,dx_1\,dx_2,$$

where $f(x_1, x_2)$ is the density function of the bivariate normal distribution. The first step is to rewrite the joint density in marginal-conditional form, $f(x_1)f(x_2|x_1)$ using the results (see Section A.9) that $x_1 \sim N(\mu_1, \sigma_{11})$ and that $x_2|x_1 \sim N(\mu_2', \sigma_{22}')$, where

$$\mu_2' = \mu_2 + (\sigma_{12}/\sigma_{11})(x_1 - \mu_1), \qquad \sigma_{22}' = \sigma_{22} - \sigma_{12}^2/\sigma_{11}.$$

Now let

$$\begin{aligned} y_1 &= (x_1 - \mu_1)/\sqrt{\sigma_{11}}, & y_2 &= (x_2 - \mu_2')/\sqrt{\sigma_{22}'}, \\ a_1' &= (a_1 - \mu_1)/\sqrt{\sigma_{11}}, & a_2' &= (a_2 - \mu_2')/\sqrt{\sigma_{22}'}, \\ b_1' &= (b_1 - \mu_1)/\sqrt{\sigma_{11}}, & b_2' &= (b_2 - \mu_2')/\sqrt{\sigma_{22}'}. \end{aligned}$$

With this transformation, y_1 and $y_2|y_1$ have standard normal distributions, enabling us to write

$$\begin{aligned} P(a_1 < x_1 \le b_1, a_2 < x_2 \le b_2) &= \int_{a_1'}^{b_1'} \int_{a_2'}^{b_2'} \phi(y_1)\phi(y_2|y_1)\, dy_1\, dy_2 \\ &= \int_{a_1'}^{b_1'} \phi(y_1) \int_{a_2'}^{b_2'} \phi(y_2|y_1)\, dy_2\, dy_1 \\ &= \int_{a_1'}^{b_1'} \phi(y_1)[\Phi(b_2'|y_1) - \Phi(a_2'|y_1)]\, dy_1. \end{aligned}$$

The algorithm is now implemented by choosing the standard normal distribution truncated to (a_1', b_1') as the importance distribution; that is, draw $Y_1^{(g)}$ from the truncated standard normal using (5.1). With this choice for the importance function, we have

$$P(a_1 < X_1 \le b_1, a_2 < X_2 \le b_2) \approx \frac{1}{G} \sum \big[\Phi\big(b_2'|Y_1^{(g)}\big) - \Phi\big(a_2'|Y_1^{(g)}\big)\big]\big[\Phi(b_1') - \Phi(a_1')\big].\text{"} \tag{5.3}$$

5.6 Verify Equation (5.3).

Programming (added, not in the book). Use the numbers of G 5.7 (next question). (1) Implement the truncated-normal draw (5.1) and compare the mean of 10,000 draws (`np.random.default_rng(21)`) with `scipy.stats.truncnorm`. (2) Compute by quadrature the exact variance of one term of (5.3) and of one crude indicator $\mathbf{1}\{a_1 < X_1 \le b_1, a_2 < X_2 \le b_2\}$, their ratio, and check the bound you prove in the solution.

Solution.

Notation.

- $p = P(a_1 < X_1 \le b_1, a_2 < X_2 \le b_2)$: target; ϕ, Φ: standard normal density and c.d.f.
- $h(y_1) = \Phi(b_2'|y_1) - \Phi(a_2'|y_1) \in [0,1]$, where $\Phi(c'|y_1) = \Phi\big((c - \mu_2'(y_1))/\sqrt{\sigma_{22}'}\big)$ and $\mu_2'(y_1) = \mu_2 + (\sigma_{12}/\sigma_{11})\sqrt{\sigma_{11}}\, y_1$.
- $m_1 = \Phi(b_1') - \Phi(a_1')$; $q(y_1) = \phi(y_1)\mathbf{1}\{a_1' < y_1 \le b_1'\}/m_1$: truncated normal (importance) density.

Step — Inner integral. Given x_1, $x_2 \sim N(\mu_2', \sigma_{22}')$, so standardizing,

$$\int_{a_2}^{b_2} f(x_2|x_1)\, dx_2 = \Phi\Big(\frac{b_2 - \mu_2'}{\sqrt{\sigma_{22}'}}\Big) - \Phi\Big(\frac{a_2 - \mu_2'}{\sqrt{\sigma_{22}'}}\Big) = h(y_1). \tag{20.1}$$

Step — Outer integral. With $x_1 = \mu_1 + \sqrt{\sigma_{11}}y_1$, $dx_1 = \sqrt{\sigma_{11}}\, dy_1$ and $f(x_1)dx_1 = \phi(y_1)dy_1$; substituting (20.1) into the marginal–conditional form:

$$p = \int_{a_1}^{b_1} f(x_1)\Big(\int_{a_2}^{b_2} f(x_2|x_1)dx_2\Big)dx_1 = \int_{a_1'}^{b_1'} \phi(y_1)\, h(y_1)\, dy_1. \tag{20.2}$$

Step — Importance-sampling form. Multiplying and dividing the integrand of (20.2) by m_1:

$$p = \int \Big(\frac{\phi(y_1)\mathbf{1}\{a_1' < y_1 \le b_1'\}}{m_1} \Big) \big(m_1\, h(y_1)\big)\, dy_1 = E_q\big[m_1\, h(Y_1)\big]. \tag{20.3}$$

The importance weight $\phi/q = m_1$ is constant on the support. If $Y_1^{(g)}$ are i.i.d. from q — which is what (5.1) delivers with $F = \Phi$, $c_1 = a_1'$, $c_2 = b_1'$ — then by (20.3) each term $m_1 h(Y_1^{(g)})$ has mean p, and by the strong law of large numbers

$$\frac{1}{G}\sum_{g=1}^{G} \big[\Phi(b_2'|Y_1^{(g)}) - \Phi(a_2'|Y_1^{(g)})\big]\big[\Phi(b_1') - \Phi(a_1')\big] \xrightarrow{a.s.} p,$$

which is (5.3), unbiased for every G. ■

Step — It beats crude Monte Carlo. Since $0 \le h \le 1$, $h^2 \le h$, so

$$\mathrm{Var}_q(m_1 h) = m_1^2 E_q[h^2] - p^2 \le m_1^2 E_q[h] - p^2 = m_1 p - p^2 = p(m_1 - p) \le p(1-p), \tag{20.4}$$

the last term being the variance of one crude indicator. The gain is large when m_1 is small (the first coordinate's constraint is handled exactly) and when h varies little.

Step — Numbers (G 5.7). $\sigma_{22}' = 3 - 1^2/2 = 2.5$; $a_1' = (0.2-1)/\sqrt{2} = -0.565685$, $b_1' = (2-1)/\sqrt{2} = 0.707107$, $m_1 = 0.474446$; $p = 0.300105$ (quadrature of (20.2)). One GHK term has variance 0.000797, below the bound (20.4) $p(m_1 - p) = 0.052321$; a crude indicator has $p(1-p) = 0.210042$: a ratio of 263.6, so one GHK draw is worth about 264 crude draws.

Code and output (programming part).

```
a1p, b1p = (0.2 - 1) / np.sqrt(2), (2 - 1) / np.sqrt(2)
rng = np.random.default_rng(21); u = rng.uniform(size=10_000)
y1 = norm.ppf(norm.cdf(a1p) + u * (norm.cdf(b1p) - norm.cdf(a1p)))   # (5.1)
y1.mean(), truncnorm(a1p, b1p).mean()       # 0.0655  0.0617 (se ~0.004)
h = lambda y: (norm.cdf((5 - mu2c(y)) / np.sqrt(2.5))
               - norm.cdf((-1 - mu2c(y)) / np.sqrt(2.5)))   # mu2c(y) = -0.5 + y/sqrt(2)
m1 = norm.cdf(b1p) - norm.cdf(a1p)                            # 0.474446
p = quad(lambda y: norm.pdf(y) * h(y), a1p, b1p)[0]           # 0.300105
v = quad(lambda y: norm.pdf(y) / m1 * (m1 * h(y))**2, a1p, b1p)[0] - p**2   # 0.000797
p * (1 - p) / v                                               # 263.6
```

Intuition. GHK integrates the second coordinate exactly, given the first, and samples the first only where it matters. Every simulation draw lands in the region and contributes a smooth probability, instead of a 0/1 indicator that wastes most draws.

Ref: Greenberg (2008), §5.1 (eq. 5.1), §5.4 (importance sampling, pp. 70–72) and §5.7 (GHK).

Question 21. G 5.7 — importance sampling for a bivariate normal rectangle.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 5, §5.8 Exercises, Exercise 5.7, p. 75 [PDF p. 90]; the book gives no answer — reproduced verbatim. The method is the one of (5.3), reproduced in Question 20.

5.7 Use importance sampling to compute $P(0.2 < x_1 \le 2,\ -1 < x_2 \le 5)$, where $(x_1, x_2) \sim N_2(\mu, \Sigma)$, $\mu = (1, -0.5)'$, and

$$\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

Programming (added, not in the book). With $G = 10{,}000$ and `np.random.default_rng(21)` (the same truncated-normal draws as Question 20), report the estimate (5.3) and its numerical standard error; compare with the exact value (one-dimensional quadrature of the last integral of the text quoted in Question 20, cross-checked with `multivariate_normal.cdf` and inclusion–exclusion) and with crude Monte Carlo using 10,000 bivariate normal draws from the same generator.

Solution.

Notation.

- As in Question 20, with $\mu_1 = 1$, $\mu_2 = -0.5$, $\sigma_{11} = 2$, $\sigma_{12} = 1$, $\sigma_{22} = 3$; $a_1 = 0.2$, $b_1 = 2$, $a_2 = -1$, $b_2 = 5$.
- $\hat{p}_G$: estimate (5.3); n.s.e. $= s/\sqrt{G}$, s the standard deviation of the G terms.

Step — Ingredients. From Question 20: $\sigma'_{22} = 2.5$, $a'_1 = -0.565685$, $b'_1 = 0.707107$, $m_1 = 0.474446$, and

$$\mu'_2(y_1) = -0.5 + \tfrac{1}{2}\big((1 + \sqrt{2}\, y_1) - 1\big) = -0.5 + \frac{y_1}{\sqrt{2}}.$$

Step — Algorithm. For $g = 1, \ldots, G$: draw $U^{(g)} \sim U(0,1)$, set $Y_1^{(g)} = \Phi^{-1}(\Phi(a'_1) + U^{(g)} m_1)$ by (5.1), compute $t_g = m_1[\Phi((5 - \mu'_2(Y_1^{(g)}))/\sqrt{2.5}) - \Phi((-1 - \mu'_2(Y_1^{(g)}))/\sqrt{2.5})]$; return $\hat{p}_G = \bar{t}$ and n.s.e. $s_t/\sqrt{G}$.

Step — Result. $\hat{p}_G = 0.300411$ with n.s.e. 0.000282; the exact value is 0.300105, about 1.1 n.s.e. away. Crude Monte Carlo with the same G gives 0.3019 with standard error 0.0046, sixteen times larger, consistent with the variance ratio 263.6 of Question 20 ($\sqrt{263.6} \approx 16.2$).

Code and output (programming part).

```
t = m1 * h(y1)                                        # y1 from (5.1), Question 20
t.mean(), t.std(ddof=1) / np.sqrt(10_000)             # 0.300411  0.000282
mvn = multivariate_normal([1, -.5], [[2, 1], [1, 3]])
mvn.cdf([2, 5]) - mvn.cdf([.2, 5]) - mvn.cdf([2, -1]) + mvn.cdf([.2, -1])  # 0.300105
x = mvn.rvs(size=10_000, random_state=rng)
ind = (.2 < x[:, 0]) & (x[:, 0] <= 2) & (-1 < x[:, 1]) & (x[:, 1] <= 5)
ind.mean(), ind.std(ddof=1) / 100                     # 0.3019  0.0046
```

Intuition. The rectangle is wide in x_2 (from -1 to 5) and narrow in x_1, so once x_1 is drawn inside its interval the conditional probability for x_2 is high and stable; nearly all the uncertainty is removed analytically.

Ref: Greenberg (2008), §5.4 (pp. 70–72).

Question 22. MS B.15 — generating Poisson values by the inverse transform.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Apêndice B (Noções de simulação), §B.8 Exercícios, Exercício 15, p. 478 [PDF p. 488]; the book gives no answer — reproduced verbatim.

15) Gere 5 valores de uma variável aleatória com distribuição de Poisson de parâmetro $\lambda = 2$.

Programming (added, not in the book). Write `rpois_inv(u, lam)` returning the smallest j with $F(j) > u$, built with the recursion $p_{j+1} = p_j \lambda/(j+1)$, and the number of comparisons it made. With `np.random.default_rng(22)`: generate the 5 values; then 100,000 more, and report their mean, variance, the average number of comparisons, and a chi-square goodness-of-fit test on the classes $0, 1, \ldots, 6, \geq 7$.

Solution.

Notation.

- $p_j = P(X = j) = e^{-\lambda}\lambda^j/j!$, $F(j) = \sum_{i \leq j} p_i$, $F(-1) = 0$; $U \sim U(0,1)$.
- N: number of comparisons "$U < F(j)$?" made by the algorithm.

Step — Algorithm. Set $X = j$ when $F(j-1) \leq U < F(j)$: start with $j = 0$, $p = e^{-\lambda}$, $F = p$; while $U \geq F$, update $p \leftarrow p\lambda/(j+1)$, $j \leftarrow j+1$, $F \leftarrow F + p$. The recursion holds because

$$\frac{p_{j+1}}{p_j} = \frac{e^{-\lambda}\lambda^{j+1}/(j+1)!}{e^{-\lambda}\lambda^j/j!} = \frac{\lambda}{j+1}. \tag{22.1}$$

Step — Correctness. The intervals $[F(j-1), F(j))$ partition $[0,1)$ and have lengths p_j, so

$$P(X=j) = P\big(F(j-1) \le U < F(j)\big) = F(j) - F(j-1) = p_j. \tag{22.2}$$

Step — Cost. The algorithm stops after checking $j = 0, 1, \ldots, X$, so $N = X+1$ and $E(N) = \lambda + 1 = 3$: the inverse transform is cheap for small λ and grows linearly in it.

Step — Five values. With $\lambda = 2$: $F(0) = 0.135335$, $F(1) = 0.406006$, $F(2) = 0.676676$, $F(3) = 0.857123$. The seed-22 uniforms 0.366347, 0.199295, 0.088558, 0.653192, 0.459337 fall in $[F(0), F(1))$, $[F(0), F(1))$, $[0, F(0))$, $[F(1), F(2))$, $[F(1), F(2))$, giving $1, 1, 0, 2, 2$.

Code and output (programming part).

```
def rpois_inv(u, lam):
    j, p = 0, np.exp(-lam); F = p
    while u >= F:
        p *= lam / (j + 1); j += 1; F += p      # recursion (22.1)
    return j, j + 1                             # value, comparisons
rng = np.random.default_rng(22)
[rpois_inv(u, 2.)[0] for u in rng.uniform(size=5)]          # [1, 1, 0, 2, 2]
big = np.array([rpois_inv(u, 2.) for u in rng.uniform(size=100_000)])
big[:, 0].mean(), big[:, 0].var(ddof=1), big[:, 1].mean()   # 1.99556 2.01146 2.99556
# chi-square GOF, classes 0..6, >=7: X2 = 12.44 on 7 df, p = 0.0869
```

Mean and variance are both close to $\lambda = 2$, the comparisons average $3 = \lambda + 1$, and the goodness-of-fit test does not reject at 5%.

Intuition. Every discrete distribution can be simulated by walking up its c.d.f. until it passes a uniform draw; the Poisson recursion makes each step a single multiplication.

Ref: Morettin & Singer (2021), Apêndice B (geração de variáveis discretas, método da transformação inversa); Greenberg (2008), §5.1.

Question 23. MS B.18 — half-normal by accept–reject, then a normal sample.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Apêndice B, §B.8 Exercícios, Exercício 18, p. 478 [PDF p. 488]; the book gives no answer — reproduced verbatim.

18) Suponha que $Z \sim N(0,1)$ e considere $X = |Z|$. Obtenha a densidade $\pi(x)$ de X. Use o método da aceitação-rejeição para obter uma amostra de $\pi(x)$ com $g(x) = \exp(-x)$. A seguir gere uma amostra de Z, levando em conta que $Z = X$ ou $Z = -X$.

Programming (added, not in the book). With `np.random.default_rng(23)`, draw proposals until 10,000 are accepted; report the empirical acceptance rate against $1/c$, the mean of X against $\sqrt{2/\pi}$, the mean and variance of the signed sample Z, and Kolmogorov–Smirnov p-values of X against the half-normal and of Z against $N(0,1)$.

Solution.

Notation.

- ϕ, Φ: standard normal density and c.d.f.; $g(x) = e^{-x}$, $x \ge 0$: proposal (exponential).
- $c = \sup_x \pi(x)/g(x)$: envelope constant; $U \sim U(0,1)$; $S = \pm 1$ with probability $1/2$ each, independent of X.

Step — Density of X. For $x \ge 0$, $P(X \le x) = P(-x \le Z \le x) = 2\Phi(x) - 1$; differentiating,

$$\pi(x) = 2\phi(x) = \sqrt{\frac{2}{\pi}}\, e^{-x^2/2}, \qquad x \ge 0. \tag{23.1}$$

Step — Envelope. Dividing (23.1) by g:

$$\frac{\pi(x)}{g(x)} = \sqrt{\frac{2}{\pi}}\, e^{x - x^2/2}. \tag{23.2}$$

The exponent $x - x^2/2$ is maximized at $x = 1$ (derivative $1 - x$), where it equals $1/2$. Substituting $x = 1$ into (23.2):

$$c = \sqrt{\frac{2}{\pi}}\, e^{1/2} = \sqrt{\frac{2e}{\pi}} = 1.315489, \qquad \frac{1}{c} = 0.760173. \tag{23.3}$$

Dividing (23.2) by (23.3), the acceptance probability of a proposal x is

$$\frac{\pi(x)}{c\, g(x)} = e^{x - x^2/2 - 1/2} = e^{-(x-1)^2/2}. \tag{23.4}$$

Step — Why accepted draws follow π. Draw $x \sim g$ and accept with probability (23.4). Then

$$P(X \le x,\ \text{accept}) = \int_0^x g(t) \frac{\pi(t)}{c g(t)}\, dt = \frac{1}{c} \int_0^x \pi(t)\, dt,$$

and letting $x \to \infty$, $P(\text{accept}) = 1/c$. Dividing, $P(X \le x \mid \text{accept}) = \int_0^x \pi$: accepted values have density π, and the number of proposals per accepted value is geometric with mean c.

Step — Back to Z. Let $Z = SX$. For $z \ge 0$, $P(Z \le z) = \frac{1}{2} + \frac{1}{2} P(X \le z) = \frac{1}{2} + \frac{1}{2}(2\Phi(z) - 1) = \Phi(z)$; for $z < 0$, $P(Z \le z) = \frac{1}{2} P(X \ge -z) = \frac{1}{2}(2 - 2\Phi(-z)) = \Phi(z)$. So $Z \sim N(0, 1)$.

Code and output (programming part).

```
rng = np.random.default_rng(23); acc, tries = [], 0
while len(acc) < 10_000:
    x, u = rng.exponential(1.0), rng.uniform(); tries += 1
    if u <= np.exp(-(x - 1)**2 / 2): acc.append(x)       # (23.4)
X = np.array(acc); Z = np.where(rng.uniform(size=X.size) <= .5, X, -X)
10_000 / tries, X.mean()             # 0.762195 (1/c = 0.760173)  0.793801 (0.797885)
Z.mean(), Z.var(ddof=1)              # 0.019998  0.990055
kstest(X, halfnorm.cdf).pvalue, kstest(Z, norm.cdf).pvalue    # 0.4935  0.3449
```

Intuition. The exponential has heavier tails than the half-normal, so a scaled copy covers it everywhere; the price is that about one proposal in four is thrown away. Symmetry then turns a half-normal into a normal with one coin flip.

Ref: Morettin & Singer (2021), Apêndice B (método da aceitação-rejeição); Greenberg (2008), §5.3 (Algorithm 5.2 and the normal-from-Laplace example, pp. 66–69).

Question 24. Original math question — importance sampling for a rare Poisson tail.
Source: Original question (not from a book).

A hospital's ICU receives $X \sim \text{Poisson}(\lambda)$ emergency admissions per night; the manager needs $p = P(X \ge k)$ for a capacity k well above λ. Monte Carlo is used with G draws.

(a) For the crude estimator $\hat{p} = \frac{1}{G} \sum_g \mathbf{1}\{X^{(g)} \ge k\}$, $X^{(g)} \sim \text{Poisson}(\lambda)$, derive its mean, variance and relative standard error, and prove that for fixed G the relative error tends to infinity as $p \to 0$.

(b) Importance sampling with proposal Poisson(μ): derive the weight $w_\mu(x)$, prove unbiasedness, and derive the exact variance of one term in closed form, $\sigma^2(\mu) = e^{\mu - 2\lambda + \lambda^2/\mu}\, P(Y_{\lambda^2/\mu} \ge k) - p^2$ with $Y_\nu \sim \text{Poisson}(\nu)$.

(c) Prove that $\mu = \lambda$ gives back the crude variance, and that the proposal $f_\lambda(x)\mathbf{1}\{x \ge k\}/p$ has zero variance. Why can it not be used?

(d) Parameter analysis: for $\mu > \lambda$, prove that w_μ decreases in x, deduce $\sigma^2(\mu) \le p\,(w_\mu(k) - p)$, and show that this bound is minimized at $\mu = k$. Interpret.

Programming (added, not in the book). With $\lambda = 2$, $k = 10$, $G = 10^5$ and `np.random.default_rng(24)`: compute p exactly; run the crude estimator and the IS estimator with $\mu = 10$ (in that order, same generator) and print estimates with standard errors; tabulate

$\sigma^2(\mu)$ for $\mu \in \{2, 5, 10, 15\}$ from (b), check it by summing $f_\lambda(x)^2/f_\mu(x)$ over $x \geq k$, and find the μ that minimizes $\sigma^2(\mu)$ on a grid.

Solution.

Notation.

- $f_\nu(x) = e^{-\nu}\nu^x/x!$: Poisson p.m.f.; $Q_k(\nu) = P(Y_\nu \geq k)$; $p = Q_k(\lambda)$.
- $w_\mu(x) = f_\lambda(x)/f_\mu(x)$: importance weight; $\hat{p}_\mu = \frac{1}{G}\sum_g \mathbf{1}\{X^{(g)} \geq k\} w_\mu(X^{(g)})$, $X^{(g)} \sim f_\mu$.
- $\sigma^2(\mu)$: variance of one IS term; relative standard error $= \sqrt{\mathrm{Var}(\hat{p})}/p$.

Step — (a) Crude. The indicators are i.i.d. Bernoulli(p), so $E\hat{p} = p$ and $\mathrm{Var}\,\hat{p} = p(1-p)/G$. The relative standard error is

$$\frac{\sqrt{p(1-p)/G}}{p} = \sqrt{\frac{1-p}{p\,G}} \xrightarrow[p\to 0]{} \infty. \tag{24.1}$$

For a relative error of 10% one needs $G = (1-p)/(0.01\,p)$ draws, i.e. $G \approx 100/p$.

Step — (b) Weight and unbiasedness. Dividing the two p.m.f.s,

$$w_\mu(x) = \frac{e^{-\lambda}\lambda^x/x!}{e^{-\mu}\mu^x/x!} = e^{\mu-\lambda}\Big(\frac{\lambda}{\mu}\Big)^x. \tag{24.2}$$

Then $E_\mu[\mathbf{1}\{X \geq k\} w_\mu(X)] = \sum_{x\geq k} f_\mu(x)\frac{f_\lambda(x)}{f_\mu(x)} = \sum_{x\geq k} f_\lambda(x) = p$.

Step — (b) Second moment. $E_\mu[\mathbf{1}\{X \geq k\} w_\mu^2] = \sum_{x\geq k} f_\lambda(x)^2/f_\mu(x)$, and

$$\frac{f_\lambda(x)^2}{f_\mu(x)} = \frac{e^{-2\lambda}\lambda^{2x}/(x!)^2}{e^{-\mu}\mu^x/x!} = e^{\mu-2\lambda}\frac{(\lambda^2/\mu)^x}{x!} = e^{\mu-2\lambda+\lambda^2/\mu}\, f_{\lambda^2/\mu}(x).$$

Summing over $x \geq k$ and subtracting p^2:

$$\sigma^2(\mu) = e^{\mu-2\lambda+\lambda^2/\mu}\, Q_k(\lambda^2/\mu) - p^2. \tag{24.3}$$

Step — (c) Two special proposals. Substituting $\mu = \lambda$ into (24.3): $e^{\lambda-2\lambda+\lambda}Q_k(\lambda) - p^2 = p - p^2 = p(1-p)$, the crude variance. For $q^*(x) = f_\lambda(x)\mathbf{1}\{x \geq k\}/p$, every term is $\mathbf{1}\{x \geq k\} f_\lambda(x)/q^*(x) = p$, a constant: zero variance. But q^* requires p, the unknown being estimated; it only tells us to put the proposal mass where $\{x \geq k\}$ and in proportion to f_λ there.

Step — (d) A bound. For $\mu > \lambda$, $\lambda/\mu < 1$, so by (24.2) $w_\mu(x)$ decreases in x; on $\{x \geq k\}$, $w_\mu(x) \leq w_\mu(k)$. Hence

$$\sigma^2(\mu) = E_\mu[\mathbf{1}\{X \geq k\} w_\mu^2] - p^2 \leq w_\mu(k)\, E_\mu[\mathbf{1}\{X \geq k\} w_\mu] - p^2 = p(w_\mu(k) - p). \tag{24.4}$$

Step — (d) Best tilt. Taking logs of (24.2) at $x = k$, $\log w_\mu(k) = \mu - \lambda + k\log\lambda - k\log\mu$; differentiating in μ:

$$\frac{d\log w_\mu(k)}{d\mu} = 1 - \frac{k}{\mu},$$

negative for $\mu < k$, positive for $\mu > k$: the bound (24.4) is minimized at $\mu = k$. Reading: centre the proposal at the threshold, so that about half the draws land in the rare region, each weighted down by roughly $w_k(k)$. As $\mu \to \infty$ the weights explode (e^μ in (24.3)) and IS becomes worse than crude.

Step — Numbers. $\lambda = 2$, $k = 10$: $p = 4.649808 \times 10^{-5}$; crude relative error with $G = 10^5$ is $\sqrt{(1-p)/(pG)} = 0.4637$ by (24.1), and 10% needs $G = 2{,}150{,}527$ draws. With $\mu = 10$: $w_{10}(10) = e^8(0.2)^{10} = 3.0525 \times 10^{-4}$, bound (24.4) 1.2031×10^{-8}, exact (24.3) $\sigma^2(10) = 9.9339 \times 10^{-9}$, relative error 0.006778. The grid minimizer of (24.3) is $\mu = 10.035$, practically k. The variance ratio crude/IS is 4680.5.

Code and output (programming part).

```
lam, k, G = 2., 10, 10**5; p = poisson.sf(k - 1, lam)          # 4.649808e-05
var_is = lambda mu: (np.exp(mu - 2*lam + lam**2/mu) * poisson.sf(k - 1, lam**2/mu)
                     - p**2)                                    # (24.3)
# mu = 2, 5, 10, 15: 4.6496e-05  8.4535e-08  9.9339e-09  2.8585e-08
rng = np.random.default_rng(24)
xn = rng.poisson(lam, G); (xn >= k).mean()        # 4.0e-05 (4 hits), se 2.0e-05
xi = rng.poisson(10., G)
w = (xi >= k) * np.exp(10. - lam) * (lam / 10.)**xi
w.mean(), w.std(ddof=1) / np.sqrt(G)              # 4.6844e-05  3.165e-07
```

The crude run saw only 4 events in 10^5 nights; IS is within about one standard error of the exact p with a relative error below 1%.

Intuition. Rare-event probabilities are hard for plain simulation because almost no draw hits the event. Importance sampling simulates a world where the event is common and then corrects each hit by how much likelier it was made; choosing that world well is the whole art.

Ref: Greenberg (2008), §5.4 (importance sampling, variance of $\hat{g}$, choice of h); Morettin & Singer (2021), Apêndice B.

Grading rubric.

- (a) mean, variance and (24.1) with the limit: 3 pts.
- (b) (24.2) and unbiasedness: 2 pts; (24.3) derived: 3 pts.
- (c) both special cases and why q^* is unusable: 3 pts.
- (d) monotone weight, (24.4), minimization at $\mu = k$, reading: 4 pts.
- Programming: exact p, both estimators with standard errors, variance table: 3 pts.

Question 25. Original conceptual question — simulation methods (10 points).
Source: Original question (not from a book).

a) (3 points) In the accept–reject algorithm with $f \le c\,g$, the number of proposals needed to obtain one accepted draw is:

A. geometric with mean $1/c$, the acceptance probability.

B. Poisson with mean c, the envelope constant.

C. geometric with mean c, since each proposal is accepted with probability $1/c$.

D. binomial with parameters c and $1/c$, one trial per unit of envelope.

Key: C

Step — Why. C: each proposal is accepted independently with probability $1/c$ (Question 23), so the count until the first success is geometric with mean c. A swaps the probability and the mean; B and D are not distributions of a waiting time for a first success.

b) (3 points) The Monte Carlo estimator of $E[h(X)]$ from G i.i.d. draws has standard error:

A. proportional to $1/G$, because averaging divides by G.

B. $\mathrm{sd}(h(X))/\sqrt{G}$, whatever the dimension of X.

C. $\mathrm{sd}(h(X))/\sqrt{G}$ only when X is scalar and h is bounded.

D. growing exponentially with the dimension of X, like quadrature.

Key: B

Step — Why. B: $\mathrm{Var}(\bar{h}) = \mathrm{Var}(h)/G$ for i.i.d. draws with finite variance, in any dimension: this is why Monte Carlo beats grids in high dimension. A forgets the square root; C adds conditions that are not needed (finite variance suffices); D describes product-rule quadrature.

c) (2 points, T or F) An importance-sampling estimator is unbiased, yet a proposal with lighter tails than the target can give it infinite variance.

Key: T

Step — Why. Unbiasedness only needs $h > 0$ where $f > 0$; the second moment is $\int h^2 f^2/g$, which diverges when f/g grows too fast in the tails (Greenberg §5.4).

d) (2 points, T or F) Accept–reject cannot be used unless the normalizing constant of the target density is known.
Key: F

Step — Why. If $f = \kappa r$ with r known, choose c with $r \leq cg$ and accept with probability $r/(cg)$: accepted values still follow f (Greenberg p. 67). The constant only affects the acceptance rate. MH shares this property.

Programming (added, not in the book). Check (a) on the half-normal sampler of Question 23: with `np.random.default_rng(25)` draw 2×10^6 exponential proposals and uniforms, mark the accepted ones, take the first 100,000 acceptances, and compare the mean and variance of the number of proposals per acceptance with c and $c(c-1)$.

Code and output (programming part).

```
rng = np.random.default_rng(25); x = rng.exponential(1, 2_000_000)
acc = rng.uniform(size=x.size) <= np.exp(-(x - 1)**2 / 2)
waits = np.diff(np.r_[-1, np.flatnonzero(acc)[:100_000]])
waits.mean(), waits.var(ddof=1)     # 1.31879 (c = 1.315489)  0.42047 (c(c-1) = 0.41502)
```

Block 6 — The bootstrap *(I 5.2, 5.4; original)*

Question 26. I 5.2 — probability that an observation is in a bootstrap sample.
Source: James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, Chapter 5, §5.4 Exercises (Conceptual), Exercise 2, pp. 224–225 [PDF pp. 232–233]; the book gives no answer — reproduced verbatim (the margin index entry "`np.empty()`" is omitted).

2. We will now derive the probability that a given observation is part of a bootstrap sample. Suppose that we obtain a bootstrap sample from a set of n observations.

(a) What is the probability that the first bootstrap observation is *not* the jth observation from the original sample? Justify your answer.

(b) What is the probability that the second bootstrap observation is *not* the jth observation from the original sample?

(c) Argue that the probability that the jth observation is *not* in the bootstrap sample is $(1-1/n)^n$.

(d) When $n = 5$, what is the probability that the jth observation is in the bootstrap sample?

(e) When $n = 100$, what is the probability that the jth observation is in the bootstrap sample?

(f) When $n = 10{,}000$, what is the probability that the jth observation is in the bootstrap sample?

(g) Create a plot that displays, for each integer value of n from 1 to 100,000, the probability that the jth observation is in the bootstrap sample. Comment on what you observe.

(h) We will now investigate numerically the probability that a bootstrap sample of size $n = 100$ contains the jth observation. Here $j = 4$. We first create an array `store` with values that will subsequently be overwritten using the function `np.empty()`. We then repeatedly create bootstrap samples, and each time we record whether or not the fifth observation is contained in the bootstrap sample.

```
rng = np.random.default_rng(10)
store = np.empty(10000)
for i in range(10000):
    store[i] = np.sum(rng.choice(100, replace=True) == 4)
          > 0
np.mean(store)
```

Comment on the results obtained.

Programming (added, not in the book). Let D be the number of *distinct* original observations in a bootstrap sample of size $n = 100$. Derive $E(D)$ and $\mathrm{Var}(D)$ from indicator variables, then check them with 10,000 bootstrap samples drawn with `np.random.default_rng(26)`.

Solution.

Notation.

- n: sample size; $j \in \{1, \dots, n\}$: a fixed original observation.
- $Z_1, \dots, Z_n$: indices of the bootstrap draws, i.i.d. uniform on $\{1, \dots, n\}$ (sampling with replacement).
- $p_n = P(j \text{ is in the bootstrap sample})$; $A_j = \mathbf{1}\{j \text{ is absent}\}$; $D = n - \sum_j A_j$.

Step — (a). Each draw picks one of the n original observations with equal probability $1/n$, so

$$P(Z_1 \neq j) = 1 - \frac{1}{n}. \tag{26.1}$$

Step — (b). Sampling is with replacement, so Z_2 has the same distribution as Z_1 and is independent of it:

$$P(Z_2 \neq j) = 1 - \frac{1}{n}. \tag{26.2}$$

Step — (c). j is absent only if every draw misses it; by independence and (26.1)–(26.2) applied to all n draws:

$$P(j \notin \text{sample}) = \prod_{m=1}^{n} P(Z_m \neq j) = \Big(1 - \frac{1}{n}\Big)^n. \tag{26.3}$$

Step — (d)–(f). Taking the complement of (26.3):

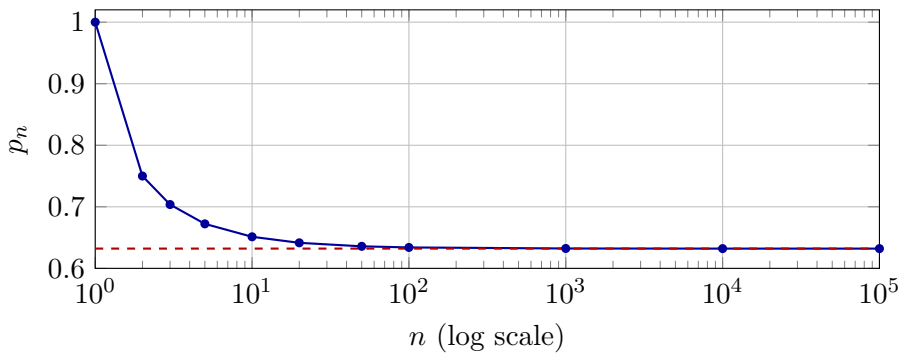
$$p_n = 1 - \Big(1 - \frac{1}{n}\Big)^n. \tag{26.4}$$

Substituting into (26.4):

$$\begin{aligned} n = 5: &\quad p_5 = 1 - (4/5)^5 = 1 - 0.32768 = 0.67232, \\ n = 100: &\quad p_{100} = 1 - (0.99)^{100} = 1 - 0.366032 = 0.633968, \\ n = 10{,}000: &\quad p_{10000} = 1 - (0.9999)^{10000} = 1 - 0.367861 = 0.632139. \end{aligned}$$

Step — (g) Plot and limit. Since $n \log(1 - 1/n) = n\big(-\frac{1}{n} - \frac{1}{2n^2} - \dots\big) \to -1$,

$$\lim_{n \to \infty} p_n = 1 - e^{-1} = 0.632121. \tag{26.5}$$



p_n falls from 1 at $n = 1$ and flattens almost immediately at $1 - e^{-1} \approx 0.632$ (dashed): for $n \geq 100$ the curve is indistinguishable from the limit.

Step — (h) The printed code. `rng.choice(100, replace=True)` without `size` returns a *single* integer, so each iteration checks whether one draw equals 4, an event of probability $1/100$. Run as

printed, `np.mean(store)` returns 0.0089, not a bootstrap probability. The intended code draws a whole bootstrap sample, `rng.choice(100, size=100, replace=True)`, and then returns 0.6362, close to $p_{100} = 0.633968$ from (e) (Monte Carlo standard error $\sqrt{0.634 \cdot 0.366/10000} \approx 0.005$). Index 4 is the fifth observation because Python counts from 0, which is why the text says "fifth".

Step — Added: distinct observations. $E(A_j) = a = (1 - 1/n)^n$ by (26.3); for $i \neq j$ both are absent iff every draw avoids two indices, $E(A_i A_j) = b = (1 - 2/n)^n$. Hence

$$E(D) = n(1-a), \qquad \mathrm{Var}(D) = \mathrm{Var}\Big(\sum_j A_j\Big) = na(1-a) + n(n-1)(b-a^2). \tag{26.6}$$

For $n = 100$: $E(D) = 63.396766$ and $\mathrm{sd}(D) = 3.120912$.

Code and output (programming part).

```
rng = np.random.default_rng(10); store = np.empty(10000)
for i in range(10000):
    store[i] = np.sum(rng.choice(100, size=100, replace=True) == 4) > 0
store.mean()                                    # 0.6362 (printed code: 0.0089)
rng = np.random.default_rng(26)
D = [np.unique(rng.choice(100, size=100, replace=True)).size for _ in range(10000)]
np.mean(D), np.std(D, ddof=1)                   # 63.3994  3.1583  (exact 63.3968, 3.1209)
```

Intuition. A bootstrap sample of size n contains only about 63.2% of the distinct original observations; the other 36.8% are "out of bag". This is why bootstrap estimates of test error that reuse the training points are optimistic, and why out-of-bag points can serve as a free validation set.

Ref: ISLP (2023), §5.2 (the bootstrap, pp. 212–215).

Question 27. I 5.4 — bootstrap standard deviation of a prediction.

Source: James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, Chapter 5, §5.4 Exercises (Conceptual), Exercise 4, p. 225 [PDF p. 233]; the book gives no answer — reproduced verbatim.

4. Suppose that we use some statistical learning method to make a prediction for the response Y for a particular value of the predictor X. Carefully describe how we might estimate the standard deviation of our prediction.

Programming (added, not in the book). Fit the Poisson regression $\log \mu_t = \beta_0 + \beta_1 x_t$, $x_t = (t - 56.5)/10$, to the coal series ($t = 1, \ldots, 112$ for 1851–1962) with `statsmodels` GLM. Predict μ for 1900 ($t = 50$). Compute the delta-method standard error of $\hat{\mu}(1900)$ and the bootstrap standard deviation from $B = 2{,}000$ pairs-bootstrap refits (`np.random.default_rng(27)`), with the 95% percentile interval. Compare the two using the Pearson dispersion statistic.

Solution.

Notation.

- (x_i, y_i), $i = 1, \ldots, n$: training data; $\hat{f}$: fitted method; x_0: the predictor value; $\hat{y}_0 = \hat{f}(x_0)$.
- B: number of bootstrap samples; $\hat{y}_0^{*b}$: prediction refitted on bootstrap sample b.
- Coal model: $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)$, $\hat{V}$: its estimated covariance; $\hat{\mu}_0 = \exp(x_0' \hat{\beta})$.

Step — The procedure. (1) Draw n indices with replacement from $\{1, \ldots, n\}$ and form the bootstrap sample $\{(x_{i^*}, y_{i^*})\}$. (2) Refit the *whole* method on it (including any tuning or variable selection) and compute $\hat{y}_0^{*b} = \hat{f}^{*b}(x_0)$. (3) Repeat for $b = 1, \ldots, B$. (4) Estimate the standard deviation of the prediction by

$$\widehat{\mathrm{SD}}(\hat{y}_0) = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} \big(\hat{y}_0^{*b} - \bar{y}_0^{*}\big)^2}, \qquad \bar{y}_0^{*} = \frac{1}{B} \sum_b \hat{y}_0^{*b}. \tag{27.1}$$

Step — What it measures. (27.1) is the sampling variability of the *fitted value* $\hat{f}(x_0)$, the analogue of $\mathrm{SE}(\hat{\alpha})$ in ISLP §5.2. For the error of predicting a new response $Y_0 = f(x_0) + \varepsilon$ one must add the irreducible noise: $\mathrm{SD}(Y_0 - \hat{y}_0) \approx \sqrt{\widehat{\mathrm{SD}}(\hat{y}_0)^2 + \widehat{\mathrm{Var}}(\varepsilon \mid x_0)}$. Resampling pairs keeps the joint behaviour of (X, Y) and does not assume the model is correct; resampling residuals instead assumes a correct mean and homoskedastic errors.

Step — Delta method (benchmark). $\hat{\mu}_0 = g(\hat{\beta})$ with $g(\beta) = \exp(x_0'\beta)$ and gradient $\nabla g = \hat{\mu}_0 x_0$, so

$$\mathrm{SE}(\hat{\mu}_0) \approx \hat{\mu}_0 \sqrt{x_0' \hat{V} x_0}. \tag{27.2}$$

Step — Coal numbers. $\hat{\beta}_0 = 0.363109$ (s.e. 0.084922), $\hat{\beta}_1 = -0.183776$ (s.e. 0.024728) per decade; $\hat{\mu}(1900) = 1.620223$; (27.2) gives 0.125930; the bootstrap (27.1) gives 0.134530 with percentile interval [1.3633, 1.8748]. The Pearson dispersion is $\hat{\phi} = 1.133236$: the counts are slightly more variable than a Poisson with a linear trend allows, and $0.125930\sqrt{1.133236} = 0.134057$, almost exactly the bootstrap value. For a new count in 1900 the predictive SD is about $\sqrt{1.620223 + 0.134530^2} = 1.2800$, dominated by Poisson noise.

Code and output (programming part).

```
t = np.arange(1, 113); X = sm.add_constant((t - 56.5) / 10)
fit = sm.GLM(y, X, family=sm.families.Poisson()).fit()
x0 = np.array([1., (50 - 56.5) / 10])
mu0 = np.exp(x0 @ fit.params)                                  # 1.620223
se = mu0 * np.sqrt(x0 @ fit.cov_params() @ x0)                 # 0.125930
rng = np.random.default_rng(27); pred = np.empty(2000)
for b in range(2000):
    i = rng.integers(0, 112, 112)
    pred[b] = np.exp(x0 @ sm.GLM(y[i], X[i], family=sm.families.Poisson()).fit().params)
pred.std(ddof=1), np.quantile(pred, [.025, .975])              # 0.134530 [1.3633 1.8748]
```

Intuition. The bootstrap answers "how much would my prediction move if I had drawn another sample of the same size?" by drawing those samples from the data itself. It agrees with the model-based formula when the model is right, and corrects it (here for mild overdispersion) when it is not.

Ref: ISLP (2023), §5.2 (pp. 212–215) and the §5.3.3 lab (`boot_SE`); Wooldridge (2016), §17.3 (Poisson regression and overdispersion).

Question 28. Original math question — nonparametric versus parametric bootstrap for a Poisson mean.

Source: Original question (not from a book).

Let $x_1, \ldots, x_n$ be counts, $\bar{x}$ their mean and $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$.

(a) Nonparametric bootstrap (X_i^* i.i.d. from the empirical distribution): find $E^*(\overline{X}^*)$ and $\mathrm{Var}^*(\overline{X}^*)$ exactly, without simulation.

(b) Parametric bootstrap (X_i^* i.i.d. Poisson($\bar{x}$)): find $\mathrm{Var}^*(\overline{X}^*)$. If the X_i are i.i.d. with mean θ and variance σ^2, prove $E(\hat{\sigma}^2) = \frac{n-1}{n}\sigma^2$ and compare the expectations of the two bootstrap variances with the true $\mathrm{Var}(\overline{X})$, both when the Poisson model holds and when $\sigma^2 = \phi\theta$ with $\phi \neq 1$.

(c) For $\psi = e^{-\theta}$ and $\hat{\psi} = e^{-\bar{x}}$, find the parametric-bootstrap bias estimate $\hat{b} = E^*(e^{-\overline{X}^*}) - e^{-\bar{x}}$ in closed form and the bias-corrected estimator.

(d) Parameter analysis: prove $\hat{b} > 0$, give its leading term in $1/n$, and say how it moves with $\bar{x}$. Interpret.

Programming (added, not in the book). For the coal counts 1851–1891 ($n = 41$): print $\bar{x}$, $\hat{\sigma}^2$, the two exact bootstrap variances, $\hat{\sigma}^2/\bar{x}$, $\hat{\psi}$, $\hat{b}$, its leading-term approximation and the corrected estimate. Check them with $B = 20{,}000$ resamples of each kind (`np.random.default_rng(28)`).

Solution.

Notation.

- E^*, Var^*: moments under the bootstrap distribution, data held fixed.
- $\theta = E(X)$, $\sigma^2 = \mathrm{Var}(X)$, $\phi = \sigma^2/\theta$: dispersion ($\phi = 1$ for Poisson).
- $\psi = e^{-\theta}$, $\hat{\psi} = e^{-\bar{x}}$, $\hat{b}$: bootstrap bias estimate, $\tilde{\psi} = \hat{\psi} - \hat{b}$.

Step — (a) One draw. X^* takes each value x_i with probability $1/n$, so

$$E^*(X^*) = \frac{1}{n}\sum_i x_i = \bar{x}, \qquad \mathrm{Var}^*(X^*) = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2. \tag{28.1}$$

The n draws are i.i.d., so by (28.1)

$$E^*(\overline{X}^*) = \bar{x}, \qquad \mathrm{Var}^*(\overline{X}^*) = \frac{\hat{\sigma}^2}{n}. \tag{28.2}$$

Step — (b) Parametric. $\mathrm{Var}^*(X^*) = \bar{x}$ for a Poisson($\bar{x}$), so

$$\mathrm{Var}^*(\overline{X}^*) = \frac{\bar{x}}{n}. \tag{28.3}$$

Step — (b) Expectation of $\hat{\sigma}^2$. Using $\sum_i (X_i - \overline{X})^2 = \sum_i X_i^2 - n\overline{X}^2$ and $E(X^2) = \sigma^2 + \theta^2$, $E(\overline{X}^2) = \sigma^2/n + \theta^2$:

$$E\Big[\sum_i (X_i - \overline{X})^2\Big] = n(\sigma^2 + \theta^2) - n\Big(\frac{\sigma^2}{n} + \theta^2\Big) = (n-1)\sigma^2.$$

Dividing by n:

$$E(\hat{\sigma}^2) = \frac{n-1}{n}\sigma^2. \tag{28.4}$$

Step — (b) Comparison. The target is $\mathrm{Var}(\overline{X}) = \sigma^2/n = \phi\theta/n$. By (28.2) and (28.4), $E[\hat{\sigma}^2/n] = \frac{n-1}{n}\cdot\frac{\sigma^2}{n}$: biased down by the factor $(n-1)/n$, whatever ϕ. By (28.3), $E[\bar{x}/n] = \theta/n = \mathrm{Var}(\overline{X})/\phi$: exact when the Poisson model holds ($\phi = 1$), too small by $1/\phi$ under overdispersion ($\phi > 1$) and too large under underdispersion ($\phi < 1$). The nonparametric bootstrap is robust to the variance model; the parametric one is more precise when the model is right.

Step — (c) Bias estimate. Under the parametric bootstrap $n\overline{X}^* \sim \mathrm{Poisson}(n\bar{x})$, and $e^{-\overline{X}^*} = (e^{-1/n})^{n\overline{X}^*}$. Substituting $s = e^{-1/n}$ into the generating function (4.1) with $n\theta$ replaced by $n\bar{x}$:

$$E^*(e^{-\overline{X}^*}) = \exp\{n\bar{x}(e^{-1/n} - 1)\}, \qquad \hat{b} = \exp\{n\bar{x}(e^{-1/n} - 1)\} - e^{-\bar{x}}, \tag{28.5}$$

and the corrected estimator is

$$\tilde{\psi} = \hat{\psi} - \hat{b} = 2e^{-\bar{x}} - \exp\{n\bar{x}(e^{-1/n} - 1)\}. \tag{28.6}$$

Step — (d) Sign and order. By (4.3), $n\bar{x}(e^{-1/n} - 1) > -\bar{x}$ (for $\bar{x} > 0$), so $\hat{b} > 0$: the bootstrap detects the upward Jensen bias of Question 4. Expanding, $n(e^{-1/n} - 1) = -1 + \frac{1}{2n} - \frac{1}{6n^2} + \ldots$, so the exponent in (28.5) is $-\bar{x} + \frac{\bar{x}}{2n} + O(n^{-2})$ and

$$\hat{b} = e^{-\bar{x}}\Big(e^{\bar{x}/(2n) + O(n^{-2})} - 1\Big) = \frac{\bar{x}\, e^{-\bar{x}}}{2n} + O(n^{-2}). \tag{28.7}$$

The leading term vanishes at rate $1/n$; as a function of $\bar{x}$ it is proportional to $\bar{x}e^{-\bar{x}}$, whose derivative $(1 - \bar{x})e^{-\bar{x}}$ is positive for $\bar{x} < 1$ and negative for $\bar{x} > 1$: the bias is largest, relative to n, when the mean is about one event.

Step — (d) Reading. The bootstrap estimates the bias of a nonlinear plug-in estimator by repeating the plug-in inside the fitted model; here it reproduces the exact Jensen gap to first order. The correction is worth making only when $\hat{b}$ is large relative to the standard error.

Step — Coal numbers. $n = 41$, $\bar{x} = 3.097561$, $\hat{\sigma}^2 = 2.478287$ ($s^2 = 2.540244$): (28.2) $= 2.478287/41 = 0.060446$; (28.3) $= 3.097561/41 = 0.075550$; $\hat{\sigma}^2/\bar{x} = 0.800077$: these years are *under*dispersed, so the Poisson bootstrap is the larger one. $\hat{\psi} = e^{-3.097561} = 0.045159$; (28.5): $\exp\{127(e^{-1/41} - 1)\} = 0.046883$, so $\hat{b} = 0.046883 - 0.045159 = 0.001724$; (28.7): $3.097561 \cdot 0.045159/82 = 0.001706$; (28.6): $\tilde{\psi} = 0.043435$.

Code and output (programming part).

```
x = y[:41]; n, xb, s2h = 41, x.mean(), x.var(ddof=0)
s2h / n, xb / n, s2h / xb              # 0.060446  0.075550  0.800077
Estar = np.exp(n * xb * (np.exp(-1 / n) - 1))
Estar - np.exp(-xb), 2 * np.exp(-xb) - Estar        # 0.001724  0.043435
rng = np.random.default_rng(28)
xnp = rng.choice(x, size=(20_000, n), replace=True).mean(1)
xpar = rng.poisson(xb, size=(20_000, n)).mean(1)
xnp.var(ddof=1), xpar.var(ddof=1)                   # 0.060686  0.076385
np.exp(-xpar).mean() - np.exp(-xb)                  # 0.001655
```

Intuition. The nonparametric bootstrap uses the data's own spread, the parametric one the spread the model implies. When they disagree, the data are telling you the variance model is off.

Ref: ISLP (2023), §5.2; Greenberg (2008), §5.6 (using simulated output); Question 4 for the exact bias of $\hat{\psi}$.

Grading rubric.

- (a) (28.1)–(28.2): 3 pts.
- (b) (28.3): 1 pt; (28.4) proved: 2 pts; comparison with ϕ: 3 pts.
- (c) (28.5)–(28.6): 3 pts.
- (d) sign, (28.7), dependence on $\bar{x}$, reading: 3 pts.
- Programming: exact values and simulation check: 3 pts.

Question 29. Original conceptual question — the bootstrap (10 points).
Source: Original question (not from a book).

a) (3 points) For large n, the fraction of the original observations that appear at least once in a bootstrap sample of size n is about:

A. 0.5, since each observation is drawn or not with equal chance.

B. $e^{-1} \approx 0.368$, the probability that an observation is never drawn.

C. 1, because the bootstrap sample has the same size as the data.

D. $1 - e^{-1} \approx 0.632$, the complement of never being drawn.

Key: D

Step — Why. D: (26.5). B is the out-of-bag fraction; A has no derivation; C ignores repetitions (Question 26 shows $E(D) = 63.4$ for $n = 100$).

b) (3 points) As $B \to \infty$, the nonparametric bootstrap standard error of $\bar{x}$ equals:

A. $\hat{\sigma}/\sqrt{n}$, with $\hat{\sigma}^2$ computed with divisor n.

B. $s/\sqrt{n}$, with s^2 computed with divisor $n - 1$.

C. $\sqrt{\bar{x}/n}$, the Poisson standard error.

D. 0, because every bootstrap sample has mean $\bar{x}$.

Key: A

Step — Why. A: (28.2). B is the usual unbiased-variance formula, not what resampling gives; C is the parametric bootstrap (28.3); D confuses $E^*(\overline{X}^*) = \bar{x}$ with each resample's mean.

c) (2 points, T or F) For a fixed sample, the bootstrap standard error converges to the true standard error of the estimator as the number of bootstrap replications B grows.

Key: F

Step — Why. As $B \to \infty$ it converges to the *ideal* bootstrap value computed from the empirical distribution (e.g. $\hat{\sigma}/\sqrt{n}$), which still differs from the true value; only $n \to \infty$ removes that gap.

d) (2 points, T or F) With overdispersed counts, a parametric Poisson bootstrap understates the standard error of $\bar{x}$.

Key: T

Step — Why. By (b) of Question 28, $E[\bar{x}/n] = \mathrm{Var}(\overline{X})/\phi < \mathrm{Var}(\overline{X})$ when $\phi > 1$.

Programming (added, not in the book). Illustrate (b)–(c) on the coal counts 1851–1891: with `np.random.default_rng(29)` compute the bootstrap standard error of $\bar{x}$ with $B = 100$, 1,000 and 10,000 resamples and compare them with the $B \to \infty$ value $\hat{\sigma}/\sqrt{n}$.

Code and output (programming part).

```
rng = np.random.default_rng(29); x = y[:41]
for B in (100, 1000, 10000):
    print(B, rng.choice(x, size=(B, 41), replace=True).mean(1).std(ddof=1))
# 100 0.259285   1000 0.242725   10000 0.244059   ; sigma_hat/sqrt(n) = 0.245858
```

Increasing B only removes the resampling noise around 0.245858; no B can tell how far that value is from the true standard error.

Block 7 — MCMC: invariance, Metropolis–Hastings, accept–reject versus MH *(G 6.8, 7.4, 7.6; original)*

Question 30. G 6.8 — an invariant density for a continuous-state kernel.

Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 6, §6.5 Exercises, Exercise 6.8, p. 89 [PDF p. 104]; the book gives no answer — reproduced verbatim.

6.8 (Bhattacharya and Waymire, 1990, p. 216) Show that

$$\pi(y) = 2(1-y), \qquad 0 \le y \le 1$$

is an invariant density for the transition kernel

$$p(x,y) = \begin{cases} e^{1-x}, & \text{if } y < x, \\ e^{1-x} - e^{y-x}, & \text{if } y > x. \end{cases}$$

Programming (added, not in the book). (1) Check by quadrature that $\int_0^1 p(x,y)\,dy = 1$ at $x = 0.3, 0.7$ and that $\int_0^1 \pi(x)p(x,y)\,dx = \pi(y)$ at $y = 0.1, 0.5, 0.9$. (2) Derive the c.d.f. of $p(x,\cdot)$, simulate the chain by inversion (`brentq` on the branch $y > x$) from $x_0 = 0.5$ with `np.random.default_rng(30)`, discard 1,000 draws, keep 20,000, and compare the sample mean and $P(Y < 0.5)$ with their values under π; report the batch-means n.s.e. and the lag-1 autocorrelation.

Solution.

Notation.

- State space $[0,1]$; $p(x,\cdot)$: density of the next state given the current state x.
- π is invariant if $\int_0^1 \pi(x)p(x,y)\,dx = \pi(y)$ for all y (Greenberg §6.3).
- $F_x(y) = \int_0^y p(x,t)\,dt$: c.d.f. of the next state.

Step — $p(x,\cdot)$ is a density. For $y > x$, $e^{1-x} - e^{y-x} = e^{-x}(e - e^{y}) > 0$ since $y < 1$. Integrating over the two branches:

$$\int_0^1 p(x,y)\,dy = \int_0^x e^{1-x}dy + \int_x^1 (e^{1-x} - e^{y-x})dy = xe^{1-x} + (1-x)e^{1-x} - e^{-x}(e - e^{x}) = 1. \quad (30.1)$$

Step — Split the invariance integral. For fixed y, the points $x < y$ use the branch "$y > x$" and the points $x > y$ use "$y < x$":

$$\int_0^1 \pi(x)p(x,y)\,dx = \int_0^y 2(1-x)(e^{1-x} - e^{y-x})dx + \int_y^1 2(1-x)e^{1-x}dx. \quad (30.2)$$

Collecting the two e^{1-x} terms into one integral over $[0,1]$:

$$\int_0^1 \pi(x)p(x,y)\,dx = \int_0^1 2(1-x)e^{1-x}dx - e^{y}\int_0^y 2(1-x)e^{-x}dx. \quad (30.3)$$

Step — First integral. With $u = 1 - x$ and $\frac{d}{du}[(u-1)e^{u}] = ue^{u}$:

$$\int_0^1 2(1-x)e^{1-x}dx = \int_0^1 2ue^{u}\,du = 2\big[(u-1)e^{u}\big]_0^1 = 2\,(0 - (-1)) = 2.$$

Step — Second integral. Since $\frac{d}{dx}[xe^{-x}] = (1-x)e^{-x}$,

$$\int_0^y 2(1-x)e^{-x}dx = 2\big[xe^{-x}\big]_0^y = 2ye^{-y}.$$

Step — Conclusion. Substituting both results into (30.3):

$$\int_0^1 \pi(x)p(x,y)\,dx = (2) - e^{y}(2ye^{-y}) = 2 - 2y = \pi(y). \qquad \blacksquare$$

Step — Simulating the kernel. From (30.1)'s pieces, the c.d.f. of the next state is

$$F_x(y) = \begin{cases} ye^{1-x}, & y \le x, \\ ye^{1-x} - e^{y-x} + 1, & y > x, \end{cases} \quad (30.4)$$

continuous at $y = x$ and equal to 1 at $y = 1$. Given $u \sim U(0,1)$: if $u \le xe^{1-x}$, $y = ue^{x-1}$; otherwise solve $F_x(y) = u$ on $(x,1)$. Under π: $E(Y) = \int_0^1 2y(1-y)dy = 1/3$ and $P(Y < 0.5) = 2(0.5) - 0.5^2 = 0.75$.

Code and output (programming part).

```
def draw(x, u):
    if u <= x * np.exp(1 - x): return u * np.exp(x - 1)
    return brentq(lambda y: y*np.exp(1 - x) - np.exp(y - x) + 1 - u, x, 1.)  # (30.4)
rng = np.random.default_rng(30); x = 0.5; ch = []
for g in range(21_000):
    x = draw(x, rng.uniform())
    if g >= 1000: ch.append(x)
ch = np.array(ch); ch.mean(), np.mean(ch < .5)      # 0.334981 (1/3)  0.74745 (0.75)
ch.reshape(50, 400).mean(1).std(ddof=1) / np.sqrt(50)   # n.s.e. 0.001868
np.corrcoef(ch[:-1], ch[1:])[0, 1]                  # lag-1 autocorrelation 0.0768
```

Quadrature confirms (30.1) and the invariance identity to 10^{-9}; the simulated mean is within one n.s.e. of 1/3.

Intuition. Invariance means that if today's state is drawn from π, tomorrow's is too: the kernel moves mass around but leaves its profile unchanged. MCMC is the art of building kernels with a prescribed invariant law; this exercise checks one by direct integration.

Ref: Greenberg (2008), §6.3 (continuous state spaces, invariant distributions, pp. 84–87).

Question 31. G 7.4 — the acceptance probability of Algorithm 7.5.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 7, §7.5 Exercises, Exercise 7.4, p. 106 [PDF p. 121]; the book gives no answer — reproduced verbatim. Algorithm 7.5 and Equation (7.6), to which it refers, are reproduced from §7.2, pp. 98–99 [PDF pp. 113–114].

Referred to (Greenberg 2008, p. 98):

$$\alpha(x,y) = \begin{cases} \min\left\{\dfrac{f(y)q(y,x)}{f(x)q(x,y)}, 1\right\}, & \text{if } f(x)q(x,y) \neq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{7.6}$$

Referred to (Greenberg 2008, p. 99): "**Algorithm 7.5: MH for Beta(3, 4) with U(0, 1) proposal**

7.1 Set $x^{(0)}$ equal to a number between zero and one.

7.2 At the gth iteration, generate U_1 and U_2 from $U(0,1)$.

7.3 If

$$U_1 \leq \alpha(x^{(g-1)}, U_2) = \frac{U_2^2(1-U_2)^3}{(x^{(g-1)})^2(1-x^{(g-1)})^3},$$

set $x^{(g)} = U_2$. Otherwise set $x^{(g)} = x^{(g-1)}$.

7.4 Go to 2 and continue until the desired number of iterations is obtained."

7.4 Derive $\alpha(x,y)$ for Algorithm 7.5.

Programming (added, not in the book). Run Algorithm 7.5 from $x^{(0)} = 0.5$ with `np.random.default_rng(31)`, discard 500 iterations and keep 5,000 (Greenberg reports an acceptance rate of 0.57 and a mean of 0.4296). Report your acceptance rate and mean against the true mean 3/7, and compute the exact long-run acceptance probability $\iint f(x)\min\{f(y)/f(x), 1\}\,dx\,dy$ by two-dimensional quadrature.

Solution.

Notation.

- Target $f(x) = x^2(1-x)^3/B(3,4)$ on $(0,1)$ (Beta$(3,4)$); proposal $q(x,y) = 1$ for $y \in (0,1)$ (independence, uniform).
- $\alpha(x,y)$: probability of moving from x to the proposal y.

Step — Apply (7.6). Both $x, y \in (0,1)$, so $f(x)q(x,y) \neq 0$. Substituting f and $q \equiv 1$ into (7.6):

$$\alpha(x,y) = \min\left\{\frac{(y^2(1-y)^3/B(3,4))\cdot(1)}{(x^2(1-x)^3/B(3,4))\cdot(1)}, 1\right\}.$$

The constant $B(3,4)$ cancels:

$$\alpha(x,y) = \min\left\{\frac{y^2(1-y)^3}{x^2(1-x)^3}, 1\right\}. \tag{31.1}$$

Step — Why the algorithm can omit the min. For $U_1 \sim U(0,1)$ and any $r \geq 0$, $P(U_1 \leq \min(r,1)) = \min(r,1) = P(U_1 \leq r)$. So step 7.3, which compares U_1 with the raw ratio, accepts with exactly the probability (31.1).

Step — Detailed balance. For $x \neq y$, by (31.1), $f(x)q(x,y)\alpha(x,y) = \min\{f(y), f(x)\} = f(y)q(y,x)\alpha(y,x)$; this symmetry is the reversibility condition from which (7.6) was derived, so Beta$(3,4)$ is invariant (Question 33(b) proves the general step).

Code and output (programming part).

```
fb = lambda v: v**2 * (1 - v)**3
rng = np.random.default_rng(31); x, acc, out = 0.5, 0, []
for g in range(5500):
    U1, U2 = rng.uniform(), rng.uniform()
    if U1 <= fb(U2) / fb(x):                  # (31.1), min not needed
        x = U2; acc += g >= 500
    if g >= 500: out.append(x)
acc / 5000, np.mean(out)                      # 0.5682  0.432394  (true 3/7 = 0.428571)
dblquad(lambda y, x: beta(3, 4).pdf(x) * min(1, fb(y) / fb(x)),
        0, 1, 0, 1)[0]                        # 0.572589
```

The exact long-run acceptance probability 0.5726 matches Greenberg's reported 0.57 and our 0.5682.

Intuition. With an independence proposal the chain compares how much more likely the target makes the new point than the current one; it always moves uphill and sometimes downhill, and the normalizing constant never matters.

Ref: Greenberg (2008), §7.2 (pp. 97–100).

Question 32. G 7.6 — accept–reject versus Metropolis–Hastings for a beta mean.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, Chapter 7, §7.5 Exercises, Exercise 7.6, p. 106 [PDF p. 121]; the book gives no answer — reproduced verbatim. Algorithm 5.2 and the condition on c, to which it refers, are reproduced from §5.3, p. 66 [PDF p. 81].

Referred to (Greenberg 2008, p. 66): "The accept–reject (AR) algorithm can be used to simulate values from a density function $f(\cdot)$ if it is possible to simulate values from a density $g(\cdot)$ and if a number c can be found such that $f(Y) \leq cg(Y)$, $c \geq 1$, for all Y in the support of $f(\cdot)$. [...]
Algorithm 5.2: Accept–reject method

5.1 Generate a value y from $g(\cdot)$.

5.2 Draw a value u from $U(0, 1)$.

5.3 Return y as a draw from $f(\cdot)$ if $u \leq f(y)/cg(y)$. If not, reject it and return to step 1. (The effect of this step is to accept y with probability $f(y)/cg(y)$.)"

7.6 Estimate the mean of a Beta(3.7, 4.8) distribution with (1) an AR algorithm and a Beta(4, 5) proposal density (you will need to determine the value of c needed in Algorithm 5.2); (2) an MH algorithm with a Beta(4, 5) proposal density. After the break-in sample, graph the values of the mean against the iteration number to monitor convergence. Compare your answers to the true value.

Programming (added, not in the book). With `np.random.default_rng(32)`: (1) evaluate f/g near 0 and at 0.5 for the Beta$(4, 5)$ proposal; (2) run AR with a proposal for which a finite c exists (justify it) until 10,000 draws are accepted; (3) run the independence MH with Beta$(4, 5)$ (1,000 burn-in, 10,000 kept), report the acceptance rate, the running mean at 1,000, 2,500, 5,000 and 10,000 kept iterations, and the batch-means n.s.e.; plot the running mean.

Solution.

Notation.

- f: Beta$(3.7, 4.8)$ density, mean $3.7/(3.7 + 4.8) = 0.435294$; $g_{a,b}$: Beta(a, b) density; $B(\cdot, \cdot)$: beta function.
- c: AR envelope constant; $w(x) = f(x)/g_{4,5}(x)$: MH importance ratio.

Step — (1) No finite c for Beta(4,5). Dividing the densities,

$$\frac{f(x)}{g_{4,5}(x)} = \frac{x^{2.7}(1-x)^{3.8}/B(3.7, 4.8)}{x^3(1-x)^4/B(4, 5)} = \frac{B(4, 5)}{B(3.7, 4.8)}\, x^{-0.3}(1-x)^{-0.2}, \tag{32.1}$$

which tends to ∞ as $x \to 0$ or $x \to 1$. No c satisfies $f \leq cg_{4,5}$ on $(0, 1)$: *the AR algorithm the exercise asks for does not exist.* The proposal has thinner tails at both ends than the target (x^3 versus $x^{2.7}$ near 0). Numerically the ratio is 0.9528 at $x = 0.5$ but 169.2 at $x = 10^{-8}$.

Step — (1) A valid AR proposal (added). A Beta(a, b) proposal works iff $a \le 3.7$ and $b \le 4.8$. With Beta$(3, 4)$, $f/g_{3,4} \propto x^{0.7}(1-x)^{0.8}$; its log-derivative $0.7/x - 0.8/(1-x)$ vanishes at

$$x^* = \frac{0.7}{0.7+0.8} = 0.466667, \qquad c = \frac{f(x^*)}{g_{3,4}(x^*)} = 1.115321, \qquad \frac{1}{c} = 0.896603. \tag{32.2}$$

Step — (2) MH with Beta(4,5). An independence kernel only needs $g_{4,5} > 0$ wherever $f > 0$, which holds. By (7.6) (Question 31) with $q(x, y) = g_{4,5}(y)$,

$$\alpha(x, y) = \min\Big\{\frac{f(y)\, g_{4,5}(x)}{f(x)\, g_{4,5}(y)}, 1\Big\} = \min\Big\{\frac{w(y)}{w(x)}, 1\Big\}. \tag{32.3}$$

Because w is unbounded the chain is not uniformly ergodic: after landing near 0 or 1 (large w) it can stay put for a while. It still converges to f, but convergence should be monitored, which is the point of the running-mean graph.

Step — Results (programming). AR with Beta$(3, 4)$: acceptance 0.894775 (theory 0.896603), mean 0.434564 (s.e. 0.001621). MH with Beta$(4, 5)$: acceptance 0.9645; running mean after 1,000, 2,500, 5,000 and 10,000 kept iterations: 0.4284, 0.4333, 0.4375, 0.437056; n.s.e. 0.001744. Both are within about one standard error of 0.435294.

Code and output (programming part).

```
f, g34, g45 = beta(3.7, 4.8), beta(3, 4), beta(4, 5)
f.pdf(1e-8) / g45.pdf(1e-8), f.pdf(.5) / g45.pdf(.5)     # 169.2  0.9528
c = f.pdf(0.7 / 1.5) / g34.pdf(0.7 / 1.5)                # 1.115321
rng = np.random.default_rng(32); dr, tries = [], 0
while len(dr) < 10_000:
    y, u = rng.beta(3, 4), rng.uniform(); tries += 1
    if u <= f.pdf(y) / (c * g34.pdf(y)): dr.append(y)
np.mean(dr), 10_000 / tries                              # 0.434564  0.894775
w = lambda v: f.pdf(v) / g45.pdf(v); x, acc, out = .5, 0, []
for g in range(11_000):
    y = rng.beta(4, 5)
    if rng.uniform() <= min(1, w(y) / w(x)): x = y; acc += g >= 1000
    if g >= 1000: out.append(x)
run = np.cumsum(out) / np.arange(1, 10_001)              # plot run against iteration
acc / 10_000, run[[999, 2499, 4999, 9999]]   # 0.9645 [0.4284 0.4333 0.4375 0.4371]
```

Intuition. Accept–reject needs the proposal to dominate the target everywhere, so it must have heavier tails; MH only needs the proposal to reach everywhere, and pays for poor tails with sticky stretches instead of impossibility. Beta$(4, 5)$ is a near-perfect body match (MH acceptance 96%) with tails that are slightly too thin.

Ref: Greenberg (2008), §5.3 (AR, Algorithm 5.2), §7.2 (independence kernels), §7.3 (convergence monitoring).

Question 33. Original math question — random-walk MH on the log scale for a Poisson rate.
Source: Original question (not from a book).

The coal posterior of Question 14 is $\pi(\theta \mid y) \propto \theta^{a-1}e^{-b\theta}$ with $a = 127.5$, $b = 42$. Pretend it is not recognized as a gamma and sample it by Metropolis–Hastings with the multiplicative random walk $\theta' = \theta e^u$, $u \sim N(0, s^2)$.

(a) Derive the proposal density $q(\theta, \theta')$ and the acceptance probability. Show that the same probability arises from a symmetric random walk on $\varphi = \log\theta$ with the target written in φ.

(b) Prove that the MH kernel (with its rejection mass) leaves π invariant, via detailed balance.

(c) Suppose the factor θ'/θ is forgotten. Find the distribution the chain then samples and the bias in the posterior mean.

(d) Parameter analysis: prove the limits of the acceptance probability as $s \to 0$ and $s \to \infty$ and explain why the numerical standard error is large at both extremes.

Programming (added, not in the book). Implement the sampler (start $\theta = 1$, 2,000 burn-in, 50,000 kept, `np.random.default_rng(33)` for each run) for $s \in \{0.02, 0.2, 1\}$, reporting acceptance rate, posterior mean and batch-means n.s.e. (50 batches) against the exact mean a/b; repeat $s = 0.2$ without the Jacobian factor.

Solution.

Notation.

- $\pi(\theta) \equiv \pi(\theta \mid y) \propto \theta^{a-1}e^{-b\theta}$, $\theta > 0$; $q(\theta, \theta')$: proposal density.
- $K(\theta, A)$: MH transition probability into a set A; $r(\theta)$: probability of staying put.
- $\varphi = \log\theta$; $\tilde{\pi}(\varphi)$: density of φ; $s > 0$: step size.

Step — (a) Proposal density. $\log\theta' = \log\theta + u \sim N(\log\theta, s^2)$, so θ' is log-normal:

$$q(\theta, \theta') = \frac{1}{s\,\theta'}\,\phi\Big(\frac{\log(\theta'/\theta)}{s}\Big), \qquad \theta' > 0. \tag{33.1}$$

Since ϕ is even, $\phi(\log(\theta/\theta')/s) = \phi(\log(\theta'/\theta)/s)$, and dividing (33.1) with the arguments swapped by (33.1):

$$\frac{q(\theta', \theta)}{q(\theta, \theta')} = \frac{1/(s\theta)}{1/(s\theta')} = \frac{\theta'}{\theta}. \tag{33.2}$$

Step — (a) Acceptance. Substituting (33.2) into (7.6) of Question 31:

$$\alpha(\theta, \theta') = \min\Big\{\frac{\pi(\theta')}{\pi(\theta)} \cdot \frac{\theta'}{\theta}, 1\Big\} = \min\Big\{\Big(\frac{\theta'}{\theta}\Big)^{a} e^{-b(\theta'-\theta)}, 1\Big\}. \tag{33.3}$$

Step — (a) Log scale. With $\theta = e^{\varphi}$, $d\theta = e^{\varphi}d\varphi$, the density of φ is $\tilde{\pi}(\varphi) = \pi(e^{\varphi})e^{\varphi}$. A symmetric walk $\varphi' = \varphi + u$ accepts with $\min\{\tilde{\pi}(\varphi')/\tilde{\pi}(\varphi), 1\} = \min\{\pi(\theta')\theta'/(\pi(\theta)\theta), 1\}$, the same as (33.3): the factor θ'/θ is the Jacobian.

Step — (b) Detailed balance. For $\theta \neq \theta'$, by (33.3) in general form,

$$\pi(\theta)q(\theta, \theta')\alpha(\theta, \theta') = \min\{\pi(\theta)q(\theta, \theta'),\ \pi(\theta')q(\theta', \theta)\} = \pi(\theta')q(\theta', \theta)\alpha(\theta', \theta). \tag{33.4}$$

The kernel is $K(\theta, A) = \int_A q(\theta, \theta')\alpha(\theta, \theta')d\theta' + r(\theta)\mathbf{1}\{\theta \in A\}$, with $r(\theta) = 1 - \int q(\theta, \theta')\alpha(\theta, \theta')d\theta'$. Integrating against π and using (33.4) to swap the roles of θ and θ' in the first term:

$$\int \pi(\theta)K(\theta, A)d\theta = \int_A \pi(\theta')\Big[\int q(\theta', \theta)\alpha(\theta', \theta)d\theta\Big]d\theta' + \int_A \pi(\theta)r(\theta)d\theta.$$

The bracket is $1 - r(\theta')$, so

$$\int \pi(\theta)K(\theta, A)d\theta = \int_A \pi(\theta')(1 - r(\theta'))d\theta' + \int_A \pi(\theta)r(\theta)d\theta = \int_A \pi = \pi(A). \qquad \blacksquare$$

Step — (c) Forgetting the Jacobian. Without θ'/θ the acceptance is $\min\{\pi(\theta')/\pi(\theta), 1\}$ while q is not symmetric. Writing it as a correct MH acceptance for some target $\pi^{\dagger}$ requires $\pi^{\dagger}(\theta')\theta'/(\pi^{\dagger}(\theta)\theta) = \pi(\theta')/\pi(\theta)$, i.e. $\pi^{\dagger}(\theta) \propto \pi(\theta)/\theta$. By (b) the chain is then invariant for

$$\pi^{\dagger}(\theta) \propto \theta^{a-2}e^{-b\theta} = G(a-1, b), \qquad \mathbb{E}^{\dagger}(\theta) - \mathbb{E}(\theta) = \frac{a-1}{b} - \frac{a}{b} = -\frac{1}{b}. \tag{33.5}$$

For the coal posterior: $126.5/42 = 3.011905$ instead of $127.5/42 = 3.035714$.

Step — (d) Limits. Write $R = (\theta'/\theta)^{a}e^{-b(\theta'-\theta)} = e^{au - b\theta(e^{u}-1)}$ with $u = sZ$, $Z \sim N(0,1)$. As $s \to 0$, $u \to 0$ and $R \to 1$ for every Z; since $\min\{R, 1\} \leq 1$, dominated convergence gives $E[\alpha] \to 1$. As $s \to \infty$, for $Z > 0$, $u \to \infty$ and $au - b\theta(e^{u} - 1) \to -\infty$; for $Z < 0$, $u \to -\infty$ and $au \to -\infty$ while $b\theta(1 - e^{u}) \leq b\theta$; so $R \to 0$ for every $Z \neq 0$ and $E[\alpha] \to 0$.

Step — (d) Reading. Tiny steps are almost always accepted but barely move, so successive draws are highly correlated; huge steps are almost always rejected and the chain sits still. Both inflate the n.s.e.; the best s is of the order of the posterior standard deviation of $\log\theta$, here $1/\sqrt{a} \approx 0.089$.

Code and output (programming part).

```
a, b = 127.5, 42.; lp = lambda t: (a - 1) * np.log(t) - b * t
def rw_log(s, jac, seed, G=50_000, burn=2_000):
    rng = np.random.default_rng(seed); th, acc, out = 1., 0, np.empty(G)
    for g in range(G + burn):
        pr = th * np.exp(s * rng.standard_normal())
        la = lp(pr) - lp(th) + (np.log(pr / th) if jac else 0.)   # (33.3)
        if np.log(rng.uniform()) <= la: th = pr; acc += g >= burn
        if g >= burn: out[g - burn] = th
    return acc / G, out.mean(), out.reshape(50, -1).mean(1).std(ddof=1) / np.sqrt(50)
# s     accept   mean      n.s.e.        (exact mean 3.035714)
# 0.02  0.9284   3.037804  0.011093
# 0.2   0.4636   3.037605  0.001913
# 1     0.1127   3.032593  0.005092
# 0.2, no Jacobian: 0.4664  3.015205  0.002112   (target of (33.5): 3.011905)
```

The middle step size has the smallest n.s.e., about six times smaller than $s = 0.02$; the Jacobian-free chain settles at 3.0152, 1.6 n.s.e. from its target 3.0119 in (33.5) and 9.7 n.s.e. below the true mean.

Intuition. A proposal that moves multiplicatively is not symmetric in θ, and MH must be told: the Jacobian is the price of proposing on one scale and targeting on another. Forgetting it does not make the sampler crash; it silently samples the wrong distribution.

Ref: Greenberg (2008), §7.2 (MH algorithm, random-walk kernels, pp. 97–100); Tsay (2010), §12.3 (Metropolis–Hastings).

Grading rubric.

- (a) (33.1)–(33.3): 4 pts; log-scale equivalence: 1 pt.
- (b) (33.4) and the integration with the rejection mass: 4 pts (detailed balance stated without the rejection term: max. 2).
- (c) (33.5): 3 pts.
- (d) both limits proved and the reading: 3 pts.
- Programming: three runs and the Jacobian-free run with correct comparisons: 3 pts.

Question 34. Original conceptual question — MCMC (10 points).
Source: Original question (not from a book).

a) (3 points) In MH with a symmetric random-walk proposal, $q(x, y) = q(y, x)$, the probability of accepting a move from x to y is:

A. $\min\{f(y)/f(x), 1\}$, the ratio of target densities capped at one.

B. $\min\{f(x)/f(y), 1\}$, the ratio of current to proposed density.

C. $f(y)/\{f(x) + f(y)\}$, the share of the proposed point in the pair.

D. 1, because a symmetric proposal needs no correction.

Key: A

Step — Why. A: (7.6) with the q's cancelling (Greenberg p. 99). B inverts the ratio, so it would favour moves downhill; C is Barker's rule, a different (valid but less efficient) acceptance; D ignores the target altogether.

b) (3 points) The Gibbs sampler is a special case of MH in which:

A. the proposal is a symmetric random walk on each block in turn.

B. every proposal is accepted with probability 1/2.

C. the draws are independent, as in accept–reject.

D. each block is proposed from its full conditional, and the acceptance probability is 1.

Key: D

Step — Why. D: with q equal to the full conditional, the ratio in (7.6) is exactly 1 (Greenberg §7.2). A and B describe other kernels; C is false: Gibbs draws are dependent (Question 11 found an inefficiency factor of 1.17).

c) (2 points, T or F) If a kernel satisfies detailed balance with respect to π, then π is an invariant distribution of the kernel.

Key: T

Step — Why. Question 33(b): integrate detailed balance over the starting point.

d) (2 points, T or F) Detailed balance is necessary for π to be invariant: every MCMC kernel with invariant distribution π is reversible.

Key: F

Step — Why. The converse of (c) fails. The systematic-scan Gibbs sampler of Question 11 (update θ_1, then θ_2, then k) leaves the posterior invariant, since each step does, but the composed kernel is not reversible in general. G 6.8's kernel was checked for invariance directly, not through detailed balance.

Programming (added, not in the book). Check (d) exactly on a toy target: for $p(x, y)$ on $\{0, 1\}^2$ with $p(0, 0) = 0.3$, $p(0, 1) = 0.2$, $p(1, 0) = 0.1$, $p(1, 1) = 0.4$, build the 4×4 transition matrices P_x (update x from $p(x \mid y)$) and P_y (update y from $p(y \mid x)$), the systematic scan $P = P_x P_y$, and report $\max |\pi P - \pi|$ and $\max_{i,j} |\pi_i P_{ij} - \pi_j P_{ji}|$.

Code and output (programming part).

```
p = np.array([[.3, .2], [.1, .4]]); st = [(0, 0), (0, 1), (1, 0), (1, 1)]
pi = np.array([p[s] for s in st])
Px = np.array([[p[a, d] / p[:, d].sum() if b == d else 0 for (a, b) in st]
               for (c, d) in st])                 # row: current (c, d)
Py = np.array([[p[c, b] / p[c, :].sum() if a == c else 0 for (a, b) in st]
               for (c, d) in st])
P = Px @ Py; F = pi[:, None] * P
np.abs(pi @ P - pi).max(), np.abs(F - F.T).max()      # 5.6e-17   0.053333
```

π is invariant to machine precision, yet the flow $\pi_i P_{ij}$ is not symmetric (largest gap 0.053333): invariance without reversibility. Each single update P_x, P_y is reversible; their product is not.